

FlyByWire feature user manuals

One manual per pull request (pr/01 to pr/26): what the feature does, how to use it, what to expect in each situation, and its current limitations with the reason

Aircraft	A32NX and A380X
Where to find it	This folder: each manual as HTML (with its <code>img/NN/</code> pictures) and as PDF; <code>FBW-user-manuals.pdf</code> holds them all
Status	Written 2026-09-28 against the builds installed in the sim (develop plus the pr/* features).
Pictures	Instrument pictures come from the local browser harness running the installed instrument bundles; the native WASM displays (weather radar, terrain, VD) are in-sim screenshots; audio features use diagrams.

1. The manuals

PR	Manual	Aircraft	Branch
01	A32NX weather radar on the ND (PDF)	A32NX	<code>pr/01-a32nx-nd-weather-radar</code> (base of PRs 2, 4, 3, 9)
02	A380X weather radar on the ND and weather on the VD (PDF)	A380X	<code>pr/02-a380x-nd-weather-radar</code> (stacked on PR 1)
03	Weather radar mode on the ND (PDF)	A32NX and A380X	<code>pr/03-nd-wxr-mode-label</code> (stacked on PR 4)
04	Terrain on the ND (TERR ON ND) and A380X VD terrain (PDF)	A32NX and A380X	<code>pr/04-nd-terrain</code> (stacked on PR 2)
05	A380X SURV controls and pedestal SURV panel (PDF)	A380X	<code>pr/05-a380x-surv-panel</code> (base of PR 6)
06	ECAM procedure lines for the SURV systems, and CABIN CREW on the LINE-UP checklist (PDF)	A380X	<code>pr/06-a380x-checklists-surv</code> (stacked on PR 5: needs its system selection)
07	TCAS in TA ONLY: intruders shown as TAs (PDF)	A32NX and A380X	<code>pr/07-tcas-ta-only-intruder-level</code> (independent)
08	Independent CPT and F/O MCDUs (PDF)	A32NX	<code>pr/08-a32nx-mcdu-independent</code> (independent; the close-up cockpit model needs a companion change in <code>aircraft-large-files</code>)
09	Radar, terrain and VD after the A380 and A320 manuals (PDF)	A380X (VD) and both aircraft (weather picture)	<code>pr/09-nd-manuals-pass</code> (top of the stack 1, 2, 4, 3)
10	A380X MFD FMS pages to the FCOM (PDF)	A380X	<code>pr/10-a380x-mfd-fcom-pages</code> (base of PRs 11-17, 20, 22)
11	A380 takeoff calculator and company T.O data (PDF)	A380X (the flyPad screen is shared with the A32NX, see the PR 12 manual)	<code>pr/11-a380-takeoff-calculator</code> (stacked on PR 10); completed by PR 12
12	A320 takeoff calculator and MCDU uplink takeoff data (PDF)	A32NX (the A380 part of the screen is in the PR 11 manual)	<code>pr/12-a320-takeoff-calculator</code> (stacked on PR 11)

13	A380X pilot-stored elements kept across sessions (PDF)	A380X	pr/13-pilot-stored-elements-persistence (stacked on PR 10)
14	A380X DIRECT TO options (PDF)	A380X	pr/14-direct-to-options (stacked on PR 10)
15	A380X fuel jettison (PDF)	A380X	pr/15-fuel-jettison (independent)
16	A380X KCCU keyboard entries (PDF)	A380X (MFD, OIT, ND OANS panel: every entry field of the displays)	pr/16-kccu-keyboard-entry (stacked on PR 10)
17	A380X ECAM procedures from the FCOM (PDF)	A380X	pr/17-ecam-procedures (independent)
18	A32NX RAAS runway awareness callouts (PDF)	A32NX	pr/18-a32nx-raas (independent)
19	A380X RWY AHEAD without an airport map (PDF)	A380X	pr/19-a380x-oans-rwy-ahead-without-map (stacked on PR 18: it uses the runway module of the RAAS PR)
20	A380X computed alternate fuel, EXTRA time and OFP fuel figures (PDF)	A380X	pr/20-a380x-alt-n-fuel-computation (stacked on PR 10)
21	A380X company datalink reply time (PDF)	A380X (the A32NX MCDU uplink stays instant)	pr/21-a380x-company-datalink-delay (stacked on PR 12)
22	A380X ATC COM pages and SD mailbox (PDF)	A380X	pr/22-a380x-mfd-atccom (stacked on PR 10)
23	flyPad landing calculator (PDF)	A380X and A32NX	pr/23-efb-landing-calculator (stacked on PR 12)
24	flyPad descent calculator (PDF)	A380X and A32NX	pr/24-efb-descent-calculator (stacked on PR 23)
25	A380X OANS default airport (PDF)	A380X	pr/25-a380x-oans-default-airport (independent)
26	flyPad taxi route (PDF)	A32NX and A380X (the same flyPad page on both)	pr/26-efb-taxi-route

2. How to read them

- **What it is for:** the problem the change solves, in one paragraph.
- **Using it** / numbered steps: where the controls are and what to press.
- **What you should see** tables: the expected behaviour in each situation; if the sim does something else, it is a bug worth reporting with the PR number.
- **Limitations and why:** what does not work yet, the reason (sim limit, missing data, not in the FCOM, out of scope) and what to do about it.
- Stacked PRs need the PRs below them installed: the branch line of each manual names them.
- Pictures that do not come from the harness (drawn diagrams, in-sim screenshots taken during development) carry an amber line under their caption: they are not representative of the in-game UI and serve as a reference for the explanation only.

A32NX weather radar on the ND

A native weather radar picture on both A32NX NDs, from MSFS's own radar data

Aircraft	A32NX
Where to find it	ND in ARC or ROSE NAV; the weather radar panel on the pedestal (WX SYS and MODE knobs)
PR branch	pr/01-a32nx-nd-weather-radar (base of PRs 2, 4, 3, 9)
Status	Tested in the sim over several sessions and weather presets (September 2026).
Pictures	In-sim screenshots taken during development (the radar is drawn by a native WASM gauge, which the browser harness cannot run).

1. What it is for

2. Using it

3. Reading the picture
4. What to expect

5. Limitations and why

1. What it is for

The A32NX ND had no working weather radar. A native gauge (ndwxr) now draws the MSFS weather radar on the CPT and F/O NDs, aligned with the range rings, in the colours of an airborne radar.



Figure 1. A32NX ND in ARC with weather ahead: green, yellow and red cells.

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

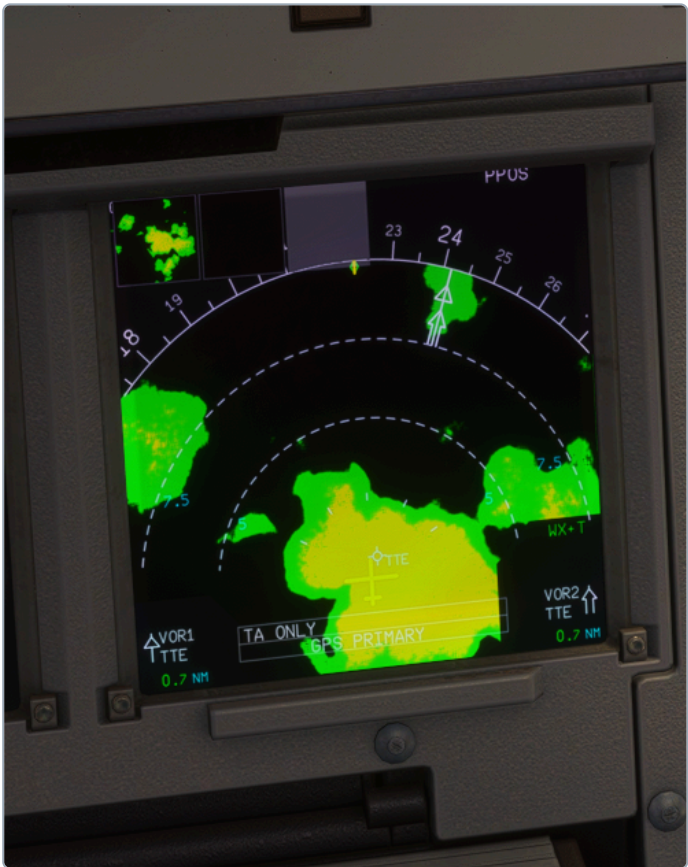


Figure 2. Another cell field, ROSE-like forward view.
In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

2. Using it

Control	Position	ND
WX SYS knob	SYS 1 / SYS 2	Radar on (system 1 or 2)

	OFF	No radar
MODE knob	WX	Precipitation
	WX+T	Precipitation and turbulence (magenta)
	TURB	Turbulence only
	MAP	Nothing (no ground map)
EFIS	ARC or ROSE NAV	Radar shown (forward half); nothing on ROSE ILS, ROSE VOR, PLAN

The selected mode is written in green on the ND (see the PR 3 manual).

3. Reading the picture

Colour	Rain rate	Meaning
Green	from 0.7 mm/h	Light precipitation
Yellow	from 4 mm/h	Moderate
Red	from 12 mm/h	Heavy: avoid
Magenta	above 20 mm/h, within 40 NM	Turbulence (WX+T, TURB)

The engine's rain rate is noisy pixel by pixel; each level is smoothed, so the picture shows solid areas with crisp edges.

4. What to expect

Situation	What you should see
On the ground	Nothing drawn: like a real radar, it does not transmit with both main gear compressed.
Clear sky	No picture (nothing to return).
WX SYS OFF, or MODE MAP	No picture.
CPT and F/O NDs	Independent: each ND uses its own IR position (IR 1 CPT, IR 2 F/O, IR 3 by the ATT HDG knob) and its own range and mode; the two knobs are shared.

5. Limitations and why

Limitation	Why	What to do
Turbulence is a proxy.	MSFS gives no turbulence data; rain above 20 mm/h within 40 NM is drawn magenta.	-
No tilt, no gain.	The MSFS radar has neither.	-
Drawn only airborne with weather around.	The MSFS radar image needs both.	Test with the MSFS weather presets.

A380X weather radar on the ND and weather on the VD

The native weather radar on both A380X NDs, driven by the SURV panel, and the weather on the vertical display

Aircraft	A380X
Where to find it	ND in ARC or ROSE NAV with the WX overlay selected on the EFIS panel; MFD SURV CONTROLS (WXR block); the VD under the ND
PR branch	pr/02-a380x-nd-weather-radar (stacked on PR 1)
Status	Tested in the sim (September 2026).
Pictures	In-sim screenshots taken during development (native WASM gauge; the browser harness cannot draw it).

1. What it is for
2. Using it
3. Weather on the VD
4. Limitations and why

1. What it is for

The A380X ND always showed WXR INOP (both WXR systems were hard-coded as failed) and had no radar picture. It now has the native radar of the A32NX, with the A380 rules, and the VD shows the weather on its vertical cut.



Figure 1. A380X ND with weather cells ahead, and the same cells on the VD below.

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

2. Using it

- EFIS control panel: select the **WX** overlay.
- MFD SURV CONTROLS (see the PR 5 manual): WXR AUTO, TURB on or off, MODE WX or MAP, WX ON VD.
- The radar is shown when the WXR & TAWS system selected on the SURV panel is not failed. A system is failed only while its AESU is reset or its power is lost (AC ESS for system 1, AC 4 for system 2).
- The ND range table is -1 (ZOOM, the airport map) and 10 to 640 NM; the radar is hidden at ZOOM.

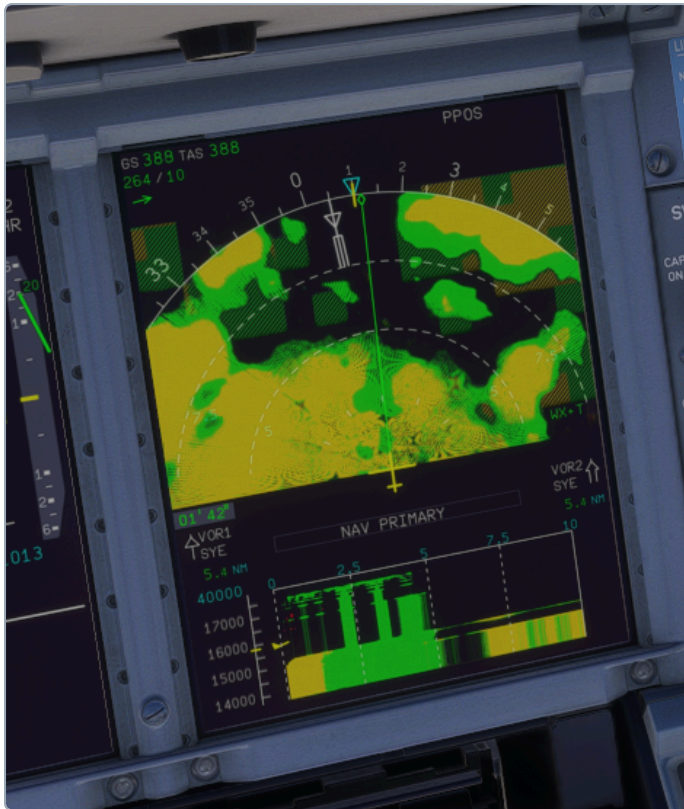


Figure 2. Moderate rain (yellow) over a wide area, with its VD cut.
In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

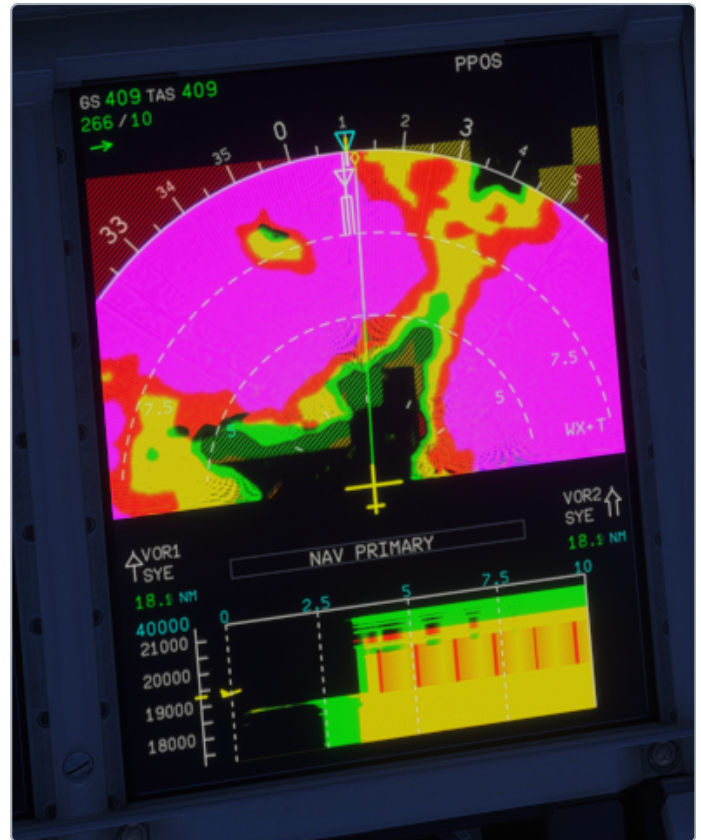


Figure 3. TURB on: heavy rain drawn magenta (turbulence proxy).
In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

3. Weather on the VD

- WX ON VD (SURV CONTROLS) shows the weather on the VD, along the same cut as the terrain (see the PR 9 manual).
- Cells are drawn green and yellow with a red core; no magenta: the real VD does not display turbulence (FCOM DSC-31-20-40-10).
- Heights are placed on the ADR baro-corrected altitude, like the VD's aircraft symbol.

4. Limitations and why

Limitation	Why	What to do
The VD weather is stylised.	The vertical MapView of MSFS is a single thin beam (tilting it crashes the gauge); the VD picture is built from the ND radar views.	-
PRED W/S and GAIN are not drawn.	The MSFS radar has no predictive windshear or gain.	-
No TURB-only mode.	The A380 has none: TURB off gives WX, on gives WX+T.	-

Weather radar mode on the ND

The selected weather radar mode (WX, WX+T, TURB, MAP) is shown in green on the ND, on both aircraft

Aircraft	A32NX and A380X
Where to find it	ND, ARC and ROSE NAV pages, right side, above the TERR box
PR branch	pr/03-nd-wxr-mode-label (stacked on PR 4)
Status	Tested in the sim (September 2026).
Pictures	Captain NDs in the local browser harness, running the builds installed in the sim (the radar picture itself is drawn by the native gauge, not by the harness).

1. What it is for

Nothing on the ND told the crew which radar mode was selected. The mode now appears in green on the right of the ARC and ROSE NAV pages while the radar is on, as on the real ND.

Label	A32NX (MODE knob)	A380X (SURV CONTROLS)
WX	WX	WX (TURB off)
WX+T	WX+T	WX with TURB on
TURB	TURB	-- (no TURB-only mode on the A380)
MAP	MAP	MODE MAP
(none)	WX SYS OFF	WXR off, no WX overlay, or the WXR & TAWS system failed



Figure 1. A32NX ARC: WX+T above the TERR box.

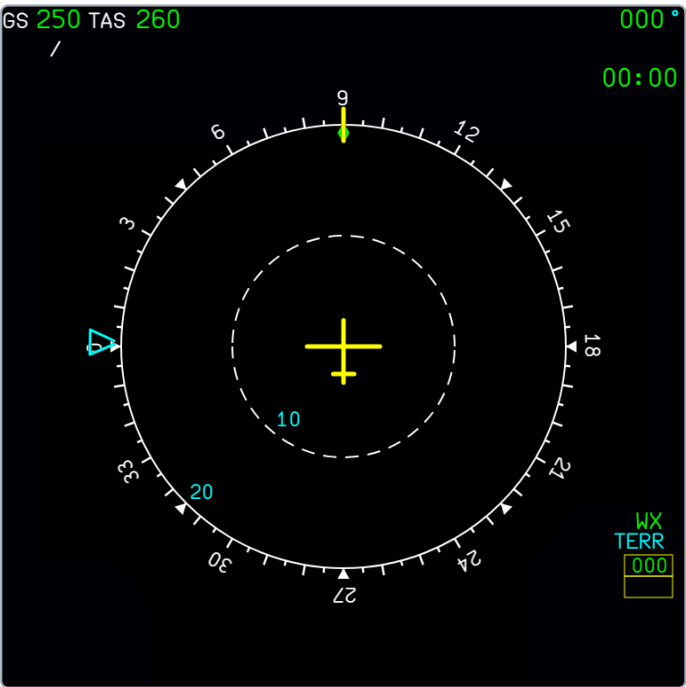


Figure 2. A32NX ROSE NAV: WX.

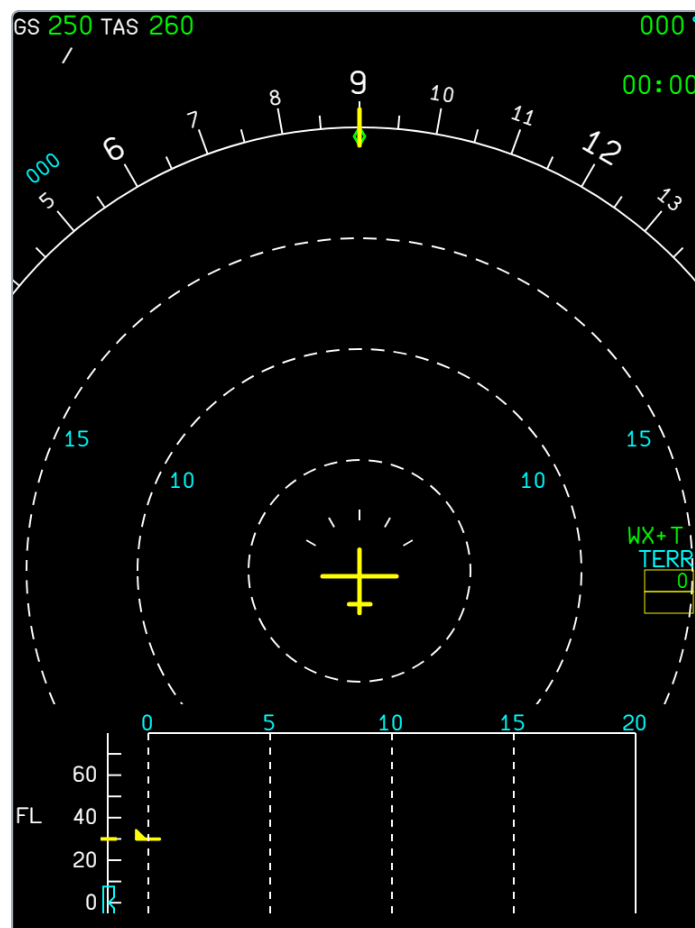


Figure 3. A380X ARC: WX+T.

- Shown only on ARC and ROSE NAV pages, and not while the terrain replaces the weather (TERR ON ND).
- Shown on the ground too, where the radar itself does not transmit (nothing is drawn).
- The native radar gauge decides and publishes the label ('A32NX_WXR_ND_{L,R}_MODE'), so the ND follows exactly what the gauge draws.

2. Limitations and why

Limitation	Why	What to do
No label in PLAN, ROSE ILS, ROSE VOR.	The radar is not shown on those pages.	-

Terrain on the ND (TERR ON ND) and A380X VD terrain

EGPWS terrain display on the ND of both aircraft without SimBridge, from MSFS's own terrain data, and the terrain profile on the A380X vertical display

Aircraft	A32NX and A380X
Where to find it	A32NX: TERR ON ND pushbutton; A380X: the TERR overlay on the EFIS panel (a WXR & TAWS system selected and not failed); the VD under the A380X ND
PR branch	pr/04-nd-terrain (stacked on PR 2)
Status	Tested in the sim (September 2026).
Pictures	In-sim screenshots at Aspen (KASE) during development (native WASM gauge).

1. What it is for

2. Reading the colours

3. A380X vertical display
4. The peaks box

5. Limitations and why

1. What it is for

TERR ON ND needed SimBridge. The native gauge now draws it from MSFS's own terrain, with the EGPWS colours, on every map page but PLAN, also on the ground.

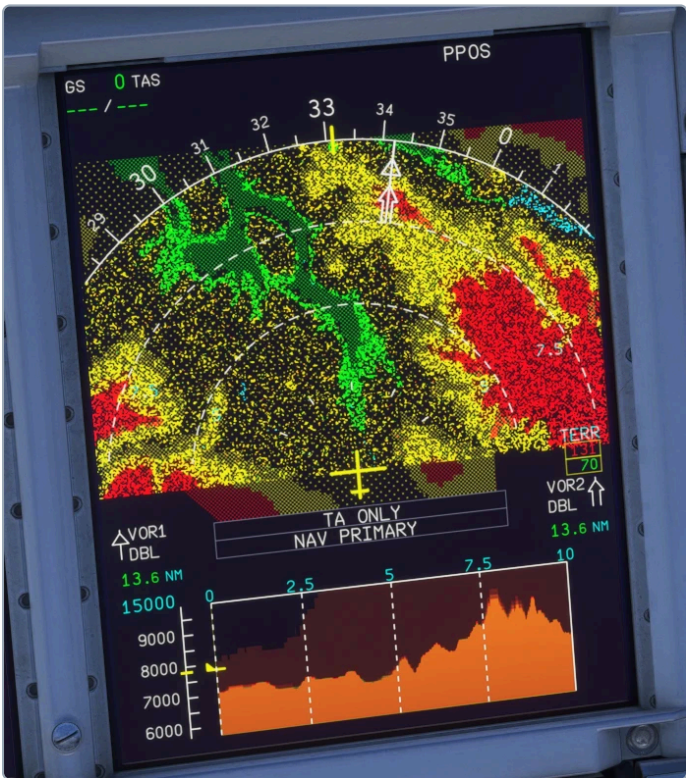


Figure 1. A380X at Aspen, 10 NM: terrain far above the aircraft in red, close in yellow, below in green; the VD terrain profile in brown.

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

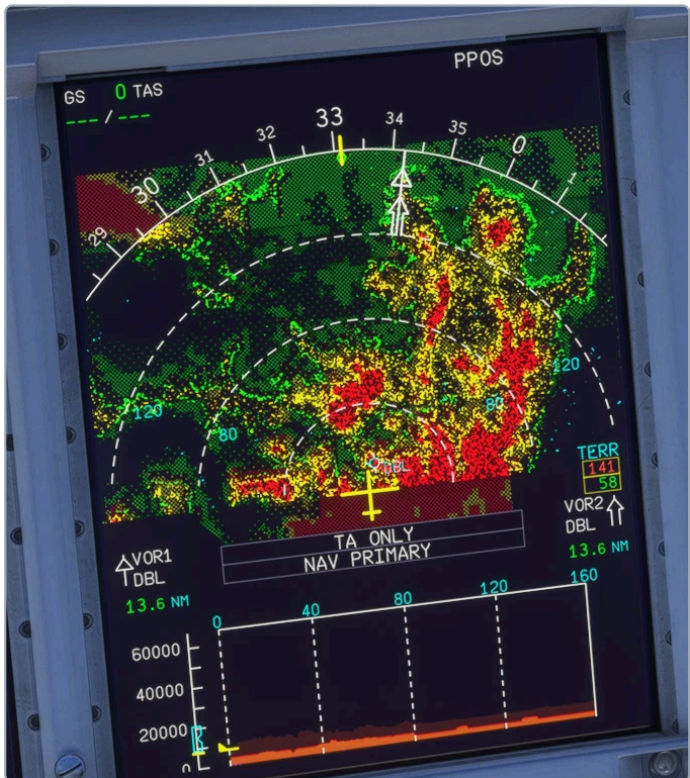


Figure 2. The same place at 160 NM.
In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

2. Reading the colours

Terrain relative to the aircraft	Colour
More than 2000 ft above	Dense red
1000 to 2000 ft above	Dense yellow
500 ft below (250 ft gear down) to 1000 ft above	Medium yellow
Down to 1000 ft below (500 ft gear down)	Dense green
1000 to 2000 ft below	Light green
More than 2000 ft below	Nothing
Water	Cyan-blue dots

The dots are a random stipple, as on the real display. The map turns with the ND's heading; ARC shows the forward half, the ROSE modes the whole disk.

3. A380X vertical display



Figure 3. The A380X VD terrain profile (brown), water in cyan from sea level down.

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

- Needs the TERR SYS of the SURV page not OFF (otherwise NO TERR DATA AVAILABLE, see the PR 5 manual).
- The cut follows the flight plan or the track (PR 9 manual).

4. The peaks box

The TERR box with the highest and lowest elevations of the range needs real elevations, which a gauge cannot read from MSFS: its figures come from **SimBridge** when it runs. Without SimBridge the box stays empty and the terrain is drawn as usual.

5. Limitations and why

Limitation	Why	What to do
At most 8 map views per module.	MSFS crashes a 9th; the A380X switches its views between weather and terrain.	-
Peaks box only with SimBridge.	No elevation readback from MSFS.	Run SimBridge.
Terrain far above the range top is shown red.	MSFS does not clamp to the first colour entry.	-

A380X SURV controls and pedestal SURV panel

The weather radar block of the MFD SURV CONTROLS page, the STATUS & SWITCHING page and the pedestal SURV panel now work and drive the same systems

Aircraft	A380X
Where to find it	MFD system menu SURV (or the KCCU SURV key): CONTROLS and STATUS & SWITCHING; the SURV panel on the pedestal
PR branch	pr/05-a380x-surv-pane1 (base of PR 6)
Status	Tested in the sim (September 2026).
Pictures	Captain MFD and ND in the local browser harness, running the build installed in the sim; figure 3 is an in-sim screenshot.

1. What it is for

2. SURV CONTROLS

3. STATUS & SWITCHING
4. The pedestal SURV panel

5. Messages on the vertical display

6. Limitations and why

1. What it is for

The A380 has no separate radar, TCAS or TAWS control panels: they are on the MFD SURV pages and the pedestal SURV panel (FCOM DSC-34-20-60-50). In FBW the weather radar block of the page was a stub (every button greyed), STATUS & SWITCHING did not reflect failures, and no button of the pedestal panel worked. Now they do, and the page and the panel stay in step.

2. SURV CONTROLS



Figure 1. SURV CONTROLS: XPDR (code, IDENT, AUTO / ON / STBY, ALT RPTG), TCAS (TA/RA, TA ONLY, STBY; NORM, ABV, BLW), WXR, TAWS, and DEFAULT SETTINGS.

WXR button	Effect
WXR AUTO / OFF	Weather radar on or off (the ND radar picture).
PRED W/S AUTO / OFF	Predictive windshear (written for the systems; MSFS has no predictive windshear).
TURB AUTO / OFF	Turbulence detection (magenta on the ND, see the radar manuals).
GAIN AUTO / MAN	Radar gain (written; MSFS has no gain).
MODE WX / MAP	Weather or ground mapping.
WX ON VD AUTO / OFF	Weather on the vertical display.
ELEVN / TILT	AUTO, or a manual elevation or tilt.

The WXR buttons are inhibited only when the selected WXR system has really failed.

3. STATUS & SWITCHING

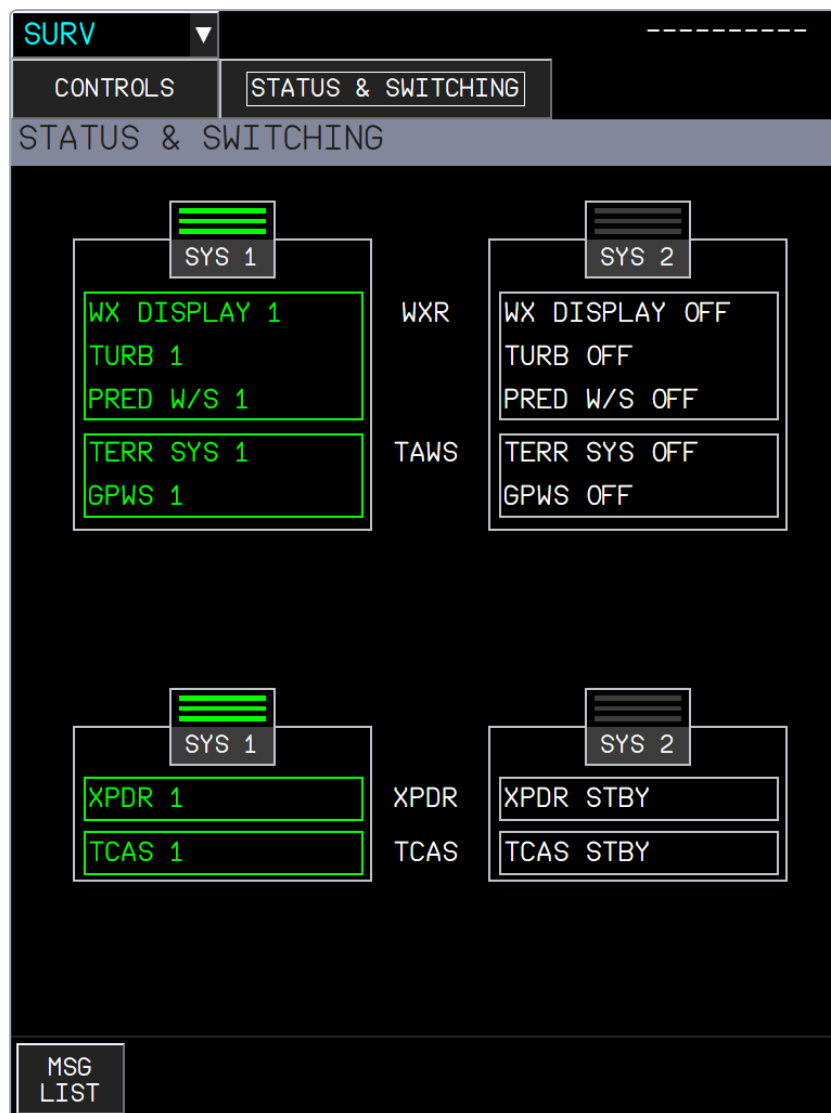


Figure 2. STATUS & SWITCHING: SYS 1 / SYS 2 of the WXR and TAWS (top) and of the XPDR and TCAS (bottom); the selected system is lit, each item shows its real state.

- WXR DISPLAY, TURB, PRED W/S, TERR SYS, GPWS items show the real failure of each system.
- XPDR / TCAS SYS 1 / SYS 2 select the transponder system (the one the TCAS uses) and light from it.



Figure 3. The SURV CONTROLS page in the sim (in-sim screenshot).

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

4. The pedestal SURV panel

The panel buttons never worked: their light template runs the button code only when the panel is powered, and the pedestal never told it so. They are powered from DC 1 now and linked to the MFD pages:

Panel button	Does	Same as
WXR TAWS SYS 1 / SYS 2	Selects the WXR / TAWS system	STATUS & SWITCHING, top
XPDR TCAS SYS 1 / SYS 2	Selects the transponder system	STATUS & SWITCHING, bottom
G/S MODE	Glide slope mode of the GPWS on / off	SURV CONTROLS, TAWS
TCAS ABV / BLW / TA ONLY	TCAS display and mode	SURV CONTROLS, TCAS

Each button lights from the same value the page uses, so pressing one on the pedestal changes the page, and the reverse.

5. Messages on the vertical display

The VD message was always "NO TERR AND WX DATA AVAILABLE" when TERR SYS was off. Now:

Situation	VD message
TERR SYS OFF only	NO TERR DATA AVAILABLE
WXR failed or OFF, or WX ON VD OFF	NO WX DATA AVAILABLE

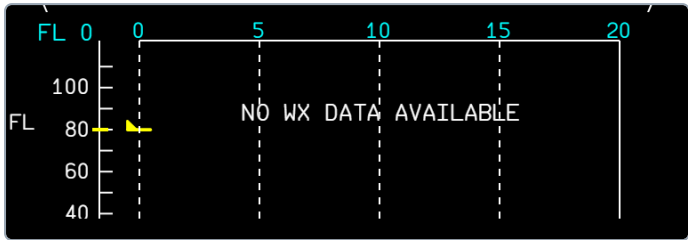


Figure 4. WXR off: NO WX DATA AVAILABLE.

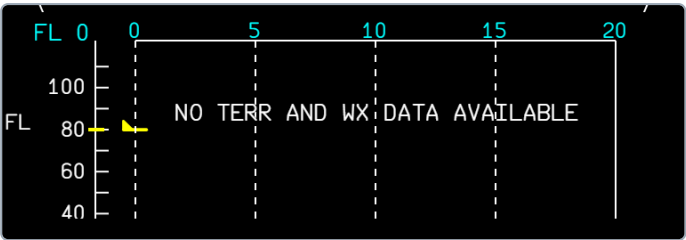


Figure 5. TERR SYS off and WXR off: NO TERR AND WX DATA AVAILABLE.

6. Limitations and why

Limitation	Why	What to do
PRED W/S and GAIN MAN do nothing visible.	The MSFS radar has no predictive windshear and no gain; the values are written for when it does.	-
One MSFS radar for both systems.	SYS 1 / SYS 2 select the failure state and the lights; the picture comes from the same sim radar.	-

ECAM procedure lines for the SURV systems, and CABIN CREW on the LINE-UP checklist

The WXR & TAWS and XPDR & TCAS lines of the ADR / IR / TCAS fault procedures tick from the real system selection

Aircraft	A380X
Where to find it	EWD procedures (ADR x+y FAULT, IR x+y FAULT, TCAS 1 / 2 FAULT); MFD SURV STATUS & SWITCHING page; EWD checklist LINE-UP
PR branch	pr/06-a380x-checklists-surv (stacked on PR 5: needs its system selection)
Status	Tested in the sim (September 2026).
Pictures	EWD and MFD in the local browser harness, running the builds installed in the sim; the EWD states are the ones the FWS sends.

1. What it is for

2. Where the systems are selected

3. What you see
4. LINE-UP checklist

5. Limitations and why

1. What it is for

The dual ADR and IR fault procedures ask the crew to move the WXR & TAWS and the XPDR & TCAS to the system that still has valid data. Those lines were hard-coded as done, so the ECAM ticked them even when the wrong system stayed selected. They now follow the selection made on the MFD SURV STATUS & SWITCHING page (PR 5), and the TCAS 1 / 2 FAULT procedures show their XPDR & TCAS line, which was hidden.

2. Where the systems are selected

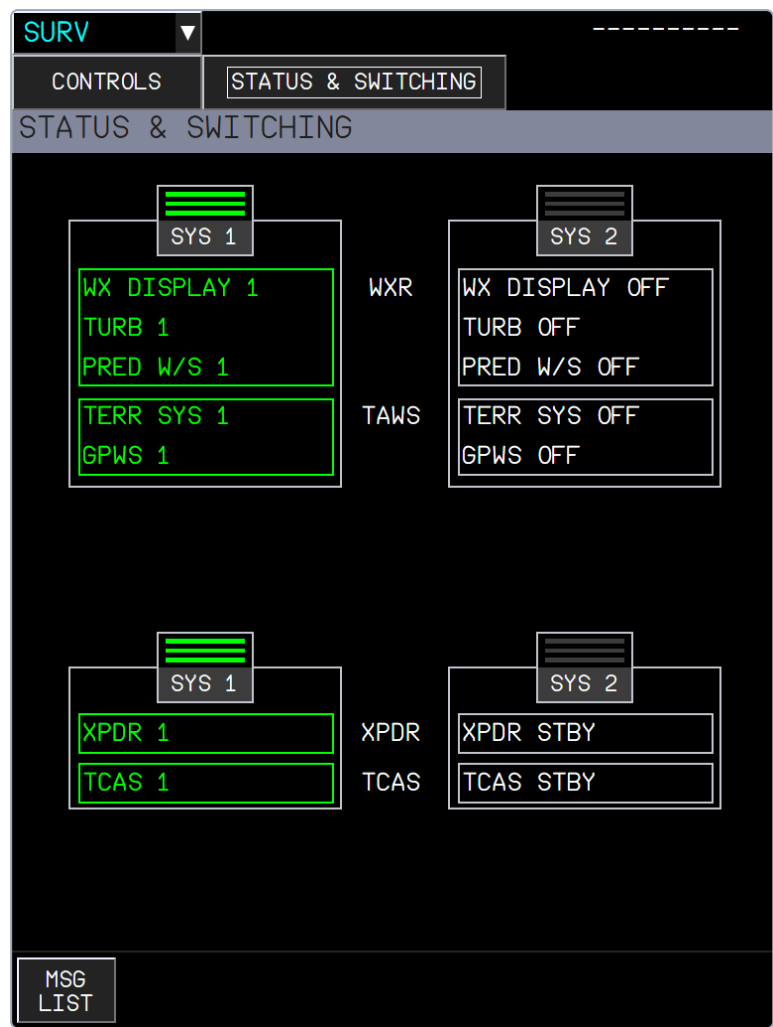


Figure 1. MFD SURV STATUS & SWITCHING: the SYS 1 / SYS 2 buttons of WXR & TAWS (top) and XPDR & TCAS (bottom).

3. What you see

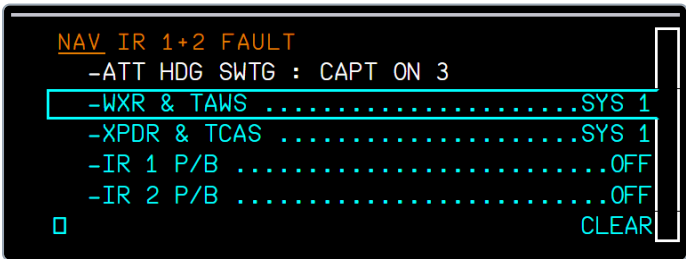


Figure 2. IR 1+2 FAULT with both systems on SYS 2: the two lines wait, in cyan.

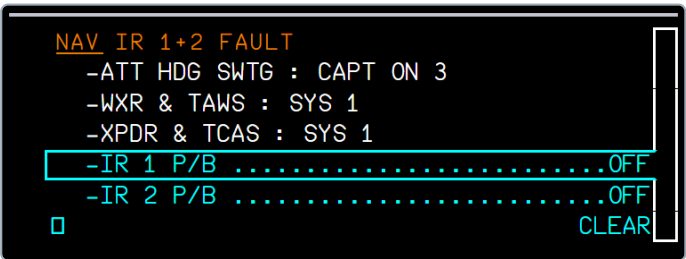


Figure 3. After selecting SYS 1 on the SURV page: the lines are done (white) and the cursor moves on.

Procedure	WXR & TAWS	XPDR & TCAS
ADR 1+2 FAULT, ADR 2+3 FAULT, IR 1+2 FAULT, IR 2+3 FAULT	SYS 1	SYS 1
ADR 1+3 FAULT, IR 1+3 FAULT	SYS 2	SYS 2
TCAS 1 FAULT	-	SYS 2
TCAS 2 FAULT	-	SYS 1

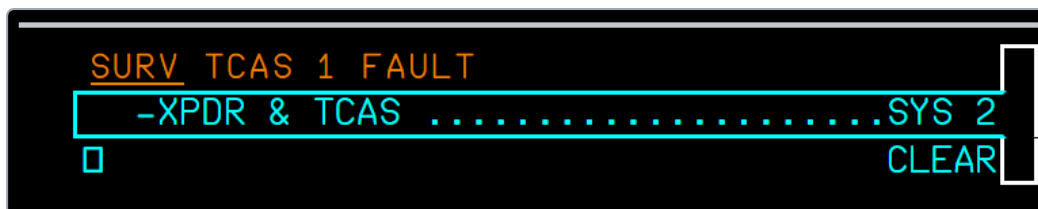


Figure 4. TCAS 1 FAULT: the XPDR & TCAS line is now shown.

Situation	What you should see
The right system is already selected when the fault appears	The line is shown already done.
The crew selects the other system	The line ticks within a second; the cursor moves to the next line.
The crew goes back to the failed side	The line becomes due again (it follows the live selection).

4. LINE-UP checklist

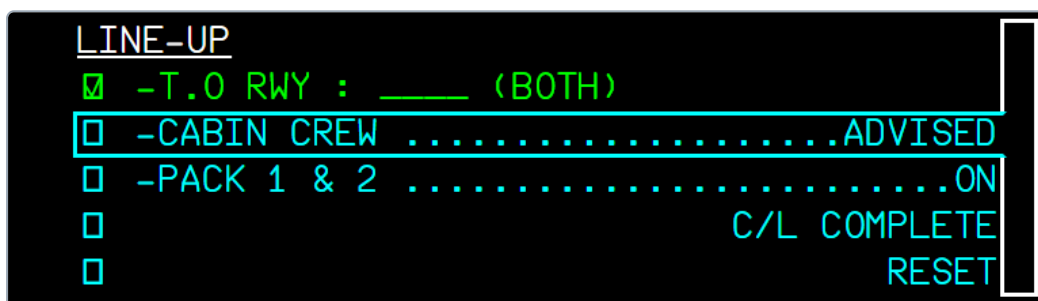


Figure 5. ECAM LINE-UP: the CABIN CREW ... ADVISED line (crew check).

The ECAM LINE-UP checklist gets the CABIN CREW line that the flyPad checklist already had (A380 FCOM BEFORE TAKEOFF checklist: CABIN CREW ... ADVISE). It is ticked by the crew.

5. Limitations and why

Limitation	Why	What to do
No SURV lines in the normal checklists.	The A380 FCOM normal checklists (PRO-NOR-C-L) have none; an earlier version of this change added some and was reverted.	-
Needs PR 5.	The system selection LVars come from the SURV panel of PR 5.	Install both.

TCAS in TA ONLY: intruders shown as TAs

While resolution advisories are inhibited, an intruder that would be an RA is shown and announced as a traffic advisory; the TA ONLY message is centred on the A32NX ND as in the FCOM

Aircraft	A32NX and A380X
Where to find it	ND traffic display; TCAS mode on the A32NX TCAS panel / A380X SURV CONTROLS
PR branch	pr/07-tcas-ta-only-intruder-level (independent)
Status	Tested in the sim (September 2026).
Pictures	Captain NDs in the local browser harness, running the builds installed in the sim; the symbol legend is a drawing (the harness has no traffic).

1. What it is for

With every RA inhibited (TA ONLY selected, or below 1000 ft radio altitude), the TCAS kept classifying an intruder that met the RA criteria as an RA. The ND drew it as a red RA square, and since it was not a TA, there was no "TRAFFIC, TRAFFIC" and no TA state. A real TCAS in TA ONLY shows it as an amber TA with the TA aural. The classification is now capped at TA while all RAs are inhibited.



Figure 1. The ND traffic symbols: other, proximate, TA, RA, with the relative altitude in hundreds of feet and the vertical trend arrow.
Diagram drawn for this manual: not representative of the in-game UI, reference for the explanation only.

2. What you see



Figure 2. A32NX: TA ONLY, centred at the bottom of the ND (A320 FCOM DSC-34-SURV-60-20 figure).

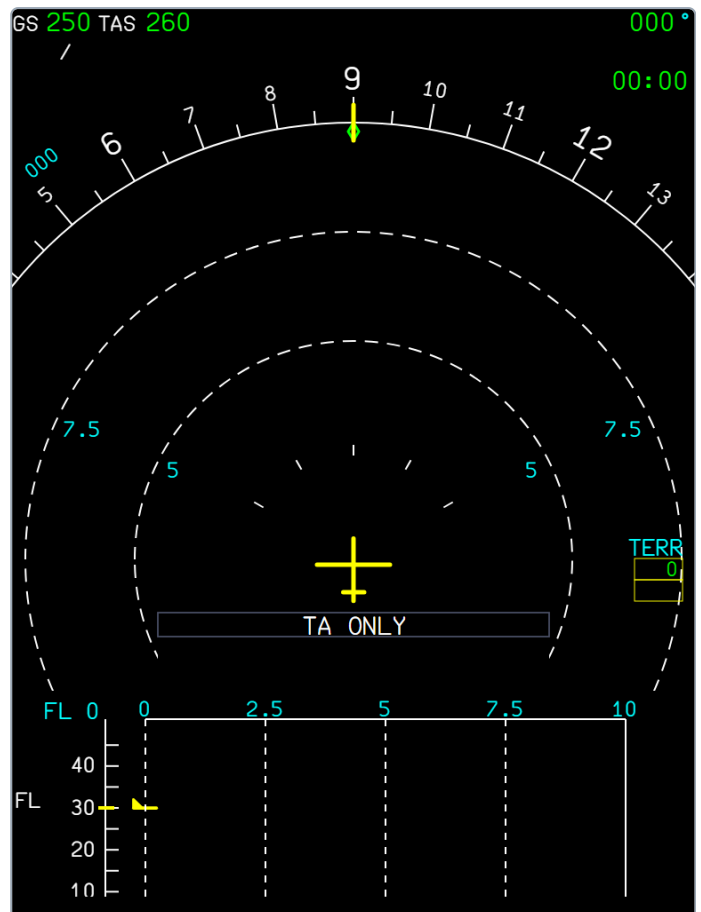


Figure 3. A380X: TA ONLY in the message box.

Situation	What you should see
TA ONLY selected, intruder meeting the RA criteria	Amber TA circle and "TRAFFIC, TRAFFIC" (before: red RA square, no aural).
Below 1000 ft radio altitude (all RAs inhibited)	Same: TAs only, TA ONLY on the ND.
Partial inhibits (no CLIMB / DESCEND / INCREASE DESCENT RA)	Unchanged: RAs are still issued.
An RA already running when the inhibit starts	It runs to its own end.

3. Limitations and why

Limitation	Why	What to do
Same change on both aircraft.	The A32NX TcasComputer and the A380X LegacyTcasComputer share the classification code.	-

Independent CPT and F/O MCDUs

Each MCDU has its own page, scratchpad and messages, on the one shared FMS

Aircraft	A32NX
Where to find it	The two MCDUs on the pedestal
PR branch	pr/08-a32nx-mcd�-independent (independent; the close-up cockpit model needs a companion change in aircraft-large-files)
Status	Tested in the sim (September 2026).
Pictures	The MCDU gauge in the local browser harness, running the build installed in the sim; the gauge draws the CPT screen in its left half and the F/O screen in its right half, shown here side by side.

1. What it is for

2. What is per MCDU and what is shared

3. Messages
4. Power

5. Limitations and why

1. What it is for

The two MCDUs mirrored each other: same page, same scratchpad. On the aircraft each MCDU is its own display, working on the shared FMGC data (A320 FCOM DSC-22_10-10). Now each pilot can work on a different page.

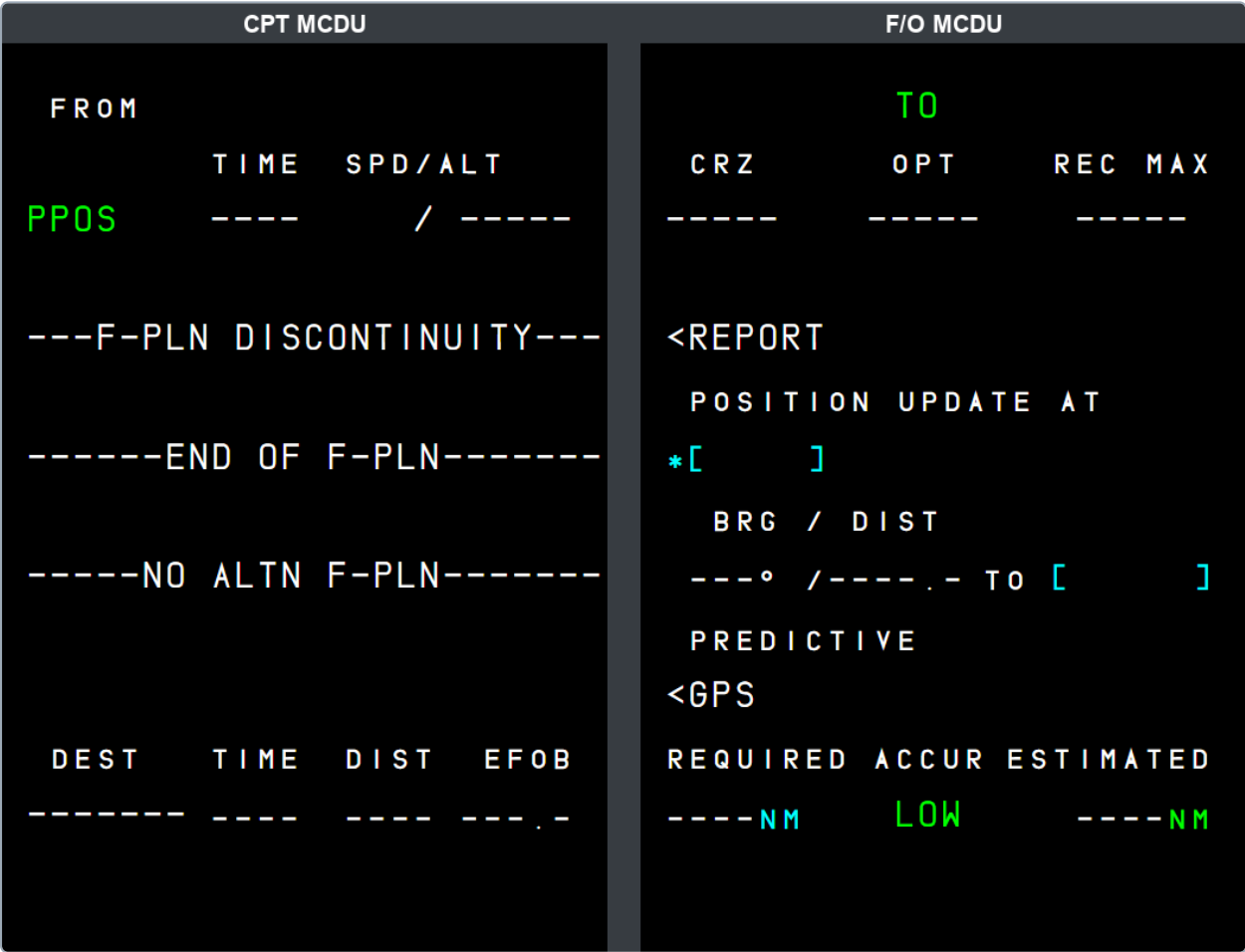


Figure 1. CPT on F-PLN, F/O on PROG at the same time.

2. What is per MCDU and what is shared

Per MCDU	Shared (one FMS)
----------	------------------

The page shown, the line keys	The flight plans, performance, fuel, navigation data
The scratchpad and what is typed in it	The data entered from either MCDU
The scratchpad messages (FORMAT ERROR, NOT ALLOWED ...) of an entry	FMGC messages (queued on both)
The keyboard entry (the MCDU that was clicked)	The EFIS symbols, the guidance

CPT MCDU	F/O MCDU
FROM	TO
TIME SPD/ALT	CRZ OPT REC MAX
PPOS --- / ---	-----
---F-PLN DISCONTINUITY---	<REPORT
-----END OF F-PLN-----	POSITION UPDATE AT
-----NO ALTN F-PLN-----	*[]
	BRG / DIST
	---° /-----.- TO []
	PREDICTIVE
DEST TIME DIST EFOB	<GPS
-----	REQUIRED ACCUR ESTIMATED
LFBO	---NM LOW ---NM
	250/12

Figure 2. Each scratchpad keeps its own text: LFBO on the CPT, 250/12 on the F/O.

3. Messages

CPT MCDU	F/O MCDU
FROM	INIT ↔
TIME SPD/ALT	CO RTE FROM/TO
PPOS --- / ---	----- REQUEST*
---F-PLN DISCONTINUITY---	ALTN/CO RTE INIT
-----END OF F-PLN-----	-----/-----
-----NO ALTN F-PLN-----	FLT NBR

	WIND/TEMP>
DEST TIME DIST EFOB	COST INDEX TROPO
-----	--- 36090
LFBO	CRZ FL/TEMP GND TEMP
	----- /---° ---°
	FORMAT ERROR

Figure 3. A wrong entry on the F/O MCDU: FORMAT ERROR there only.

CPT MCDU	F/O MCDU
FROM	INIT ↔
TIME SPD/ALT	CO RTE FROM/TO
PPOS --- / ---	----- REQUEST*
---F-PLN DISCONTINUITY---	ALTN/CO RTE INIT
-----END OF F-PLN-----	-----/-----
-----NO ALTN F-PLN-----	FLT NBR

	WIND/TEMP>
DEST TIME DIST EFOB	COST INDEX TROPO
-----	--- 36090
LFBO	CRZ FL/TEMP GND TEMP
	----- /---° ---°
	XYZ

Figure 4. CLR on the F/O MCDU: its message clears and its entry comes back; the CPT scratchpad is untouched.

Situation	What you should see
An entry is rejected	The message shows on the MCDU where the entry was made.
An FMGC message (e.g. CHECK TAKE OFF DATA)	Queued on both MCDUs; CLR on one clears it there only.
The FMS changes pages by itself (flight phase change on PROG / PERF / IDENT, engine-out page, INIT B to FUEL PRED)	Applies to both MCDUs.
ATSU system status (not the answer to an entry)	On the ATSU scratchpad of both MCDUs.
Typing with the PC keyboard	Goes to the MCDU that was clicked last; only that MCDU animates its keys.
SimBridge remote MCDU	CPT screen on its left side, F/O screen on its right side.

4. Power

- CPT MCDU: AC ESS SHED bus. F/O MCDU: AC 2 bus.
- An unpowered MCDU ignores its keys; the other one keeps working.

5. Limitations and why

Limitation	Why	What to do
No return to the MENU page after a power loss.	Same as the CPT MCDU before this change.	-
Close-up cockpit model: both screens show the whole texture until the companion change is merged.	The 4 UV values of the high-detail model (LOD00) live in the aircraft-large-files repository; this PR only changes LOD01.	Merge the aircraft-large-files PR first.

Radar, terrain and VD after the A380 and A320 manuals

The vertical cut, the VD scale and terrain placement, and the weather picture reworked to the FCOM / FCTM

Aircraft	A380X (VD) and both aircraft (weather picture)
Where to find it	A380X VD; both NDs with the radar
PR branch	pr/09-nd-manuals-pass (top of the stack 1, 2, 4, 3)
Status	Tested in the sim (September 2026).
Pictures	In-sim screenshots taken during development (native WASM gauge).

1. The vertical cut (A380X VD)
2. The VD scale and the terrain height
3. The weather picture (both aircraft)
4. Limitations and why

1. The vertical cut (A380X VD)

- In the managed lateral modes the VD shows the terrain and weather **along the flight plan** (a 200 NM window from just behind the aircraft, with the turns); in the selected modes **along the track** (FCOM: VIEW ALONG ACFT TRK). More than 5 NM off the plan: the track.
- The terrain is the highest across a width of $2 \times \text{RNP}$ (1 NM terminal, 2 NM en route above 18 000 ft).
- The grey area starts at the next track change of more than 3° (the VD no longer matches the ND beyond it).

2. The VD scale and the terrain height

- With a trajectory, the highest of the aircraft and the path sits at three quarters of the window: **in cruise the ground stays on the plot**, as on the real aircraft at FL400.
- The gap under the aircraft symbol is the true height, whatever the altimeter setting (FCOM / FCTM).
- Water in the ND teal; no magenta on the VD; no VD weather when TERR is not available.



Figure 1. A380X VD with the terrain profile and water (in-sim).

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

3. The weather picture (both aircraft)

- The radar scans on the ground too (A320 FCOM), but draws nothing there.
- MAP mode is a ground map from the terrain.
- The picture **builds up over 30 s** after the radar starts, sweeping left to right (A380 FCOM).
- The ARC picture keeps the ND bottom corners free, as on the real display.

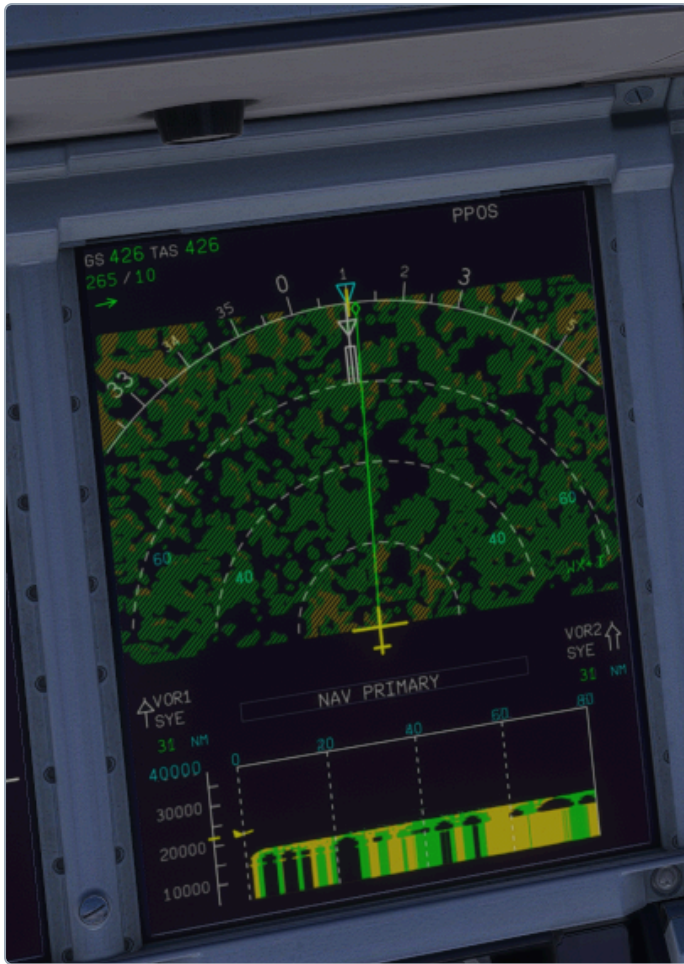


Figure 2. Just after WX is selected: the picture is building up.

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

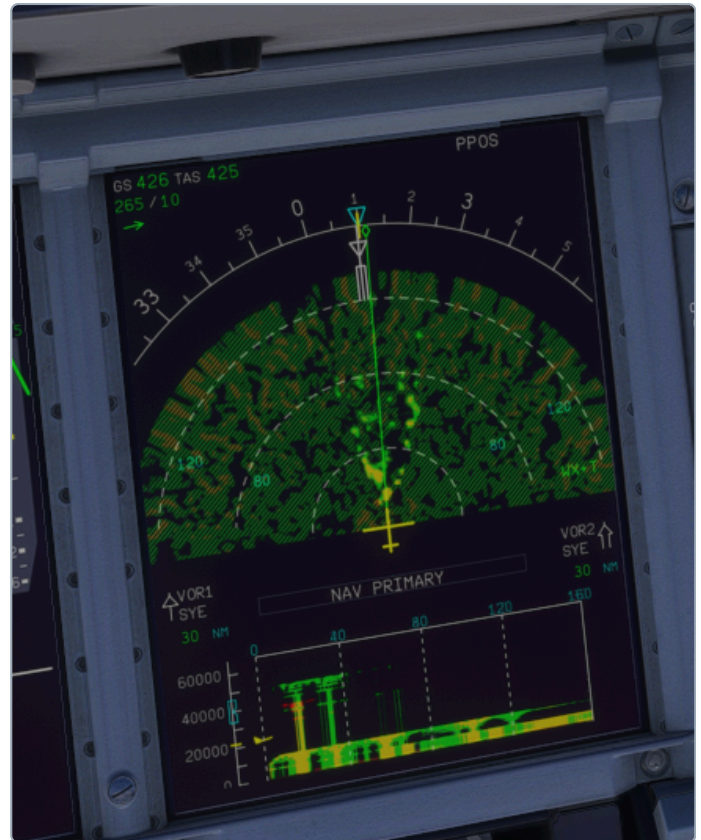


Figure 3. A few seconds later.

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

4. Limitations and why

Limitation	Why	What to do
No A380 AUTO off-path display.	MSFS runs one radar per aircraft; a third view changed what the beam views delivered, so it was removed.	-

A380X MFD FMS pages to the FCOM

Every FMS page of the MFD laid out on its A380 FCOM figure, the FCOM entry formats, and the pages and functions FBW did not have

Aircraft	A380X
Where to find it	MFD (both sides), FMS pages: ACTIVE , POSITION , SEC INDEX , DATA menus; SURV pages (see also the PR 5 manual)
PR branch	pr/10-a380x-mfd-fcom-pages (base of PRs 11-17, 20, 22)
Status	Built in September 2026 and used in the sim since; individual pages were fixed after the in-sim checks (button widths, SEC page return, printer checkboxes).
Pictures	Captain MFD in the local browser harness, running the build installed in the sim. The harness has no navigation database and no flight plan, so most values are dashes or amber boxes: the pictures show the layout and the functions of each page.

1. What changed, in short

2. ACTIVE menu

3. POSITION menu

4. SEC INDEX
5. DATA menu

6. What to expect

7. Limitations and why

1. What changed, in short

- **Layout:** each page is placed on its figure of the A380 FCOM (DSC-22-FMS-20-30, the 2011 KAL FCOM): field positions, 40 px entry fields, 22.5 px labels, 38 px tab bar, the title bar. Where FBW followed a newer FMS standard than this FCOM, the FCOM figure was followed.
- **Entry formats:** every field accepts the FCOM entry formats (DSC-22-FMS-20-100) and answers with the FCOM messages (FORMAT ERROR, ENTRY OUT OF RANGE, NOT IN DATABASE...).
- **New pages:** ALTERNATE, CLOSEST AIRPORTS, OFFSET, EQUI-TIME POINT, LL XING / TIME MKR, WIND, the POSITION pages, DATA / WAYPOINT, NAVAID, ROUTE, PRINTER, company requests (wind, flight plan, report, takeoff data), TRANSFER TO MAILBOX, REJECTED ATC INFO.
- **New functions:** FUEL PLANNING with CONFIRM BLOCK, LRC and MAX TURB speeds, VERT REV RTA, FMS P/N, pilot-stored elements, automatic airway connection, printing.

The menus at the top of the FMS pages give access to everything: **ACTIVE** (the active flight plan and its pages), **POSITION**, **SEC INDEX** (the secondary flight plans) and **DATA**.

2. ACTIVE menu

INIT

The screenshot shows the 'INIT' menu in a flight simulator. At the top, there's a dropdown menu for 'FMS 1' and a series of tabs: 'ACTIVE', 'POSITION', 'SEC INDEX', and 'DATA'. Below these is a header 'ACTIVE/INIT'. The main area contains several input fields and buttons. 'FLT NBR' is a 10-digit amber box (0000000000). 'ACFT STATUS' is a button. 'CPNY F-PLN REQUEST' is a button. 'FROM' and 'TO' are 4-digit amber boxes (0000), followed by 'ALTN' (4 dashes). 'CPNY RTE' is a 10-digit amber box (0000000000), followed by 'RTE SEL' button. 'ALTN RTE' is a 10-digit box (-----), followed by 'ALTN RTE SEL' button. Below a horizontal line, 'CRZ FL' is a box (FL----), 'CRZ TEMP' is a box (--- °C), 'CI' is a 3-digit box (---), and 'TROPO' is a box (36090 FT). 'TRIP WIND' is a 5-digit box (-----), followed by 'WIND' button. 'CPNY WIND REQUEST' is a button. Below another horizontal line, there are buttons for 'IRS', 'DEPARTURE', 'NAVAIDS', 'FUEL&LOAD', 'T.O. PERF', 'RTE SUMMARY', and 'CPNY T.O. REQUEST'. At the bottom left is a 'MSG LIST' button.

Figure 1. INIT: every button is live. CPNY F-PLN REQUEST (flight plan from the airline, i.e. SimBrief), RTE SEL / ALTN RTE SEL (route selection from the stored and company routes), CPNY WIND REQUEST, CPNY T.O. REQUEST, WIND, and the shortcuts IRS, DEPARTURE, NAVAIDS, FUEL&LOAD, T.O. PERF and RTE SUMMARY.

- Amber boxes are mandatory entries; blue dashes are optional ones.
- **CPNY RTE:** type a stored or company route of the city pair: it is inserted; otherwise NOT IN DATABASE.
- **RTE SUMMARY** stores the active route as a pilot-stored route (DATA / ROUTE).

FUEL&LOAD and FUEL PLANNING

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/FUEL&LOAD

GW

.

T

CG

.

%

FOB

.

T

ZFW

000.0

T

ZFWCG

00.0

%

BLOCK

000.0

T

FUEL
PLANNING *

TAXI

1.5

T

PAX NBR

000

TRIP

.

T

:

CI

RTE RSV

.

T

5.0

%

JTSN GW

.

T

ALTN

0.0

T

:

TOW

.

T

FINAL

6.0

T

00:30

LW

.

T

UTC

EF0B

MIN FUEL AT DEST

DEST

NONE

:

.

T

6.0

T

ALTN

NONE

:

.

T

EXTRA

.

T

:

RETURN

MSG
LIST

Figure 2. FUEL&LOAD. FUEL PLANNING (next to BLOCK) is available when the flight plan and the weights allow the predictions.

FUEL PLANNING computes the minimum block fuel from the FMS predictions (trip, reserves, alternate, final, taxi) and offers **CONFIRM BLOCK**, which inserts it (FCOM P 179). The EXTRA line shows what is left above the minimum.

PERF

FMS 1

ACTIVEPOSITIONSEC INDEXDATA

ACTIVE/PERF

CRZFL 000OPT FL ---REC MAX FL ---

T.OCLB

CRZDESAPPRGA

RWY ---T.O SHIFT --- M

V1 000 KT

F --- KT

TOGA

FLEX

DERATED

VR 000 KT

S --- KT

V2 000 KT

O --- KT

FLAPS

THS FOR

PACKS

ANTI-ICE

-

00.0 %

ON

OFF

THR RED

ACCEL

NOISE

----- FT

----- FT

TRANS

EO ACCEL

CPNY T.O REQUEST

----- FT

----- FT

RETURN

POS MONITOR

MSG LIST

Figure 3. PERF T.O with the company takeoff data button (see the PR 11 manual).

FMS 1

ACTIVEPOSITIONSEC INDEXDATA

ACTIVE/PERF

CRZFL 000OPT FL ---REC MAX FL ---

T.OCLB

CRZDESAPPRGA

CI ---

MODE	MACH	SPD	AT STEP TO FL
PRESEL	.--	--- KT	
MANAGED	.85	--- KT	---:-- --- NM
LRC	.--	--- KT	
MAX TURB	.--	--- KT	

DEST

CMS

STEP ALTs

---:--

---.T

RETURN

POS MONITOR

MSG LIST

Figure 4. PERF CRZ: MANAGED, PRESEL, and the LRC and MAX TURB speeds.

- **LRC** (long range cruise): the Mach giving 99 % of the maximum specific range (FCOM PER-IFT).
- **MAX TURB**: the recommended speed for flying in turbulence.

F-PLN and its revisions

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/F-PLN

FROM

UTC

SPD

ALT

TRK

DIST

FPA

P.POS

--:--

DISCONTINUITY

END OF F-PLN

NO ALTN F-PLN

--:--

DEST

F-PLN INFO

DIR TO

MSG LIST

Figure 5. F-PLN without a flight plan: P.POS, DISCONTINUITY, END OF F-PLN, NO ALTN F-PLN.

Click a waypoint name to open the lateral revision menu, a speed/altitude to open the vertical revision. The new pages below open from these menus or from F-PLN INFO.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/F-PLN/ALTERNATE

DATA FROM

TO ALTN

CO RTE

DIR TRK

DIST

EXTRA

SELECTED

DATABASE

NO ALTN

OTHER ALTN

RETURN

MSG LIST

Figure 6. ALTERNATE: the selected alternate, the database alternates with their company route, track, distance and EXTRA fuel, and two OTHER ALTN fields.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/F-PLN/CLOSEST AIRPORTS

DATA FROM P.POS

TO ARPTs

DIST

BRG

UTC

EFOB

EFF WIND

CLOSEST ARPTs

OTHER ARPT

RETURN

MSG LIST

Figure 7. CLOSEST AIRPORTS: the 4 closest airports from P.POS with distance, bearing, UTC, EFOB and effective wind, and up to three OTHER ARPTs.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/F-PLN/OFFSET

START WPT

END WPT

INTERCEPT ANGLE

P.POS

30°

OFFSET DIST

-- NM

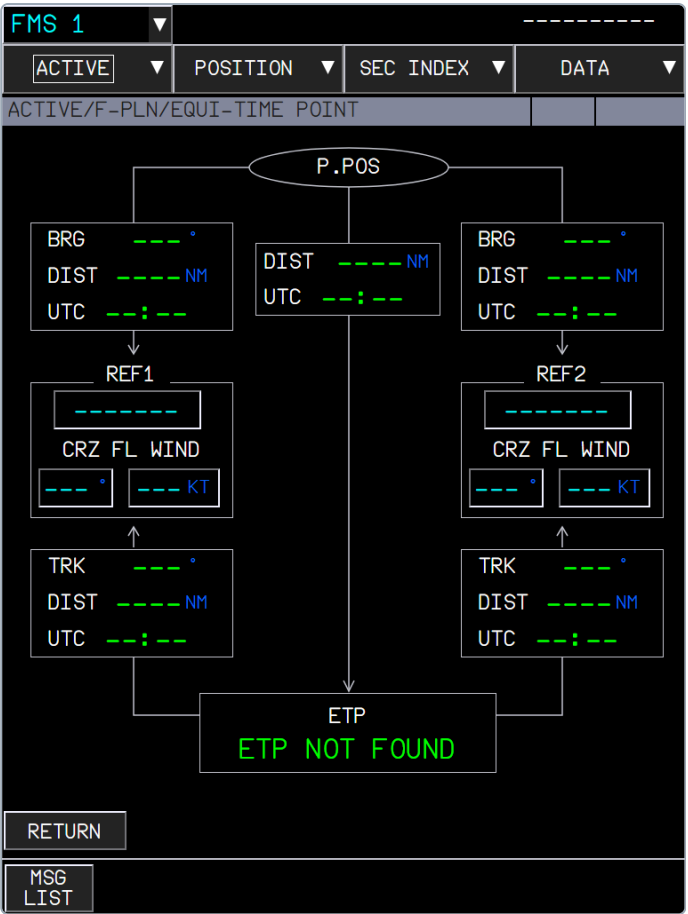
☐ LEFT
☐ RIGHT

RETURN

CANCEL OFFSET*

MSG LIST

Figure 8. OFFSET: start and end waypoints, distance LEFT / RIGHT, intercept angle (30° by default), CANCEL OFFSET.



FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/F-PLN/LL XING-TIME MKR

LAT/LONG XING

START WPT

▼

☒ LAT

☐ LONG

INCREMENT

1

NUMBER

INSERT AS WPT*

TIME MKR

UTC

REMAINING TIME

AURAL ALERT

☐

RETURN

MSG LIST

Figure 10. LL XING / TIME MKR: a waypoint at every LAT or LONG crossing (start, increment, number, INSERT AS WPT); a time marker with remaining time and an aural alert.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/F-PLN/VERT REV

RTA

SPD

CMS

ALT

STEP ALTs

RTA AT

▼

DIST

NM

ETT

ETA

RTA

☐ AT

☐ AT OR BEFORE

☐ AT OR AFTER

RTA SPD

.

KT

RETURN

MSG LIST

Figure 11. VERT REV, RTA tab: time at a waypoint (AT, AT OR BEFORE, AT OR AFTER), ETA and the speed to meet it; the time error thresholds of the FCOM apply.

FMS 1 ▼		-----	
ACTIVE ▼	POSITION ▼	SEC INDEX ▼	DATA ▼
ACTIVE/F-PLN/AIRWAYS			
AIRWAYS FROM			
		▲	▼
RETURN			
MSG LIST			

Figure 12. AIRWAYS: up to 31 segments; typing the next airway after a waypoint connects the two airways automatically at their intersection.

WIND and the company requests

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/WIND

HISTORY

CLB

CRZ

DES

CLB WIND

----- FT

*

 KT

----- FT

*

 KT

----- FT

*

 KT

----- FT

*

 KT

----- FT

*

 KT

CPNY WIND REQUEST

MSG LIST

Figure 13. WIND: HISTORY (winds recorded in the last flight), CLB, CRZ (per waypoint, 4 levels and the SAT), DES; CPNY WIND REQUEST at the bottom.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

COMPANY WIND DATA REQUEST

FREE TEXT

SEND WIND REQUEST *

RETURN

MSG LIST

Figure 14. COMPANY WIND DATA REQUEST: free text and SEND WIND REQUEST.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

COMPANY F-PLN REQUEST

FLT NBR

CPNY RTE

FREE TEXT

SEND F-PLN REQUEST *

RETURN

MSG LIST

Figure 15. COMPANY F-PLN REQUEST: flight number, company route, free text.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

COMPANY F-PLN REPORT

FREE TEXT

SEND REPORT TO CPNY *

RETURN

MSG LIST

Figure 16. COMPANY F-PLN REPORT: free text, SEND REPORT TO CPNY.

NOTE

The company functions use SimBrief (flight plan) and the flyPad (takeoff data). A request without a company able to answer ends after a timeout with the FCOM message.

3. POSITION menu

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

POSITION/MONITOR

ACCURACY LOWEPU --.---RNP ---.-

FMS1 00°00.0N/000°00.0E

RADIO --°--.-/-°--.-

MIXIRS --°--.-/-°--.-

GPIRS --°--.-/-°--.-

FMS2 00°00.0N/000°00.0E

DEVIATION FROM FMS1

IRS1 --°--.-/-°--.-

IRS2 --°--.-/-°--.-

IRS3 --°--.-/-°--.-

GPS1 00°00.0N/000°00.0E

GPS2 00°00.0N/000°00.0E

POSITION UPDATE

BRG / DIST TO --- / ---.-

RETURN

NAVAIDS

GPS

IRS

MSG LIST

Figure 17. MONITOR: FMS1/2, RADIO, MIXIRS, GPIRS, IRS and GPS positions, accuracy, EPU and RNP, FREEZE POSITION, BRG / DIST TO.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

POSITION/IRS

IRS ALIGNING ON GPS POS: 00°00.0N/000°00.0E

ALIGN ON OTHER REF

IRS 1 OFF

IRS 2 OFF

IRS 3 OFF

IRS1

IRS2

IRS3

FREEZE ALL IRS *

POSITION 00°00.0N/000°00.0E

T.TRK 0.0°T

GND SPD 0.0KT

T.WIND 0.0° / 0.0KT

T.HDG 0.0°T

MAG HDG 0.0°

MAG VAR 0.0°E

GPIRS POSITION 00°00.0N/000°00.0E

ACCURACY ---.-FT

RETURN

MSG LIST

Figure 18. IRS: alignment, the three IRS with their data, FREEZE ALL IRS.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

POSITION/NAVAIDS

TUNED FOR DISPLAY

SELECTED FOR FMS NAV

VOR 1

IDENT

----.

FREQ

----°

CRS

CLASS

VOR 2

IDENT

----.

FREQ

----°

CRS

CLASS

ADF 1

IDENT

----.

FREQ

☐ BFO

LS

IDENT

FREQ/CHAN

----.

CRS

----°

SLOPE

----°

CLASS

DESELECT GLIDE *

RETURN

MSG LIST

Figure 19. NAVAIDS: tuned for display (VOR, ADF, LS) and selected for FMS NAV, DESELECT GLIDE.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

POSITION/GPS

GPS 1

GPS 2

00°00.0N/000°00.0E

00°00.0N/000°00.0E

NAV

MODE

NAV

9

NBR OF SAT

10

40FT

ACCURACY

43FT

0.0°T

TRK

0.0°T

00:00:00

UTC

00:00:00

0FT

ALT

0FT

0KT

GND SPD

0KT

RETURN

MSG LIST

Figure 20. GPS: both receivers: position, mode, satellites, accuracy, track, UTC, altitude, ground speed.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

POSITION/TIME

UTC

00:00:00

DATE

FLT TIME

BLK TIME

OUT

OFF

ON

IN

PERM DATA SETUP

UTC ONLY

MSG LIST

Figure 21. TIME: UTC, date, flight and block time, OUT / OFF / ON / IN, and PERM DATA SETUP (what the permanent data shows).

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

POSITION/REPORT

UTC

ALT

LAST SEQD WPT

TO WPT

NEXT WPT

LAST SEQD WPT

SAT

°C

T.WIND

°

/

KT

FOB

.

T

NEXT CRZ PROFILE CHANGE

NONE

DEST

NONE

FREE TEXT

SEND REPORT TO CPNY*

MSG LIST

Figure 22. REPORT: last, to and next waypoints with time and altitude, the last sequenced waypoint data, the next cruise profile change, destination.

NOTE

The TIME page is not in the 2011 FCOM; it follows a later FMS standard at the user's request.

4. SEC INDEX



Figure 23. SEC INDEX: SEC 1 / 2 / 3, IMPORT, CPNY F-PLN REQUEST into a SEC, the pages of the SEC, DELETE, XFER TO MAILBOX and PRINT.



Figure 24. TRANSFER TO MAILBOX: sends the SEC flight plan to the ATC mailbox with a free text.

REJECTED ATC INFO (FCOM P 321-323) lists the elements of an ATC flight plan inserted in SEC 3 that the FMS rejected, 4 per page. MSFS cannot send an ATC flight plan, so the list stays empty.

5. DATA menu

FMS 1

ACTIVEPOSITIONSEC INDEXDATA

DATA/STATUS

ACFT STATUS

FMS P/N

A380-800

ENGINETRENT 972

IDLE +0.0

PERF +0.0

MODIFY

NAV DATABASE

ACTIVE

SWAP *

SECOND

PILOT STORED ELEMENTS

WAYPOINTS 00

ROUTES 00

NAVAIDS 00

RUNWAYS 00

DELETE ALL *

MSG LIST

Figure 25. STATUS: aircraft and engine, IDLE / PERF factors (FMS P/N tab), navigation database SWAP, and the count of the pilot-stored elements with DELETE ALL.

FMS 1

ACTIVEPOSITIONSEC INDEXDATA

DATA/PRINTER

ACTIVE DATA

CPNY DATA

AUTO PRINT ON RECEPTION

F-PLN INITPRINT*

T.O DATAPRINT*

WIND DATAPRINT*

F-PLN INIT

T.O DATA

WIND DATA

ACTIVE F-PLN REPORT

AUTO PRINT

PRE-FLIGHTPRINT*

IN-FLIGHTPRINT*

POST-FLIGHTPRINT*

PRE-FLIGHT

IN-FLIGHT

POST-FLIGHT

SEC F-PLN REPORT

SEC 1PRINT*

SEC 2PRINT*

SEC 3PRINT*

SEC 1

SEC 2

SEC 3

MSG LIST

Figure 26. PRINTER: print the F-PLN INIT, T.O and wind data, the active and SEC flight plan reports, and the AUTO PRINT options.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

DATA/WAYPOINT

DATABASE WPTs

PILOT STORED WPTs

WPT IDENT

0000000

MSG

LIST

Figure 27. WAYPOINT: database waypoints by ident; pilot-stored waypoints (lat/long, place/bearing/distance, place-bearing/place-bearing), NEW WPT.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

DATA/NAVAID

DATABASE NAVAIDS

PILOT STORED NAVAIDS

NAVAID IDENT

0000

MSG

LIST

Figure 28. NAVAID: database nav aids (class, position, frequency, ILS course and slope...); pilot-stored nav aids, NEW NAVAID.

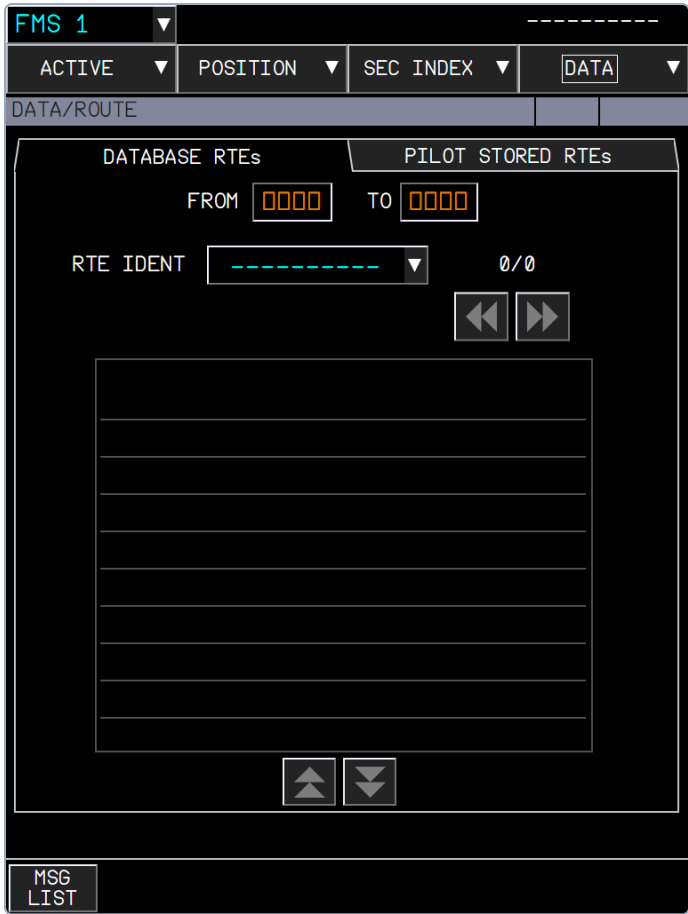


Figure 29. ROUTE: database (company) routes of a city pair, pilot-stored routes with their summary, NEW RTE, DELETE.

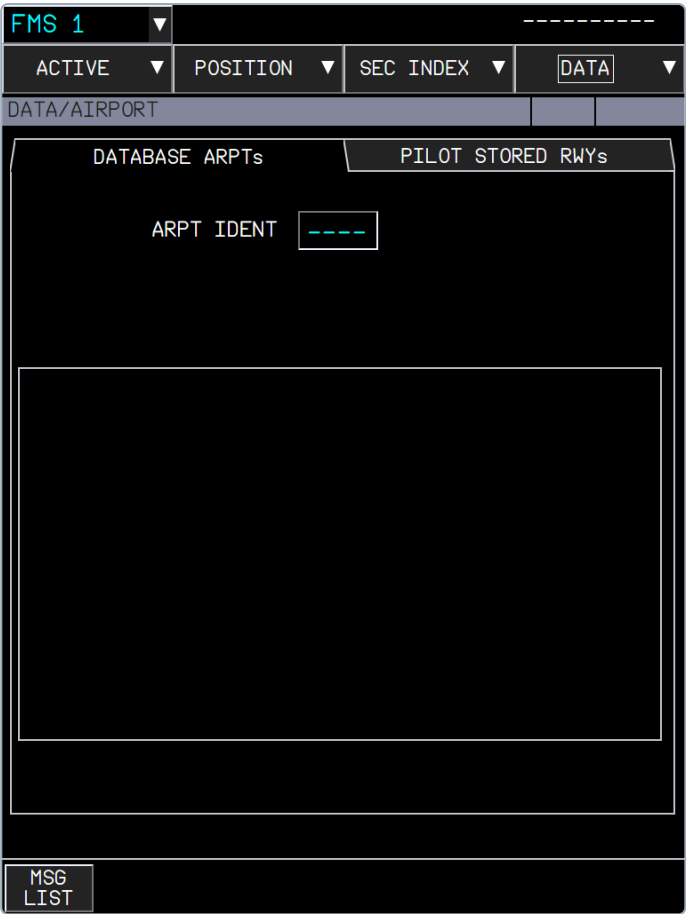


Figure 30. AIRPORT: database airports, and the PILOT STORED RWYs tab.

The printouts appear in the flyPad printouts list.

6. What to expect

Situation	What you should see
An entry in a wrong format	FORMAT ERROR in the message area; the field keeps its value.
A value out of the FCOM range	ENTRY OUT OF RANGE.
A waypoint or navaid not found	NOT IN DATABASE (or the NEW WPT / NEW NAVAID page when the FCOM says so).
A company request with no company answer	The FCOM timeout message after a few minutes.
A page not available in the flight phase	Its button is greyed.

7. Limitations and why

Limitation	Why	What to do
The layout follows the 2011 A380 FCOM.	It is the reference available; later FMS standards differ in places.	Report a difference with its FCOM page.
Some values are FBW model stand-ins.	For example GPS satellites and accuracy, like on the A32NX MCDU GPS page; MSFS does not simulate them.	-

Company functions answer only through SimBrief and the flyPad.	There is no airline datalink in MSFS.	-
Pilot-stored nav aids cannot be tuned.	The radio tuning uses the MSFS nav aids only.	-

A380 takeoff calculator and company T.O data

Takeoff weight limit, FLEX temperature, V-speeds and takeoff run on the flyPad, sent to the MFD PERF T.O page through the company takeoff data function of the FCOM

Aircraft	A380X (the flyPad screen is shared with the A32NX, see the PR 12 manual)
Where to find it	flyPad > Performance > Takeoff ; MFD > ACTIVE > PERF > T.O tab > CPNY T.O REQUEST / RECEIVED CPNY T.O
PR branch	pr/11-a380-takeoff-calculator (stacked on PR 10); completed by PR 12
Status	Used in the sim during development (September 2026); the final version with the PR 12 changes is still to re-test in the sim.
Pictures	flyPad and MFD pages in the local browser harness, running the build installed in the sim. The harness has no navigation database: the runway is entered by hand and the company uplink is simulated on the MFD.

1. What it is for

2. The flyPad screen

3. A FLEX takeoff

4. TOGA or another FLEX temperature

5. Options
6. Errors

7. On the MFD: company T.O data

8. What to expect

9. Limitations and why

1. What it is for

Before each takeoff the crew needs the takeoff weight limit of the runway, the FLEX temperature (reduced thrust), the speeds V1, VR, V2 and the flaps. On the real A380 they come from the OIS takeoff application or from the airline (company takeoff data, sent to the FMS by datalink). The flyPad calculator does the computation; SEND TO FMS plays the airline: the MFD receives the data as company T.O data, and the crew checks and inserts it, as in FCOM DSC-22-FMS-20-30.

2. The flyPad screen

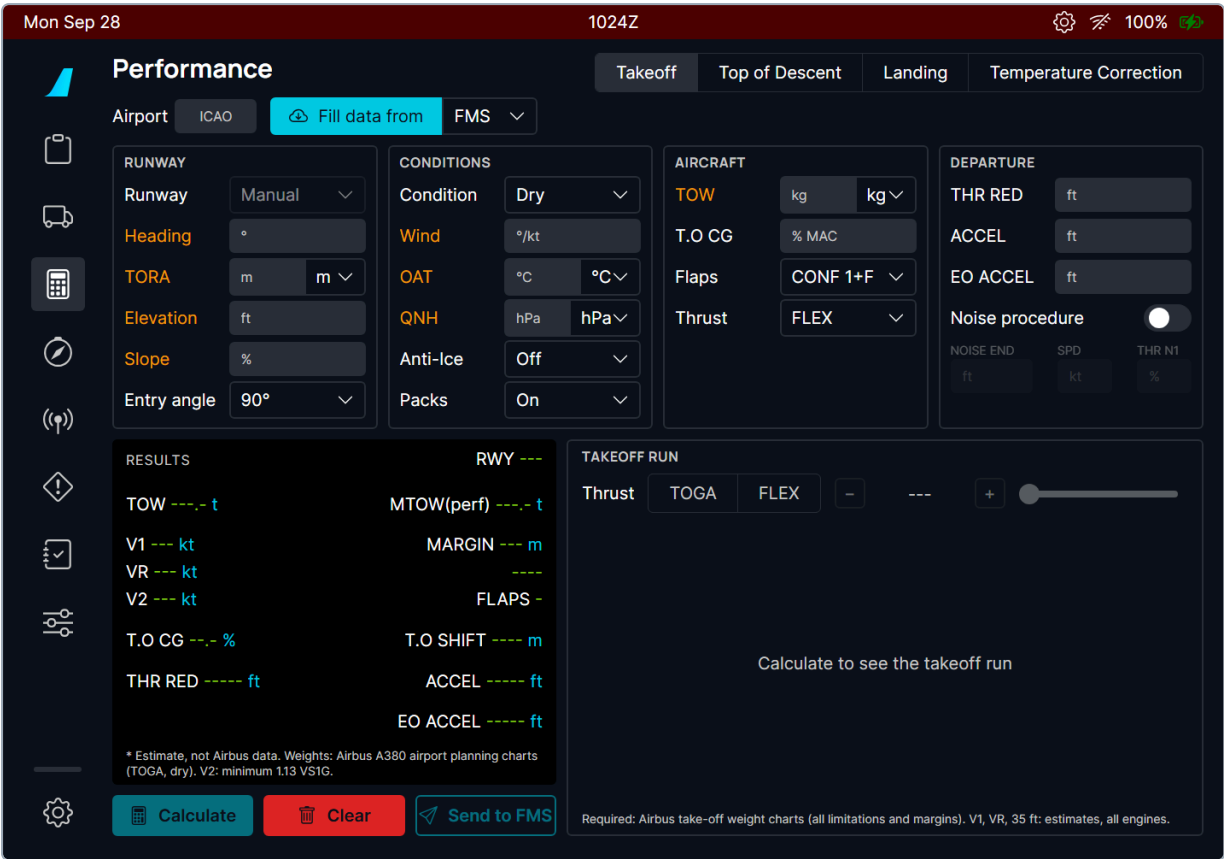


Figure 1. The takeoff screen. Amber labels: inputs still missing (Calculate stays dim).

Part	Contents
Airport, Fill data from	ICAO code; imports from the FMS (departure, runway, TOW, CG, THR RED / ACCEL), the OPF (SimBrief) or the METAR (wind, OAT, QNH).
RUNWAY	Runway from the list (or Manual), heading, TORA, elevation, slope, and the entry angle (90° or 180° turn onto the runway, or 0° when rolling straight on: it sets the line-up distance lost).
CONDITIONS	Runway condition, wind, OAT, QNH, anti-ice, packs.
AIRCRAFT	TOW in t (or klb), one decimal, T.O CG in % MAC, flaps (CONF 1+F, 2, 3 or OPT = the best one), thrust (TOGA or FLEX).
DEPARTURE	THR RED, ACCEL, EO ACCEL altitudes (they default to the elevation + 1500 ft), and the noise abatement procedure (NOISE END altitude, speed, N1).
RESULTS	The results in the layout of the OIS takeoff application (FCOM PER-TOF-TOR-SRS).
TAKEOFF RUN	The runway with the V1, VR and 35 ft points, the required distance and the margin, for TOGA or any FLEX temperature of the allowed range.

3. A FLEX takeoff

1. Fill the airport and runway (or **Fill data from FMS / OPF**), the conditions (METAR), the TOW and the T.O CG.
2. Keep **Thrust FLEX** and press **Calculate**.
3. Read the FLEX temperature and the speeds in RESULTS; check the takeoff run and its margin.
4. **Send to FMS**, then insert the data on the MFD (section 6).

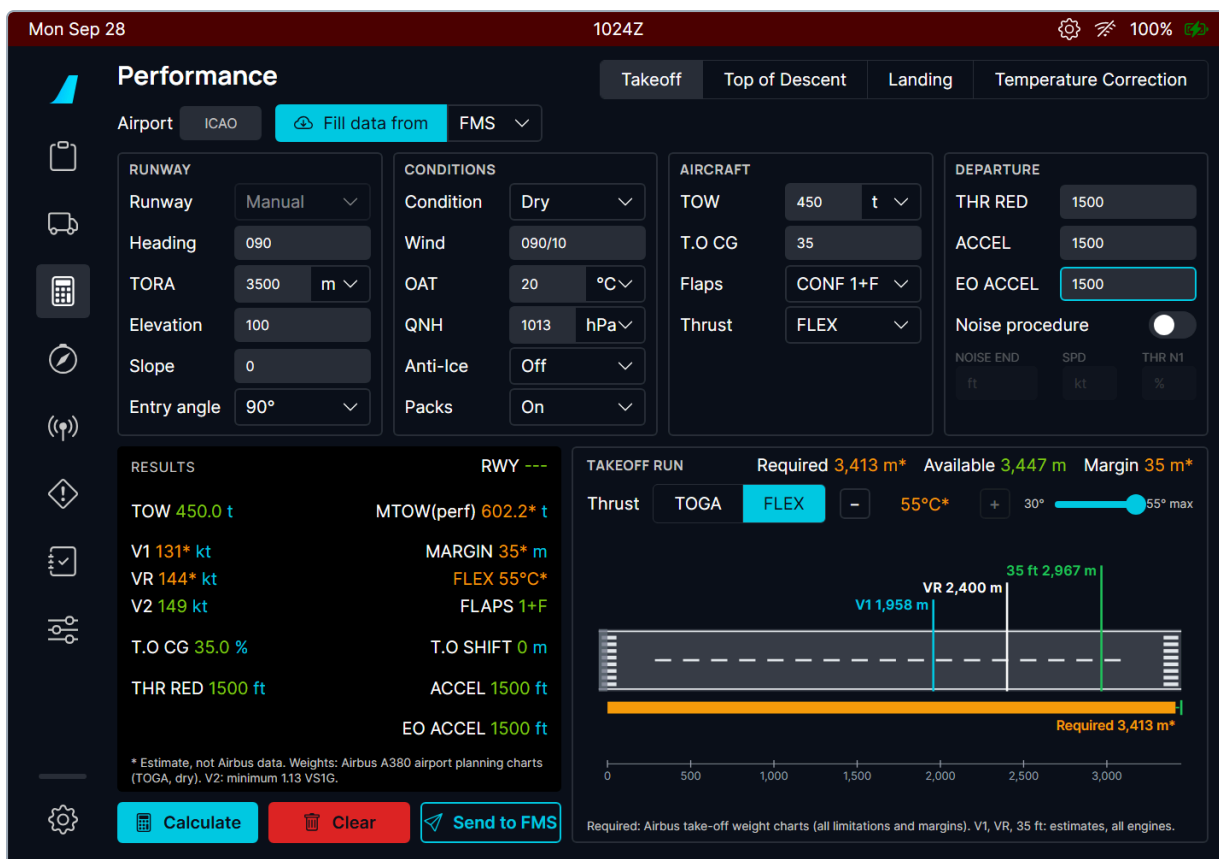


Figure 2. 450 t, 3500 m, 20 °C, CONF 1+F: FLEX 55 °C (the maximum, amber: estimate), V1 131 / VR 144 / V2 149 kt, MTOW(perf) 602 t. The takeoff run needs 3413 m of the 3447 m available after line-up: 35 m margin.

Reading the results

Line	Meaning
TOW / MTOW(perf)	Your takeoff weight, and the maximum the runway and conditions allow at TOGA.
V1, VR, V2	The takeoff speeds. V2 is at least 1.13 VS1G and 1.10 VMCA, VR at least 1.05 VMCA, V1 at least VMCG (FCOM PER-TOF-TOR-SRS).
MARGIN	Runway left beyond the required distance.
FLEX / TOGA	The FLEX temperature, or TOGA.
FLAPS, T.O CG, T.O SHIFT	The configuration, the CG entered, the takeoff shift (distance lost at line-up or an intersection).
THR RED, ACCEL, EO ACCEL	The altitudes of the DEPARTURE section.

NOTE

Green values come from Airbus data (the takeoff weight charts of the A380 Aircraft Characteristics); **amber with a star** are estimates of the FBW model (see Limitations).

4. TOGA or another FLEX temperature

The TAKEOFF RUN panel lets you look at the run for TOGA, or for any FLEX temperature between the OAT and the maximum (FCOM PER-TOF-THR-FLX). Use - / + or the slider; the speeds and the run follow.

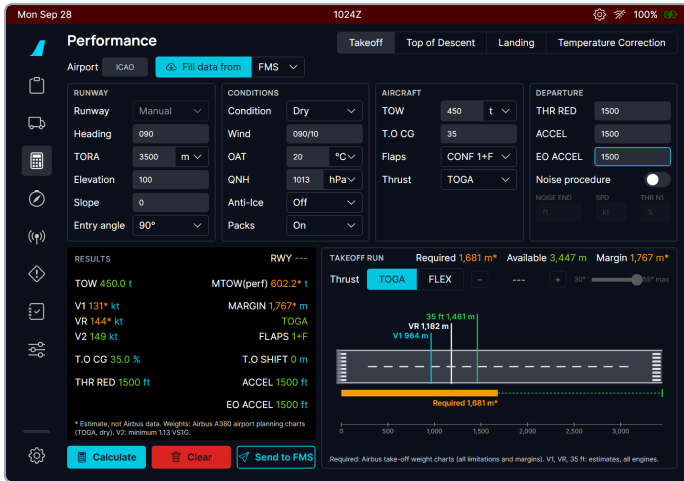


Figure 3. TOGA: the run needs 1681 m, margin 1767 m.

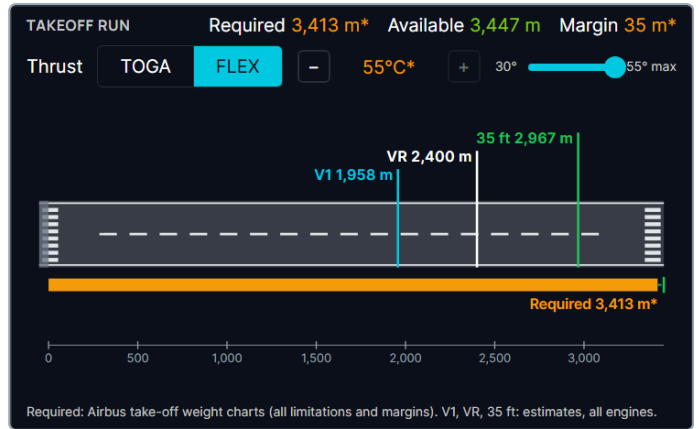


Figure 4. FLEX range 30 °C to 55 °C max: the selected temperature is the one sent to the FMS.

Send to FMS sends the thrust selected for the takeoff run only (TOGA, or FLEX at its temperature), with its own speeds.

5. Options

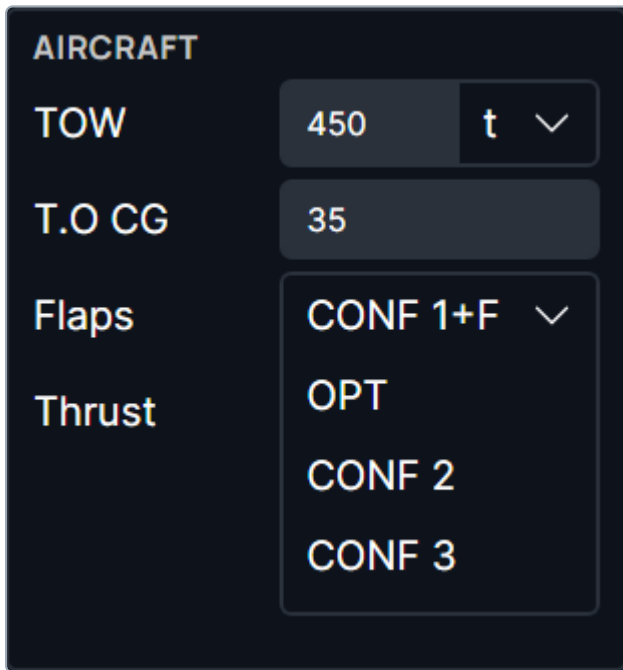


Figure 5. Flaps: CONF 1+F, 2, 3, or OPT (the calculator picks the best).

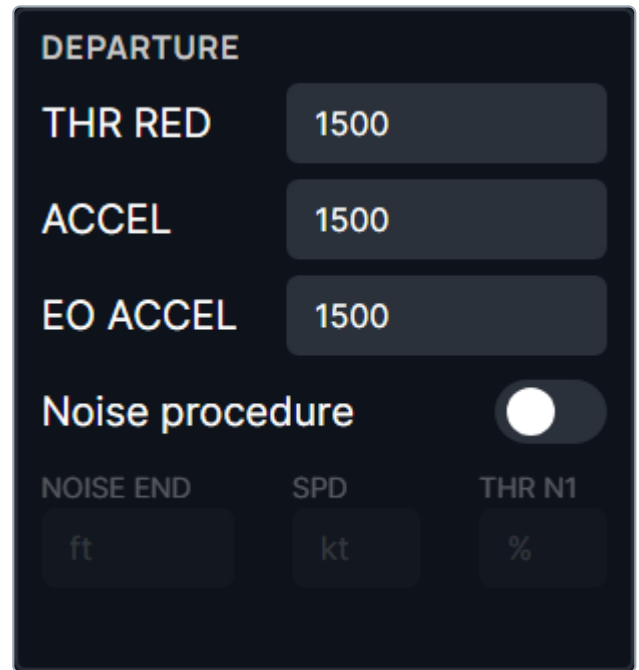


Figure 6. Noise procedure: NOISE END altitude, speed and N1 become available; they are sent with the data.

6. Errors

When the takeoff is not possible, a message says why and there is no result:

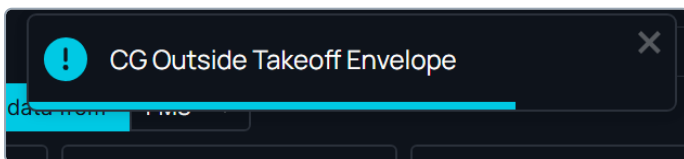


Figure 7. CG outside the takeoff envelope (here 30 % at 450 t: the forward limit moves aft with the weight).

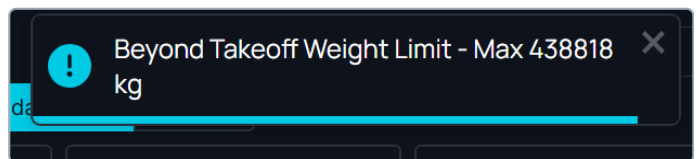


Figure 8. 510 t on 2100 m at 40 °C: the maximum takeoff weight is 438.8 t.

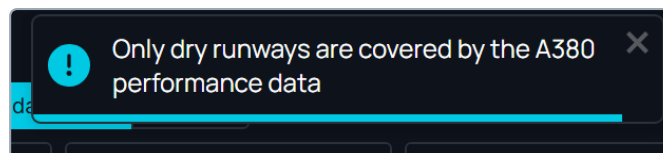


Figure 9. A wet or contaminated runway: the A380 data covers dry runways only.

Message	Meaning
Beyond Takeoff Weight Limit - Max ...	TOW above the runway limit: reduce the weight or use another runway.
CG Outside Takeoff Envelope	The CG is outside the A380 takeoff CG envelope for this weight.
Above Structural MTOW	TOW above 510 t.
Only dry runways are covered...	The A380 data has no wet or contaminated takeoff.
Runway shorter than the A380 performance data (1700 m)	The charts start at 1700 m.
Above Maximum Tailwind / OAT / Pressure Altitude	Outside the A380 limits (FCOM LIM-12).
Flex Not Possible	No FLEX temperature is possible at this weight: take off at TOGA.

7. On the MFD: company T.O data

The MFD PERF T.O page gets two buttons: **CPNY T.O REQUEST** (ask the airline) and, once data has arrived, **RECEIVED CPNY T.O.**

 The MFD PERF T.O page. At the top, it says "FMS 1" and "ACTIVE/PERF". Below that are tabs for "T.O", "CLB", "CRZ", "DES", "APPR", and "GA". The "T.O" tab is selected. It shows fields for "RWY", "T.O SHIFT", "V1", "VR", "V2", "F", "S", "O", "TOGA", "FLEX", "DERATED", "FLAPS", "THS FOR", "PACKS", "ANTI-ICE", "THR RED", "ACCEL", "NOISE", "TRANS", "EO ACCEL", and "CPNY T.O REQUEST". There are also "RETURN" and "POS MONITOR" buttons at the bottom.

Figure 10. PERF T.O before any data: the CPNY T.O REQUEST button (bottom right).

 The MFD COMPANY T.O DATA REQUEST page. At the top, it says "FMS 1" and "COMPANY T.O DATA REQUEST". Below that are fields for "REQUEST FOR RWY", "RWY 1/2", "T.O SHIFT", "T.O LIMIT", "TOW", "MAG WIND", "T.O CG", "RWY COND", "TEMP", "QNH", "TOGA", "FLEX", "DERATED", "FLAPS", "FREE TEXT", "SEND T.O REQUEST *", "RETURN", and "RECEIVED T.O DATA".

Figure 11. COMPANY T.O DATA REQUEST: up to 2 runways, TOGA / FLEX / DERATED, free text; SEND T.O REQUEST.

- After SEND T.O REQUEST the page shows *REQUEST PENDING...*; without an answer after 4 minutes, *NO COMPANY REPLY*. The flyPad **Send to FMS** is that answer.
- When data arrives: the message *COMPANY T.O DATA RECEIVED WAITING FOR INSERTION* and the button **RECEIVED CPNY T.O** on PERF.

Figure 12. Data received: the message and the RECEIVED CPNY T.O button.

Figure 13. RECEIVED COMPANY T.O DATA: runway 05 in amber (not the flight plan runway), TOW in amber with UPLINK/ACTIVE TOW DISAGREE: INSERT is not available.

The RECEIVED COMPANY T.O DATA page

- Up to 4 runways (arrows, RWY 1/4) × 2 thrust ratings (the TOGA / FLEX list at the top).
- **Runway in amber:** not the departure runway of the active flight plan.
- **TOW in amber** with *UPLINK/ACTIVE TOW DISAGREE*: the uplink TOW is more than 2 t below or 7 t above the FMS TOW (FUEL&LOAD).
- TEMP shows dashes for a FLEX takeoff; T.O LIMIT is the TORA; the noise fields are filled when the procedure was sent.
- **INSERT** (only when runway and TOW agree) fills PERF T.O: speeds, flaps, thrust, shift, THR RED, ACCEL, EO ACCEL and noise. It does not fill THS FOR: the crew enters the CG there. **CLEAR** deletes the data.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

RECEIVED COMPANY T.O DATA

DATA FOR RWY 05

FLEX 59

◀

▶

RWY 1/1

T.O WEIGHT 371 T

MAG WIND 058 °

6 KT

T.O CG 32.0%

RWY COND DRY

TEMP --- °C

QNH 1017

V1 119 KT

VR 131 KT

V2 136 KT

FLAPS 1

TOGA

FLEX

DERATED

T.O SHIFT --- M

T.O LIMIT 8166 FT

THR RED 1530 FT

ACCEL 1530 FT

EO ACCEL 1530 FT

NOISE END 3000 FT

SPD 180 KT

THR 85 %

INSERT *

CLEAR *

T.O PERF

CPNY T.O REQUEST

CLEAR INFO

COMPANY T.O DATA RECEIVED
WAITING FOR INSERTION

Figure 14. An uplink with the noise procedure: NOISE END 3000 ft, SPD 180 kt, THR 85 %.

TIP

The flyPad warns you right after Send to FMS if INSERT will not work: another departure runway in the FMS, no TOW in the FMS, or a TOW outside -2 t / +7 t of the FMS one.

8. What to expect

Situation	What you should see
Inputs missing	Amber labels; Calculate stays dim.
Calculate with valid inputs	RESULTS and the takeoff run; FLEX at its maximum by default.
TOW too high, CG out of envelope, wet runway...	A message with the reason; no result.
Send to FMS with a runway from the list	"Takeoff data sent to the FMS: RECEIVED CPNY T.O on the PERF page"; the MFD shows the data a moment later.
Send to FMS with a manual runway	Nothing is sent: the FMS needs the runway ident.
RECEIVED page, other runway than the flight plan	Runway in amber; INSERT not available.
INSERT	PERF T.O filled; THS FOR still to enter.

9. Limitations and why

Limitation	Why	What to do
Dry runways only.	The only public A380 takeoff data is the takeoff weight charts of the Airbus A380 Aircraft Characteristics (dry, ISA and ISA+15, runway 1700-5100 m, altitude 0-8000 ft).	On a wet runway, use the dry result with extra margin, as a training estimate only.
Speeds, FLEX, wind, slope and temperature effects are estimates (amber *).	The charts give weights only. The rest is a model fitted on them and checked on the FCOM examples (FLEX 38 at 560 t / 614 t).	-
The run distances (V1, VR, 35 ft points) are all-engine estimates.	No A380 distance data is public.	-
Send to FMS needs a runway from the list.	The FMS checks the uplink against its departure runway.	Choose the runway in the list.
Not certified data.	For simulation and training.	-

A320 takeoff calculator and MCDU uplink takeoff data

The flyPad takeoff screen of the A380, with the FBW A320neo performance model, sending its results to the MCDU as uplink takeoff data (FCOM DSC-22_20-50-10-28)

Aircraft	A32NX (the A380 part of the screen is in the PR 11 manual)
Where to find it	flyPad > Performance > Takeoff ; MCDU > PERF TAKE OFF > <TO DATA (6L)
PR branch	pr/12-a320-takeoff-calculator (stacked on PR 11)
Status	The flyPad calculation was used in the sim; the MCDU uplink pages are checked in the harness; the full flow is still to re-test in the sim.
Pictures	flyPad and MCDU in the local browser harness, running the build installed in the sim. The harness has no navigation database and no flight plan: the runway is typed and the uplink is sent to the MCDU by the harness (the same message the flyPad sends).

1. What it is for
2. The flyPad screen (A320)
3. Sending to the MCDU
4. What to expect
5. Limitations and why

1. What it is for

The A320 takeoff speeds, FLEX temperature and THS come from the airline or the crew's own tool. Here the flyPad computes them with the FlyByWire A320neo performance model (built from Airbus data) and sends them to the FMGC as an **uplink takeoff data** message, which the crew inserts from the MCDU PERF TAKE OFF page as on the real aircraft.

2. The flyPad screen (A320)

The screen is the one of the A380 (see the PR 11 manual) with these A320 differences:

- No noise procedure fields; T.O CG is optional: below 27 % MAC the forward CG performance applies (A320 FCOM PER-LOD) and the THS is shown in RESULTS.
- All values come from the FBW A320 model: they are shown in green (no estimates).
- Wet and contaminated runways are computed (the A380 page accepts dry only).

Mon Sep 28 1024Z 100%

Performance

Takeoff Top of Descent Landing Temperature Correction

Airport ICAO Fill data from FMS

RUNWAY

Runway Manual

Heading °

TORA m m

Elevation ft

Slope %

Entry angle 90°

CONDITIONS

Condition Dry

Wind °/kt

OAT °C °C

QNH hPa hPa

Anti-Ice Off

Packs On

AIRCRAFT

TOW t t

T.O CG % MAC

Flaps CONF 1+F

Thrust FLEX

DEPARTURE

THR RED ft

ACCEL ft

EO ACCEL ft

RESULTS

TOW --- t MTOW(perf) --- t

V1 --- kt MARGIN --- m

VR --- kt

V2 --- kt FLAPS -

T.O CG --- % T.O SHIFT --- m

THR RED ----- ft ACCEL ----- ft

EO ACCEL ----- ft

Weights, speeds, FLEX and THS: FlyByWire A320neo performance model (from Airbus data).

Calculate Clear Send to FMS

TAKEOFF RUN

Thrust TOGA FLEX - --- +

Calculate to see the takeoff run

Required: FlyByWire A320neo performance model (all limitations). V1, VR, 35 ft: estimates, all engines.

Figure 1. The A320 takeoff screen before any input.

Mon Sep 28 1024Z 100%

Performance

Takeoff Top of Descent Landing Temperature Correction

Airport ICAO Fill data from FMS

RUNWAY

Runway Manual

Heading 143

TORA 3500 m m

Elevation 499

Slope 0

Entry angle 90°

CONDITIONS

Condition Dry

Wind 140/8

OAT 22 °C °C

QNH 1015 hPa hPa

Anti-Ice Off

Packs On

AIRCRAFT

TOW 68 t t

T.O CG % MAC

Flaps CONF 1+F

Thrust FLEX

DEPARTURE

THR RED 2000

ACCEL 2000

EO ACCEL 2000

RESULTS

TOW 68.0 t MTOW(perf) 86.5 t

V1 148 kt MARGIN 519 m

VR 148 kt FLEX 64°C

V2 148 kt FLAPS 1+F

T.O CG --- % T.O SHIFT 0 m

THR RED 2000 ft ACCEL 2000 ft

EO ACCEL 2000 ft

Weights, speeds, FLEX and THS: FlyByWire A320neo performance model (from Airbus data).

Calculate Clear Send to FMS

TAKEOFF RUN

Required 2,961 m Available 3,480 m Margin 519 m

Thrust TOGA FLEX - 64°C + 44° 64° max

VR 2,243 m V1 2,243 m

35 ft 2,575 m

Required 2,961 m

Required: FlyByWire A320neo performance model (all limitations). V1, VR, 35 ft: estimates, all engines.

Figure 2. LFBO 14R-like runway (3500 m, 499 ft), 68 t, 22 °C: FLEX 64 °C (maximum), required 2961 m of 3480 m, margin 519 m.

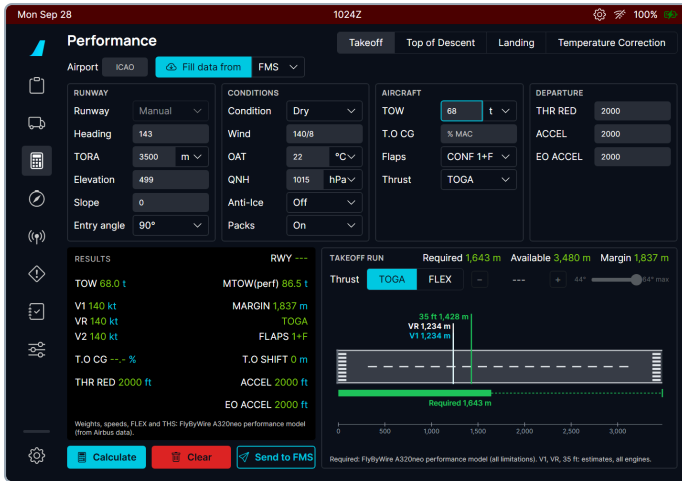


Figure 3. The TOGA run at the same weight: 1643 m.

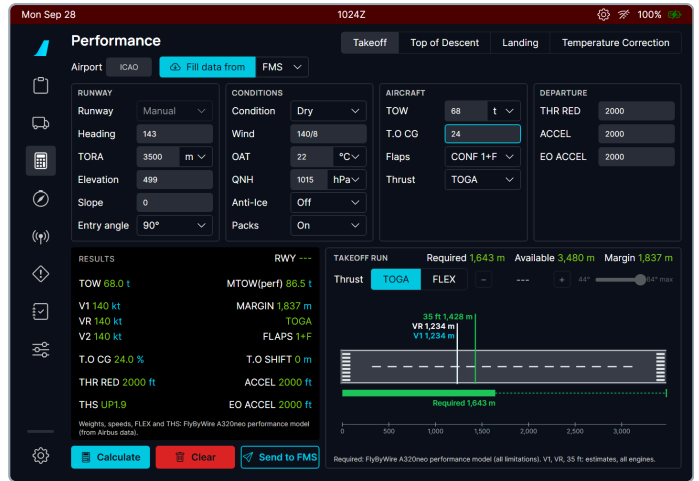


Figure 4. T.O CG 24 % (forward CG) at TOGA: the THS (UP1.9) appears in RESULTS.

NOTE

At light weights the FBW A320 model gives **V1 = VR = V2** (the pictures show 148/148/148 at FLEX and 140/140/140 at TOGA for 68 t). This comes from the FlyByWire model itself (the same with its version before this PR), not from this feature.

3. Sending to the MCDU

1. On the flyPad, choose the **runway from the list** (the uplink carries the runway ident), calculate, choose TOGA or the FLEX temperature for the takeoff run, and press **Send to FMS**.
2. The flyPad says "Takeoff data sent to the FMS: MCDU PERF TAKE OFF, UPLINK TO DATA, RECEIVED TO DATA", and warns right away if the FMGC departure runway or TOW will prevent the insertion.
3. On the MCDU the scratchpad shows **TAKEOFF DATA UPLINK**.
4. PERF TAKE OFF > <**TO DATA** (6L) > **RECEIVED** <**TO DATA** (6L): the UPLINK MAX TO DATA page (or FLX TO DATA with 4R). Check it, then **INSERT UPLINK**.

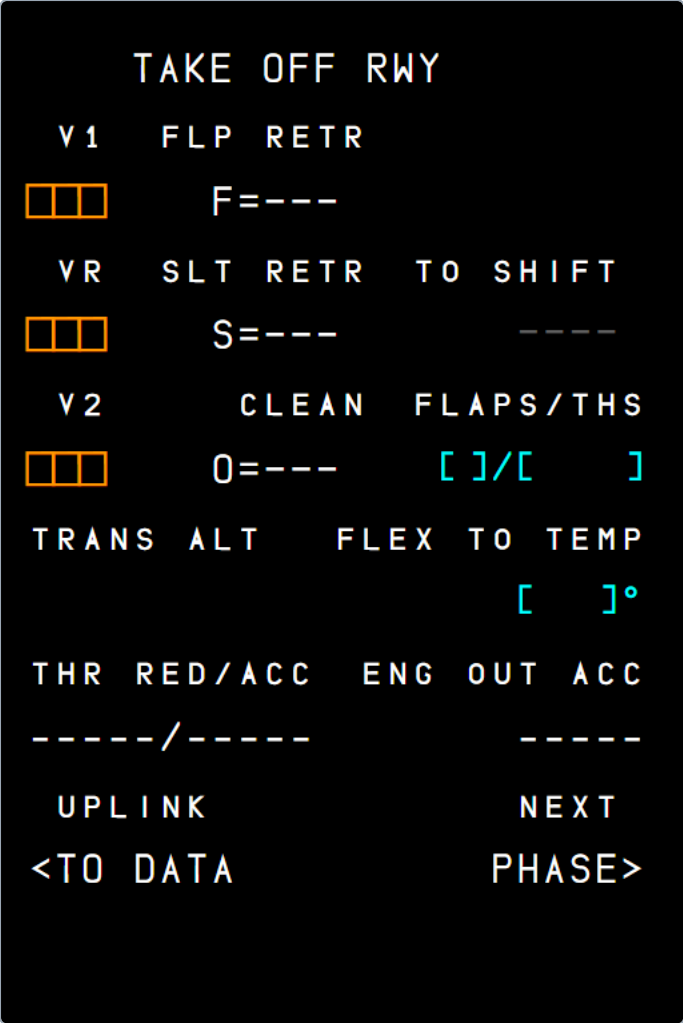


Figure 5. PERF TAKE OFF: the new UPLINK <TO DATA prompt (6L), in PREFLIGHT and DONE.

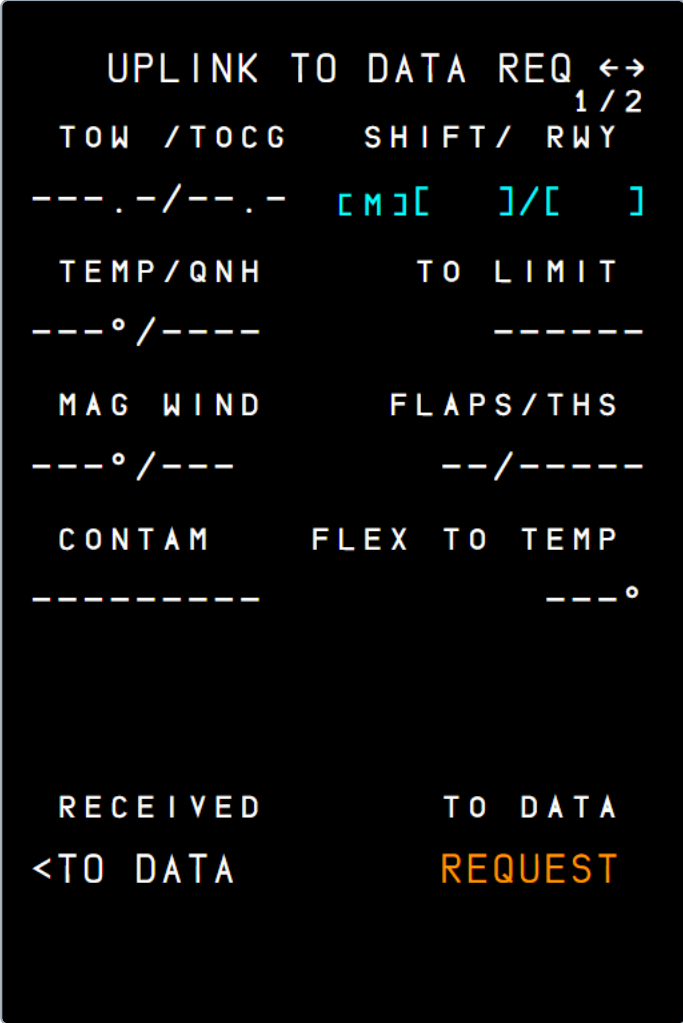


Figure 6. UPLINK TO DATA REQ: the request page (two runways, 1/2). RECEIVED <TO DATA (6L), TO DATA REQUEST (6R).

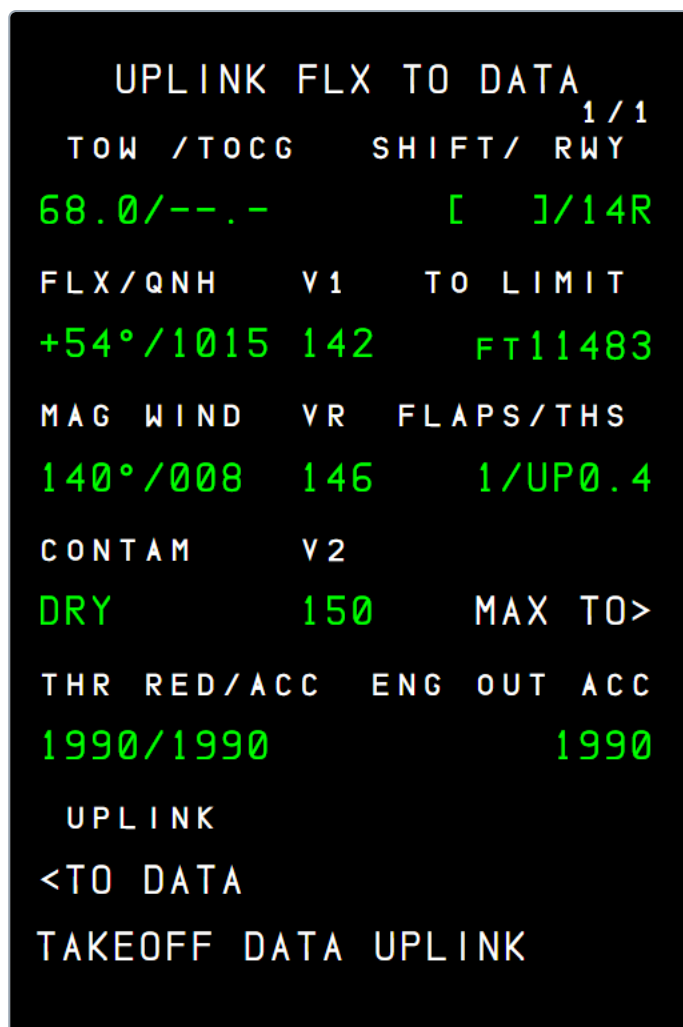
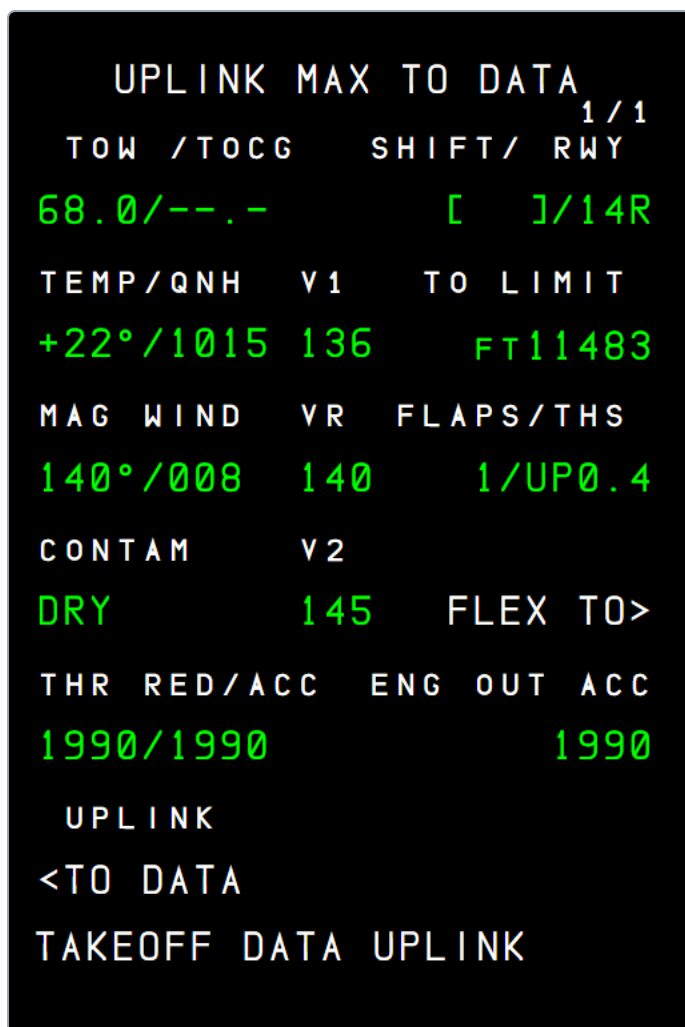


Figure 7. UPLINK MAX TO DATA after TAKEOFF DATA UPLINK: TOW, runway 14R, temperature and QNH, V1 / VR / V2, TO LIMIT, flaps / THS, THR RED / ACC, ENG OUT ACC. FLEX TO> (4R) shows the FLEX data.

Figure 8. UPLINK FLX TO DATA: FLX +54 °C with its own speeds.

There is **no INSERT UPLINK prompt** in these pictures because the harness FMGC has no flight plan. The prompt appears only when the uplink is complete (V1 and VR present) and the checks below pass.

The insertion rules (A320 FCOM)

- The uplink runway must be the departure runway of the flight plan.
- The uplink TOW must be **not more than 1 t below or 3 t above** the FMGC TOW; otherwise there is no insertion.
- INSERT UPLINK fills V1, VR, V2, FLAPS/THS, FLEX TO TEMP (cleared for a MAX takeoff), THR RED/ACC, ENG OUT ACC and the TO SHIFT.
- An incomplete uplink (for example without V1 or VR) gives no INSERT UPLINK prompt (DSC-22_20-70 P 6).



Figure 9. An uplink the FMGC cannot read: INVALID TAKEOFF UPLINK in the scratchpad.

MCDU message	Meaning
TAKEOFF DATA UPLINK	Uplink takeoff data received.
INVALID TAKEOFF UPLINK	The uplink is not valid (missing data).
REQUEST IS PENDING	A request was sent and has no answer yet.
NO ANSWER TO REQUEST	No answer 4 minutes after the request.

PERF TAKE OFF also gets the **TO SHIFT** field (1 m up to the runway length, CLR to delete): the distance lost when taking off from an intersection.

4. What to expect

Situation	What you should see
Send to FMS with a runway from the list	Success message; TAKEOFF DATA UPLINK on the MCDU.
Send to FMS with a manual runway	Nothing is sent: choose the runway in the list.
FMGC departure runway differs	flyPad warning; no INSERT UPLINK on the MCDU.
FMGC TOW differs by more than -1 t / +3 t	flyPad warning "INSERT UPLINK needs a TOW within -1 t / +3 t"; no INSERT UPLINK.
No TOW in the FMGC	flyPad warning "No TOW in the FMS (INIT B: ZFW and BLOCK)": fill INIT B first.
INSERT UPLINK	PERF TAKE OFF filled with the uplink.

5. Limitations and why

Limitation	Why	What to do
Only the thrust selected for the takeoff run is sent.	The airline sends what the crew asked for; the MCDU holds MAX and FLEX pages, the flyPad fills the one you use.	Select TOGA or FLEX before sending.
$V1 = VR = V2$ at light weights.	The FlyByWire A320 model (upstream).	Check against your own tools.
No real airline datalink.	The flyPad plays the airline.	-

A380X pilot-stored elements kept across sessions

An optional flyPad setting that keeps the waypoints, nav aids, routes and runways you stored in the MFD for the next flight

Aircraft	A380X
Where to find it	flyPad > Settings > Realism > <i>Keep MFD Pilot Stored Waypoints, Nav aids, Routes and Runways Between Sessions</i> ; MFD > DATA > WAYPOINT / NAVAID / ROUTE / AIRPORT, DATA / STATUS
PR branch	pr/13-pilot-stored-elements-persistence (stacked on PR 10)
Status	Built and tested with 17 unit tests; checked in the sim during development.
Pictures	flyPad and captain MFD in the local browser harness, running the build installed in the sim, with a few stored elements created by the harness.

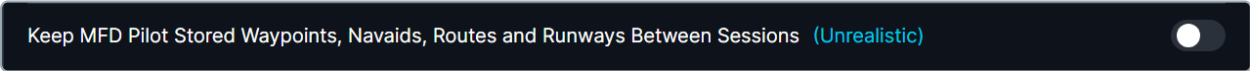
1. What it is for
2. The setting
3. Storing elements in the MFD
4. What to expect
5. Limitations and why

1. What it is for

On the A380 the crew can store its own waypoints, nav aids, routes and runways in the FMS (DATA pages). The FCOM says they are all deleted when all FMCs are shut down (DSC-22-FMS), and that is what happens by default: a new flight starts with an empty list.

In the simulator you may want to keep them from one session to the next (your own departure procedures, a favourite route). The setting below does that. It is marked **Unrealistic** because the real aircraft does not keep them.

2. The setting



Keep MFD Pilot Stored Waypoints, Nav aids, Routes and Runways Between Sessions **(Unrealistic)**

Figure 1. flyPad > Settings > Realism: the last line. Off by default.

- **Off** (default, as the FCOM): the stored elements are deleted at the end of the session.
- **On**: they are saved in the simulator storage and restored at the next session, before the FMS loads them.
- A reload of the flight (same session) never restores them twice.
- Turning the setting off drops the kept copy: the next session starts empty.
- The A380 OIT flyPad has the same option.

3. Storing elements in the MFD

The DATA menu pages have a PILOT STORED tab each. The DATA / STATUS page counts them and DELETE ALL clears everything.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

DATA/STATUS

ACFT STATUS

FMS P/N

A380-800

ENGINE TRENT 972

IDLE +0.0

PERF +0.0

MODIFY

NAV DATABASE

ACTIVE

SWAP *

SECOND

PILOT STORED ELEMENTS

WAYPOINTS 02

ROUTES 01

DELETE ALL *

NAVAIDS 02

RUNWAYS 02

MSG LIST

Figure 2. DATA / STATUS: PILOT STORED ELEMENTS: 2 waypoints, 2 nav aids, 1 route, 2 runways.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

DATA/WAYPOINT

DATABASE WPTs

PILOT STORED WPTs

WPT IDENT GOUGOU

1/2

LAT

LONG

43°33.5N/001°33.3E

PLACE

BRG

PLACE

BRG

AGN - 123° / GAI - 250°

DELETE STORED WPT *

DELETE ALL STORED WPTs *

NEW WPT

MSG LIST

Figure 3. PILOT STORED WPTs: GOUGOU, defined by place/bearing - place/bearing (AGN 123° / GAI 250°), 1/2.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

DATA/NAVAID

DATABASE NAVAIDS

PILOT STORED NAVAIDS

NAVAID IDENT

TBS

1/2

CLASS

ILS

LAT/LONG

43°36.0N/001°22.5E

ELEVATION

500FT

RWY IDENT

LFB014R

FREQ

110.7

CAT

3

CRS

144°

FIG OF MERIT

0

DELETE
STORED NAVAID *

DELETE ALL
STORED NAVAIDS *

NEW NAVAID

MSG LIST

Figure 4. PILOT STORED NAVAIDS: an ILS with its class, position, elevation, runway, frequency, category, course and figure of merit.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

DATA/ROUTE

DATABASE RTEs

PILOT STORED RTEs

RTE IDENT

NEWROUTE

1/1

FROM

LFB0

TO

WSSS

RWY	14L	AMOL5H	AMOLO
UT24	STP	UM985	NOSTA
DIR	GEN	DIR	LUPOS
DIR	BOA	DIR	FER
UM859	CHI	DIR	LABIN
DIR	PUL	UL614	KEB
UL5	VENIN	UT85	CND
UM747	SOBLO	B143	IDLER

DELETE
STORED RTE *

DELETE ALL
STORED RTEs *

NEW RTE ▾

MSG LIST

Figure 5. PILOT STORED RTEs: NEWROUTE LFB0 to WSSS with its summary (runway, SID, airways and waypoints).

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

DATA/AIRPORT

DATABASE ARPTs

PILOT STORED RWYs

ARPT IDENT

LFFF

1/2

RWY

15R

LAT/LONG

48°35.1N/002°05.1W

ELEVATION

207.0FT

LENGTH

4918FT

CRS

145°

LS IDENT

TG

DELETE
STORED RWY *

DELETE ALL
STORED RWYs *

NEW RWY

MSG LIST

Figure 6. PILOT STORED RWYs (DATA / AIRPORT): runway 15R of LFFF with its position, elevation, length, course and LS ident.

Limits of the database (FCOM)

Element	Limit	When full
Nav aids	20	The first one created is deleted.
Runways	10	The first one created is deleted.
Routes	5	PILOT RTEs LIST FULL.
Waypoints	the FMS list of 99	LIST OF 99 IN USE.

Storing an element with the name of an existing one replaces it (no duplicates).

4. What to expect

Situation	What you should see
Setting off, new session	DATA / STATUS shows 00 everywhere.
Setting on, new session	The elements of the last session are back.
Setting on, flight reloaded	The elements are there once (no duplicates).
Setting turned off	The kept copy is deleted; the current elements stay until the end of the session.
Stored route of another city pair	Not offered in INIT / RTE SEL (routes are per city pair).

5. Limitations and why

Limitation	Why	What to do
Pilot-stored nav aids cannot be tuned.	The radio tuning uses the MSFS nav aids only.	-
The waypoint limit is 99, not the FCOM 50.	The FMS shares one list of 99 waypoints with its other functions.	-
Keeping the elements is not realistic.	The real FMS deletes them at shutdown.	Leave the setting off for realism.

A380X DIRECT TO options

DIRECT WITH ABEAM, CRS IN and CRS OUT on the MFD DIRECT TO page, as in the A380 FCOM

Aircraft	A380X
Where to find it	MFD > ACTIVE > F-PLN > DIR TO (or the KCCU DIR key)
PR branch	pr/14-direct-to-options (stacked on PR 10)
Status	Built with 10 unit tests on the flight plan geometry; checked in the sim during development.
Pictures	Captain MFD in the local browser harness, running the build installed in the sim. The harness has no navigation database and no flight plan, so the geometry is shown with a diagram (figure 3).

1. What it is for
2. Using the page
3. What to expect
4. Limitations and why

1. What it is for

The DIRECT TO page of the A380 (FCOM DSC-22-FMS-20-30, DIRECT TO) has four options. FBW showed them but only the first one worked. All four work now:

Option	What the FMS builds
DIRECT	A direct leg from the aircraft to the target waypoint (as before).
DIRECT WITH ABEAM	The direct leg, plus an abeam point for each waypoint it bypasses: the point of the direct leg abeam that waypoint, named AB + the first 5 letters of the waypoint (BERGI gives ABBERGI).
CRS IN	An inbound course to the target: the aircraft continues on its current track to the intercept point INTCPT on that course, then flies the course to the target.
CRS OUT	An outbound course from the target: the aircraft continues on its current track to the intercept point INTCPT on the outbound course, then follows it (without overflying the target).

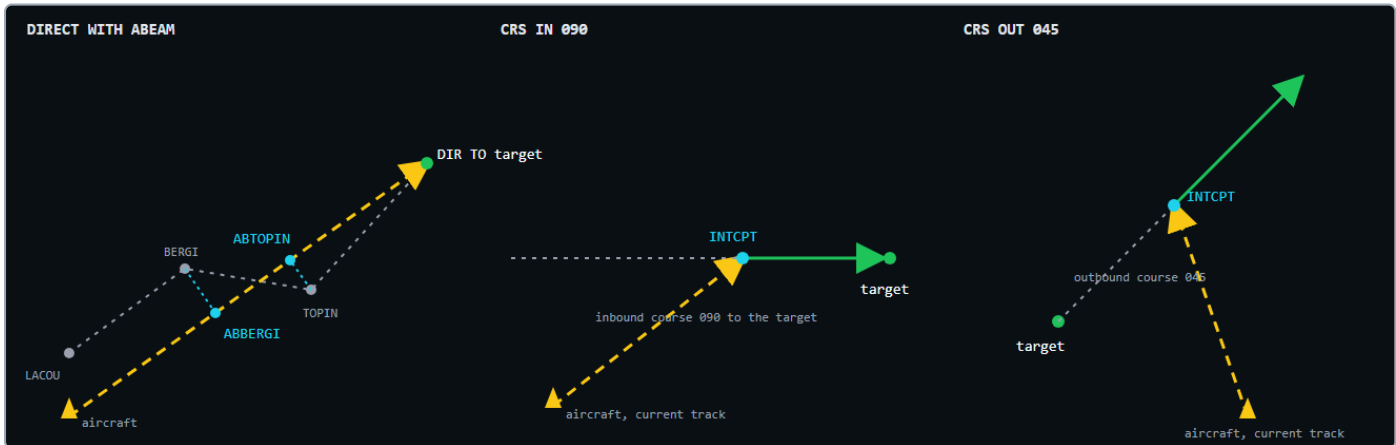


Figure 1. The three new options. Yellow: the aircraft and its path to the first point; green: the course flown after it; blue: the points the FMS creates.

Diagram drawn for this manual: not representative of the in-game UI, reference for the explanation only.

2. Using the page

1. Choose the target in **DIRECT TO** (a flight plan waypoint from the list, or type any fix).
2. Choose the option. For CRS IN / CRS OUT a course field appears next to the option.
3. For a flight plan waypoint the course is already filled: CRS IN = the course from the previous waypoint to the target, CRS OUT = the course from the target to the next waypoint (FCOM P 131). Type another one if ATC asks

(**NNN** magnetic, **NNNT** true).

4. Check the temporary flight plan (TMPY, yellow) and press **INSERT DIR TO**.

The screenshot shows the 'DIRECT TO' page. At the top, there's a header with 'FMS 1' and a dropdown menu. Below it are tabs: 'ACTIVE', 'POSITION', 'SEC INDEX', and 'DATA'. The main area is divided into two sections. On the left, there's a 'DIRECT TO' dropdown menu showing '-----'. Below it, 'UTC' is displayed as '---:--' and 'DIST' as '--- NM'. On the right, there's an 'OPTIONS' box with four radio buttons: 'DIRECT' (selected), 'DIRECT WITH ABEAM', 'CRS IN', and 'CRS OUT'. At the bottom left, there's a 'RETURN' button, and at the bottom right, there's a 'MSG LIST' button.

Figure 2. The DIRECT TO page: target, UTC and distance, and the four OPTIONS.

The screenshot shows the 'DIRECT TO' page with 'CRS IN' selected. The layout is identical to Figure 2, but the 'CRS IN' radio button is selected. The 'CRS IN' option is highlighted in blue. To the right of the 'CRS IN' option, there's a small amber box containing '000°'. The 'DIRECT' option is now unselected.

Figure 3. CRS IN selected: the course field (amber boxes until a course is known).

With DIRECT WITH ABEAM, the temporary F-PLN shows (**ABEAM PTS**) in place of the abeam points until the direct is inserted (FCOM P 162).

3. What to expect

Situation	What you should see
DIRECT WITH ABEAM, waypoints bypassed	One ABxxxxx point per bypassed waypoint, on the direct leg.
DIRECT WITH ABEAM, a waypoint whose abeam is not between the aircraft and the target	No abeam point for it.
CRS IN / OUT with a course that can be intercepted ahead	INTCPT on the course, then the course.
CRS IN more than 160° from the current track, parallel, or behind	No intercept point: a course to the target.
CRS OUT parallel to the track	No intercept point: a course from the target.
CRS IN / OUT without a course	No DIR TO until a course is entered.

4. Limitations and why

Limitation	Why	What to do
Abeam point names can repeat letters of the waypoint.	The FCOM rule is AB + 5 characters; two waypoints starting with the same 5 letters give the same name.	-
No intercept point when the geometry does not allow one.	Following the FCOM limits (160°, parallel, behind).	Use DIRECT, or another course.

A380X fuel jettison

Jettison down to a gross weight set on the MFD, with the ECAM memo, the ABN PROC procedure and the COMPLETED alert (FCOM DSC-28-40)

Aircraft	A380X
Where to find it	Overhead FUEL panel: JETTISON ARM and ACTIVE pushbuttons; MFD FMS ACTIVE/FUEL&LOAD: JTSN GW ; ECP ABN PROC > FUEL > JETTISON
PR branch	pr/15-fuel-jettison (independent)
Status	Tested in the sim (September 2026).
Pictures	MFD and EWD in the local browser harness, running the builds installed in the sim; the EWD states are the ones the FWS sends in flight.

1. What it is for
2. 1. Set the jettison gross weight
3. 2. Run the procedure
4. 3. During the jettison
5. 4. When it stops
6. Limitations and why

1. What it is for

FBW had the overhead JETTISON ARM and ACTIVE pushbuttons, but they did nothing. The fuel management (FQMS) now jettisons the fuel of the transfer tanks down to the gross weight entered on the MFD, and the ECAM shows the memo and the procedures of the FCOM.

2. 1. Set the jettison gross weight

1. MFD FMS **ACTIVE/FUEL&LOAD** page. **JTSN GW** is shown when the active flight plan has its mandatory data.
2. Enter the gross weight at which the jettison must stop, in tonnes (or 1000 lb with the LBS unit).
3. Allowed range: from **ZFW + 11 000 lb** (5.0 t) to the maximum gross weight (FCOM DSC-22-FMS-20-100). Outside it: ENTRY OUT OF RANGE with the range.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/FUEL&LOAD

GW

.-

T

CG

.-

%

FOB

.-

T

ZFW

300.0

T

ZFWCG

36.0

%

BLOCK

150.0

T

FUEL PLANNING *

TAXI

1.5

T

PAX NBR

000

TRIP

.-

T

.-

CI

RTE RSV

.-

T

5.0

%

JTSN GW

400.0

T

ALTN

0.0

T

.-

TOW

448.5

T

FINAL

6.0

T

00:30

LW

.-

T

TIME

EFOB

MIN FUEL AT DEST

DEST

NONE

.-

.-

T

6.0

T

ALTN

NONE

.-

.-

T

EXTRA

.-

T

.-

RETURN

MSG LIST

Figure 1. JTSN GW 400.0 t entered (ZFW 300.0 t).

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/FUEL&LOAD

GW

.-

T

CG

.-

%

FOB

.-

T

ZFW

300.0

T

ZFWCG

36.0

%

BLOCK

150.0

T

FUEL PLANNING *

TAXI

1.5

T

PAX NBR

000

TRIP

.-

T

.-

CI

RTE RSV

.-

T

5.0

%

JTSN GW

400.0

T

ALTN

0.0

T

.-

TOW

448.5

T

FINAL

6.0

T

00:30

LW

.-

T

TIME

EFOB

MIN FUEL AT DEST

DEST

NONE

.-

.-

T

6.0

T

ALTN

NONE

.-

.-

T

EXTRA

.-

T

.-

RETURN

CLEAR INFO

ENTRY OUT OF RANGE
RNG: 305.0 TO 512.0 T

Figure 2. 290 t is below ZFW + 5.0 t: the range is shown in the message area.

CAUTION

Without a JTSN GW the FQMS has no target weight: the jettison then runs until the transfer tanks are empty (or a pushbutton is set OFF).

3. 2. Run the procedure

The ABN PROC procedure takes the crew through the steps. ECP **ABN PROC**, then FUEL, then JETTISON:

SMOKE / FUMES

EMER EVAC

EMER DESCENT

DITCHING

FORCED LANDING

UNRELIABLE AIRSPEED INDICATION

ENG

F/CTL

L/G

NAV

FUEL

MISCELLANEOUS

Figure 3. ABN PROC menu: FUEL selected.

ABNORMAL PROC

FUEL

JETTISON

FUEL LEAK

MAN BALANCING PROCEDURE

Figure 4. FUEL sub-menu: JETTISON.

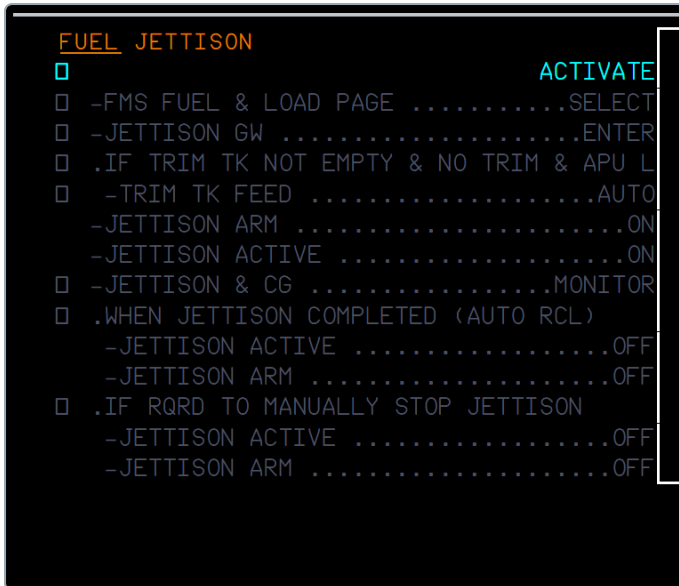


Figure 5. The procedure before ACTIVATE: all lines in grey.

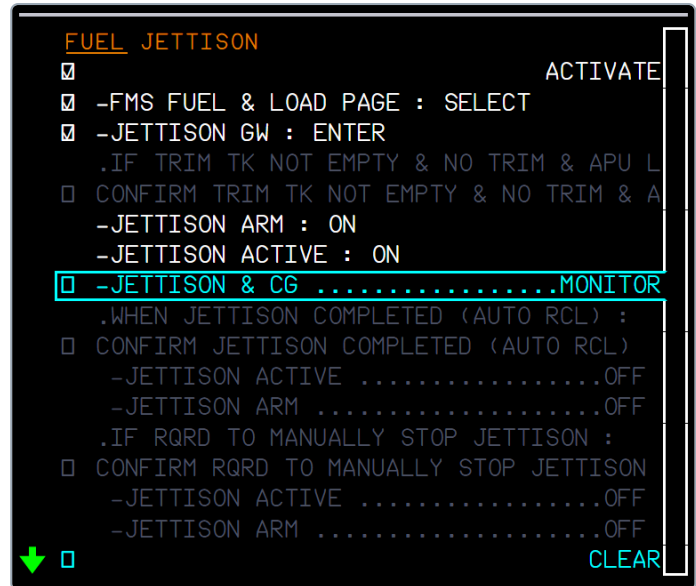


Figure 6. Activated: the FMS lines checked by the crew, JETTISON ARM and ACTIVE ticked by the FWS as the pushbuttons are set (sensed lines).

- FMS FUEL & LOAD PAGE ... SELECT, JETTISON GW ... ENTER: crew check lines.
- IF TRIM TK NOT EMPTY & NO TRIM & APU LINE FAULT: TRIM TK FEED AUTO (condition line, confirmed by the crew).
- **JETTISON ARM ON, JETTISON ACTIVE ON:** sensed, they tick themselves.
- JETTISON & CG ... MONITOR: watch the SD FUEL page and the CG.
- WHEN JETTISON COMPLETED (AUTO RCL) / IF RQRD TO MANUALLY STOP JETTISON: ACTIVE OFF then ARM OFF, sensed.

4. 3. During the jettison

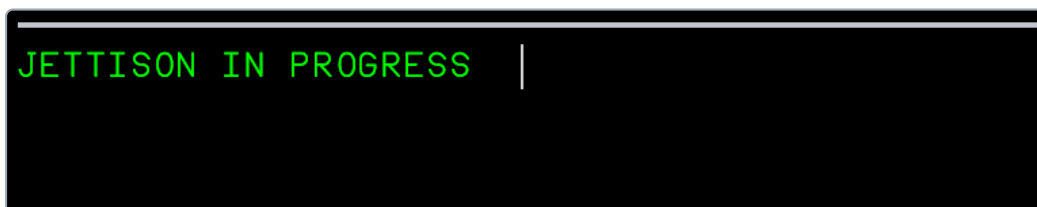


Figure 7. The EWD memo while the fuel is jettisoned.

- Memo **JETTISON IN PROGRESS** (green).
- Fuel leaves the **transfer tanks** (outer, mid, inner and trim) at the FCOM rate of **330 693 lb/h** (about 2.5 t per minute), in proportion to their content, so they empty together. The **feed tanks are never jettisoned**.
- The trim tank is held when the CG reaches the forward T.O/LDG limit, and resumes when the CG is 0.5 % aft of it.
- The other automatic transfers pause during the jettison.
- The SD FUEL page shows the two jettison valves open and the quantities decreasing.

5. 4. When it stops

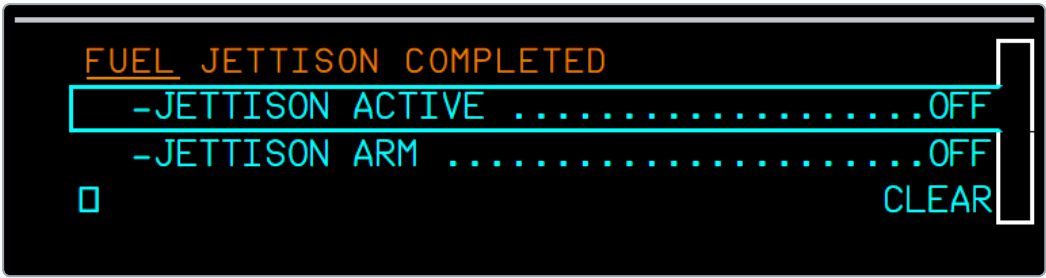


Figure 8. FUEL JETTISON COMPLETED: set ACTIVE then ARM OFF; the lines tick as the pushbuttons are released.

Situation	What you should see
Gross weight reaches the JTSN GW	The jettison stops, FUEL JETTISON COMPLETED shows, the ACTIVE ON light goes off.
All transfer tanks empty (or no JTSN GW entered)	Same: stops and COMPLETED.
ACTIVE or ARM set OFF by the crew	Stops at once (manual stop). No COMPLETED alert.
Both pushbuttons OFF after COMPLETED	The alert and the COMPLETED state clear; a new jettison can be started.
ACTIVE ON without ARM	Nothing: ARM is needed.
AC 1 bus not powered	No jettison (FQMS channel unpowered).

6. Limitations and why

Limitation	Why	What to do
JETTISON FAULT, VLV NOT CLOSED and the INOP SYS are not shown.	FBW has no jettison valve failures to trigger them.	-
The fuel does not leave through an outlet.	The MSFS fuel system has no overboard dump: the FQMS sets the tank quantities directly and opens valves 57 / 58 for the SD picture.	-
After an update, the first flight needs a full MSFS restart.	The change touches the behaviour XML of the overhead, which MSFS loads at start.	Restart MSFS once after installing.

A380X KCCU keyboard entries

Values typed on the KCCU keyboard are entered with ENT, and ESC cancels the edition, as the FCOM describes the KCCU

Aircraft	A380X (MFD, OIT, ND OANS panel: every entry field of the displays)
Where to find it	KCCU (keyboard and cursor control unit) on the pedestal, KBD switch ON; any MFD entry field
PR branch	pr/16-kccu-keyboard-entry (stacked on PR 10)
Status	Built and used in the sim (September 2026).
Pictures	Captain MFD in the local browser harness, running the build installed in the sim; the KCCU keys are sent to the MFD as the sim sends them.

1. What it is for
2. Entering a value
3. Wrong entries
4. Cancelling with ESC
5. What to expect
6. Limitations and why

1. What it is for

On the A380 the crew types on the KCCU keyboard. Before this change a value typed on the 3D KCCU keys was often lost: the field was validated only when the display lost the focus, and clicking a 3D key ends the focus before the key reaches the field. Now the field follows the FCOM (DSC-31-30-20, KCCU):

- **ENT** validates the entry directly.
- **ESC** *"cancels the current field edition. The field returns to the last valid value"*.
- With the **KBD** switch ON, the edition continues whatever happens to the display focus: only ENT, ESC or the selection of another field end it (the previous field is then validated). One field is in edition per side.
- A click on a field selects it even when it already had the focus.
- KBD switch OFF: the PC keyboard works as before.

2. Entering a value

1. Set the **KBD** switch of your KCCU ON.
2. Click the field with the cursor (trackball or mouse).
3. Type the value on the KCCU keys (or the PC keyboard).
4. Press **ENT**. The value is checked with the FCOM entry format of the field.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/INIT

FLT NBR

0000000000

ACFT STATUS

CPNY F-PLN REQUEST

FROM

0000

TO

0000

ALTN

CPNY RTE

0000000000

RTE SEL

ALTN RTE

ALTN RTE SEL

CRZ FL

FL----

CRZ TEMP

--- °C

CI

TROPO

39000 FT

CPNY WIND REQUEST

TRIP WIND

WIND

IRS

DEPARTURE

NAVAIDS

FUEL&LOAD

T.O. PERF

RTE SUMMARY

CPNY T.O. REQUEST

MSG LIST

Figure 1. TROPO selected, 39000 typed on the KCCU keys.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/INIT

FLT NBR

0000000000

ACFT STATUS

CPNY F-PLN REQUEST

FROM

0000

TO

0000

ALTN

CPNY RTE

0000000000

RTE SEL

ALTN RTE

ALTN RTE SEL

CRZ FL

FL----

CRZ TEMP

--- °C

CI

TROPO

39000 FT

CPNY WIND REQUEST

TRIP WIND

WIND

IRS

DEPARTURE

NAVAIDS

FUEL&LOAD

T.O. PERF

RTE SUMMARY

CPNY T.O. REQUEST

MSG LIST

Figure 2. After ENT: TROPO 39000 FT entered (big cyan: a pilot entry).

3. Wrong entries

ENT on a wrong value leaves the field unchanged and shows the FCOM message in the message area; the message gives the expected format. CLR INFO (KCCU key) clears it.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/INIT

FLT NBR

0000000000

ACFT STATUS

CPNY F-PLN REQUEST

FROM

0000

TO

0000

ALTN

CPNY RTE

0000000000

RTE SEL

ALTN RTE

ALTN RTE SEL

CRZ FL

FL----

CRZ TEMP

--- °C

CI

TROPO

39000 FT

CPNY WIND REQUEST

TRIP WIND

WIND

IRS

DEPARTURE

RTE SUMMARY

NAVAIDS

FUEL&LOAD

T.O. PERF

CPNY T.O. REQUEST

CLEAR INFO

FORMAT ERROR

FORMAT: FOR ALT XXXXX FOR FL XXX

Figure 3. "ABC" in TROPO: FORMAT ERROR, FORMAT: FOR ALT XXXXX FOR FL XXX.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/INIT

FLT NBR

0000000000

ACFT STATUS

CPNY F-PLN REQUEST

FROM

0000

TO

0000

ALTN

CPNY RTE

0000000000

RTE SEL

ALTN RTE

ALTN RTE SEL

CRZ FL

FL----

CRZ TEMP

--- °C

CI

TROPO

39000 FT

CPNY WIND REQUEST

TRIP WIND

WIND

IRS

DEPARTURE

RTE SUMMARY

NAVAIDS

FUEL&LOAD

T.O. PERF

CPNY T.O. REQUEST

CLEAR INFO

ENTRY OUT OF RANGE

Figure 4. 99999 in TROPO: ENTRY OUT OF RANGE.

4. Cancelling with ESC

TROPO

41000 FT

Figure 5. 41000 typed in TROPO, not validated...

TROPO

39000 FT

Figure 6. ... ESC: the field returns to its last valid value, 39000.

5. What to expect

Situation	What you should see
Value typed, ENT	The value is entered (or an FCOM error message).
Value typed, ESC	The field returns to its last valid value.
Value typed, another field clicked	The first value is validated, the new field is in edition.
Value typed, click on a 3D KCCU key or elsewhere in the cockpit	KBD ON: the field stays in edition, the value is not lost.
KBD switch OFF	The PC keyboard types in the field; the field is validated when it loses the focus, as before.

6. Limitations and why

Limitation	Why	What to do
One field in edition per side.	As on the aircraft: each KCCU drives its own side.	-

A380X ECAM procedures from the FCOM

The ABN PROC procedures that showed nothing, and the lines of the ENG FAIL, ALL ENGINES FAILURE and COND alerts, written from the A380 FCOM

Aircraft	A380X
Where to find it	ECP ABN PROC (not-sensed procedures, EWD); the sensed alerts appear by themselves on the EWD
PR branch	pr/17-ecam-procedures (independent)
Status	Tested in the sim (September 2026).
Pictures	EWD in the local browser harness, running the build installed in the sim; the EWD states are the ones the FWS sends.

1. What it is for

2. Opening a not-sensed procedure

3. The procedures added
4. Sensed alerts with FCOM lines

5. Reading the lines

6. Limitations and why

1. What it is for

Many ABN PROC procedures were marked WIP: they had no lines and no FWS entry, so selecting one showed nothing. They now carry the FCOM text (PRO-ABN-ECAM-10 and -20). Some sensed alerts also got their FCOM lines, with the lines the FWS can check ticking by themselves.

2. Opening a not-sensed procedure

1. Press **ABN PROC** on the ECP. The EWD lists the red emergency procedures, then the categories (ENG, F/CTL, L/G, NAV, FUEL, MISCELLANEOUS).

2. Move the cursor with the ECP **UP / DOWN** keys and press **CHECK**: the category opens, then the procedure.

3. The procedure shows in grey with **ACTIVATE**: a preview. CHECK on ACTIVATE runs it: the lines turn cyan and the cursor walks through them.

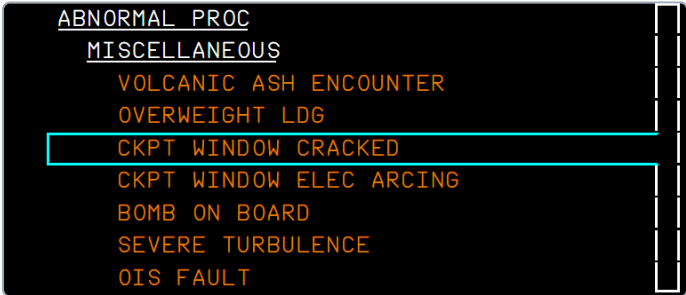


Figure 1. ABN PROC > MISCELLANEOUS.

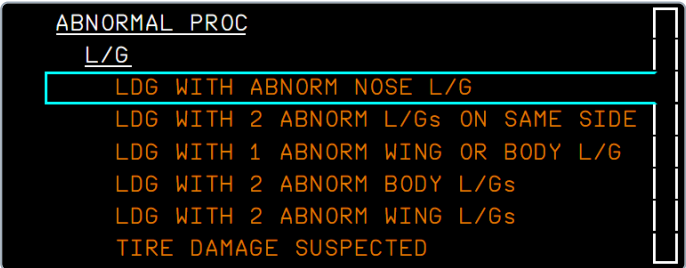


Figure 2. ABN PROC > L/G: the landing gear procedures.

```

MISC DITCHING
[ ] ACTIVATE
[ ] -ATC .....NOTIFY & SQUAWK 7700
[ ] -CABIN & CKPT .....PREPARE
[ ] -JETTISON ARM .....ON
[ ] -JETTISON ACTIVE .....ON
    -[IMFD SURV] GPWS .....OFF
    -[IMFD SURV] TERR SYS .....OFF
    -SIGNS .....ON
[ ] -EMER EXIT LT .....ON
[ ] -COMMERCIAL 1+2 .....OFF
[ ] DISREGARD NORM C/Ls
    AT TOP OF DESCENT :
[ ] -CABIN ALT MODE .....MAN
[ ] -CABIN ALT TRGT .....ZERO
    FOR APPROACH & LANDING :
[ ] FOR DITCHING : KEEP L/G UP
[ ] -BARO REF (IF AVAIL) .....SET

```

Figure 3. DITCHING before ACTIVATE (grey preview).

```

MISC DITCHING
☑ ACTIVATE
☑ -ATC : NOTIFIED & SQUAWKED
☑ -CABIN & CKPT : PREPARED
☐ -JETTISON ARM .....ON
☐ -JETTISON ACTIVE .....ON
  -[MFD SURV] GPWS .....OFF
  -[MFD SURV] TERR SYS .....OFF
  -SIGNS .....ON
☐ -EMER EXIT LT .....ON
☐ -COMMERCIAL 1+2 .....OFF
☐ DISREGARD NORM C/Ls
  _AT TOP OF DESCENT :
☐ -CABIN ALT MODE .....MAN
☐ -CABIN ALT TRGT .....ZERO
  _FOR APPROACH & LANDING :
☐ FOR DITCHING : KEEP L/G UP
☐ -BARO REF (IF AVAIL) .....SET

```

Figure 4. DITCHING activated: ATC and CABIN & CKPT checked by the crew; the cursor on JETTISON ARM.

3. The procedures added

Menu	Procedures
Emergency list (red)	DITCHING, FORCED LANDING
MISCELLANEOUS	CKPT WINDOW CRACKED, CKPT WINDOW ELEC ARCING, OIS FAULT
FUEL	FUEL LEAK (the FUEL LEAK DETECTED procedure, with LAND ANSA), MAN BALANCING PROCEDURE
L/G	LDG WITH ABNORM NOSE L/G, WITH 2 ABNORM L/Gs ON SAME SIDE, WITH 1 ABNORM WING OR BODY L/G, WITH 2 ABNORM BODY L/Gs, WITH 2 ABNORM WING L/Gs (they call the WHEEL page)
NAV	IR ALIGNMENT IN ATT MODE
ENG	HI VIBRATIONS (was missing from the list)

L/G LDG WITH ABNORM NOSE L/G

☐ ACTIVATE

☐ -CABIN CREWADVISE

☐ -ATCNOTIFY

☐ LDG LIMITED TO MAX LDG WEIGHT

☐ .IF GW DECREASE RQRD

☐ -JETTISON ARMON

☐ -JETTISON ACTIVEON

☐ -OXYGEN CREW SUPPLYOFF

☐ -CG AFT SHIFT (50 PAX = 2% CG) CONSIDER

☐ DISREGARD NORM APPR & LDG C/Ls

☐ -SIGNSON

☐ -CABIN & CKPT (LOOSE EQPT)SECURE

☐ LDG DIST AFFECTED

FOR APPROACH :

☐ -ECAM STSCHECK

☐ -BARO REFSET

☐ -MINIMASET

Figure 5. LDG WITH ABNORM NOSE L/G (preview).

```

NAV IR ALIGNMENT IN ATT MODE
[ ] ACTIVATE
[ ] -IR MODE SEL (AFFECTED) .....ATT
[ ] SPEED, HDG & FL : KEEP CONST FOR 30 S
[ ] -FMS POSITION / IRS PAGE .....SELECT
[ ] -HDG .....ENTER

```

Figure 6. IR ALIGNMENT IN ATT MODE (preview).

4. Sensed alerts with FCOM lines

These appear by themselves when the failure is detected.

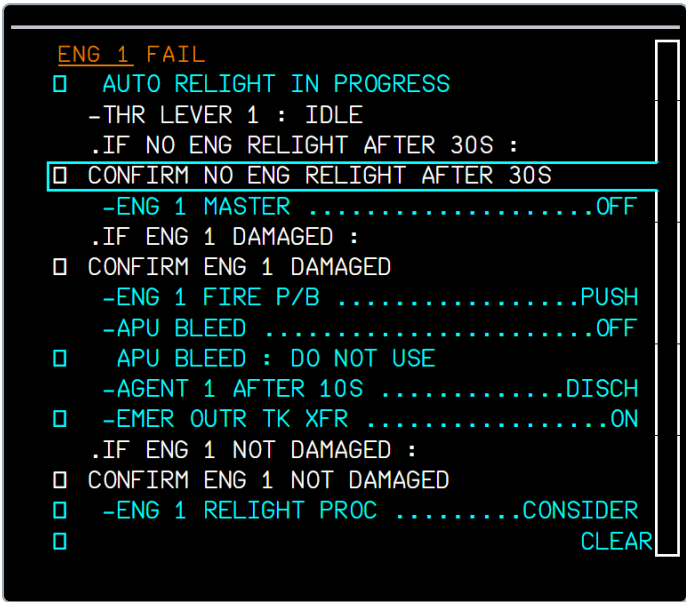


Figure 7. ENG 1 FAIL in flight: AUTO RELIGHT IN PROGRESS, THR LEVER 1 IDLE (ticked, the lever is at idle), then the relight / damaged branches.

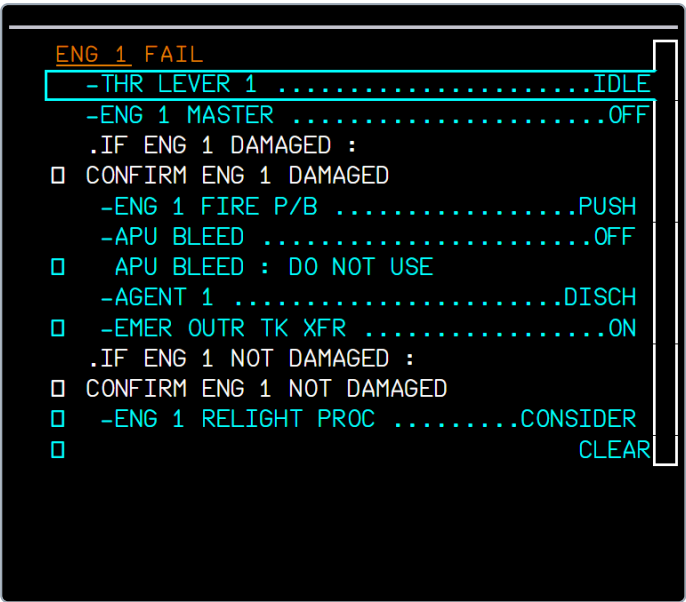


Figure 8. ENG 1 FAIL on the ground: the ground lines only.

Situation	What you should see
ENG x FAIL just after lift-off	The first line is L/G ... UP, until the gear lever is up.
ENG x FAIL in flight	AUTO RELIGHT IN PROGRESS; THR LEVER x IDLE, ENG x MASTER OFF, FIRE P/B, APU BLEED, AGENT: these tick from the cockpit controls.
ENG x FAIL on the ground	Only the ground part of the procedure.
All four engines fail in flight	ALL ENGINES FAILURE (red, LAND ASAP): the FCOM ALL ENG FLAME OUT procedure, fuel remaining / no fuel, relight, then forced landing or ditching; it scrolls with the cursor.
HOT AIR 1 (2) pushbutton OFF	COND HOT AIR 1 (2) FAULT: CARGO TEMP MONITOR, HOT AIR 1 ON.
COND TEMP CTL FAULT	Now with its FCOM lines.

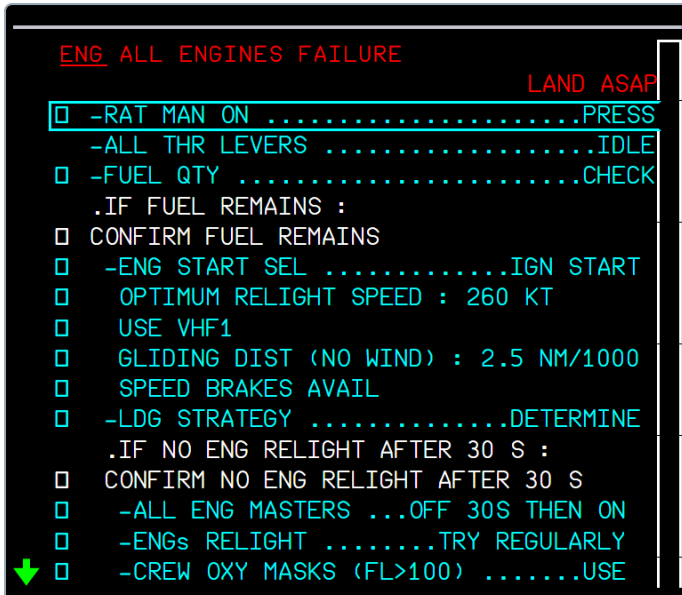


Figure 9. ALL ENGINES FAILURE, first page.

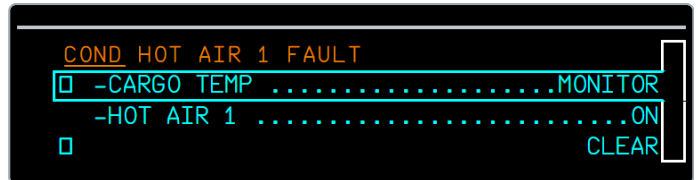


Figure 10. COND HOT AIR 1 FAULT.

5. Reading the lines

- Upper case **.IF ...** lines are ECAM conditions: the crew confirms them (CONFIRM ...), then the lines below become active.
- Headers such as **AT TOP OF DESCENT :** are centred sub-headings.
- Sensed lines tick by themselves once the action is done; the others are ticked by the crew with the ECP CHECK key.
- A line fits the 41 characters of the EWD; nesting is capped at three levels.

6. Limitations and why

Limitation	Why	What to do
About 85 sensed alerts show their title only.	The FCOM gives no lines for them ("Crew awareness").	-
No deferred procedures, limitations or STATUS for the new procedures.	Not in this change.	-
FLUCTUATING VERTICAL SPEED stays WIP; HOT AIR 1(2) OFF / VLV OPEN, T.O V1/VR/V2 DISAGREE, OVERSPEED LOAD ANALYSIS REQUIRED have no lines.	They are not in the 2011 FCOM used as the source.	-
ALL ENGINES FAILURE is long.	It is the full FCOM procedure; the EWD scrolls with the cursor.	-

A32NX RAAS runway awareness callouts

The Honeywell Runway Awareness and Advisory System option of the EGPWS: spoken advisories and cautions about runways, from the sim's own airport data

Aircraft	A32NX
Where to find it	flyPad Settings > Aircraft Options / Pin Programs > RAAS : Off, Feet or Meters
PR branch	pr/18-a32nx-raas (independent)
Status	Tested in the sim (September 2026): the voice and the callouts were heard in flight and on the ground.
Pictures	The flyPad setting from the local browser harness; RAAS is audio only, so the callouts are shown on diagrams drawn from the code (Honeywell product description 060-4564-001, section 4).

1. What it is for
2. Turning it on
3. On the ground
4. Approach and landing
5. Limitations and why

1. What it is for

RAAS is an option of the A320's Honeywell EGPWS (A320 FCOM GEN aircraft configuration, used as a line-up aid in PRO-NOR-SOP-11 TAKEOFF RUNWAY CONFIRM). It tells the crew which runway they are approaching or are on, warns of a taxiway takeoff or landing, and calls the distance remaining. It needs no airport map: the runways come from the sim's airport database.

2. Turning it on

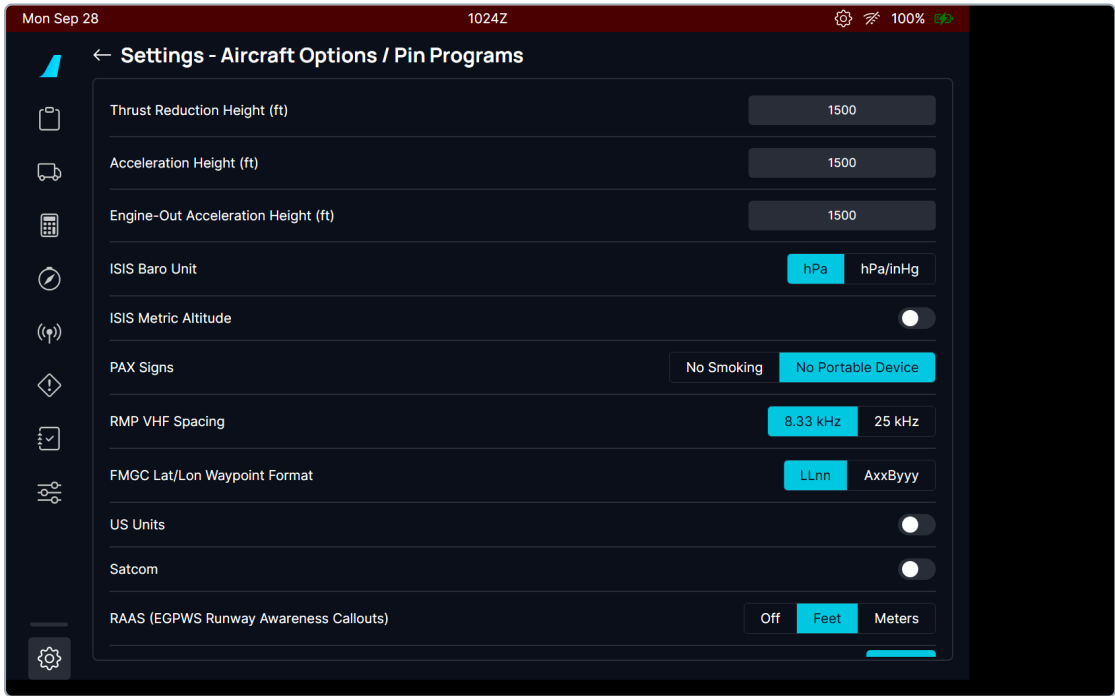


Figure 1. flyPad Settings > Aircraft Options / Pin Programs.



Figure 2. RAAS: Off, Feet (default) or Meters. The unit is the one of the distance callouts.

- The callouts run while the EGPWC is powered (AC BUS 1) and the **GPWS SYS** pushbutton is not OFF.

- The EGPWS aural alerts (e.g. "TOO LOW TERRAIN", "GLIDE SLOPE") have priority: RAAS waits for them.
- The voice is played word by word.

3. On the ground

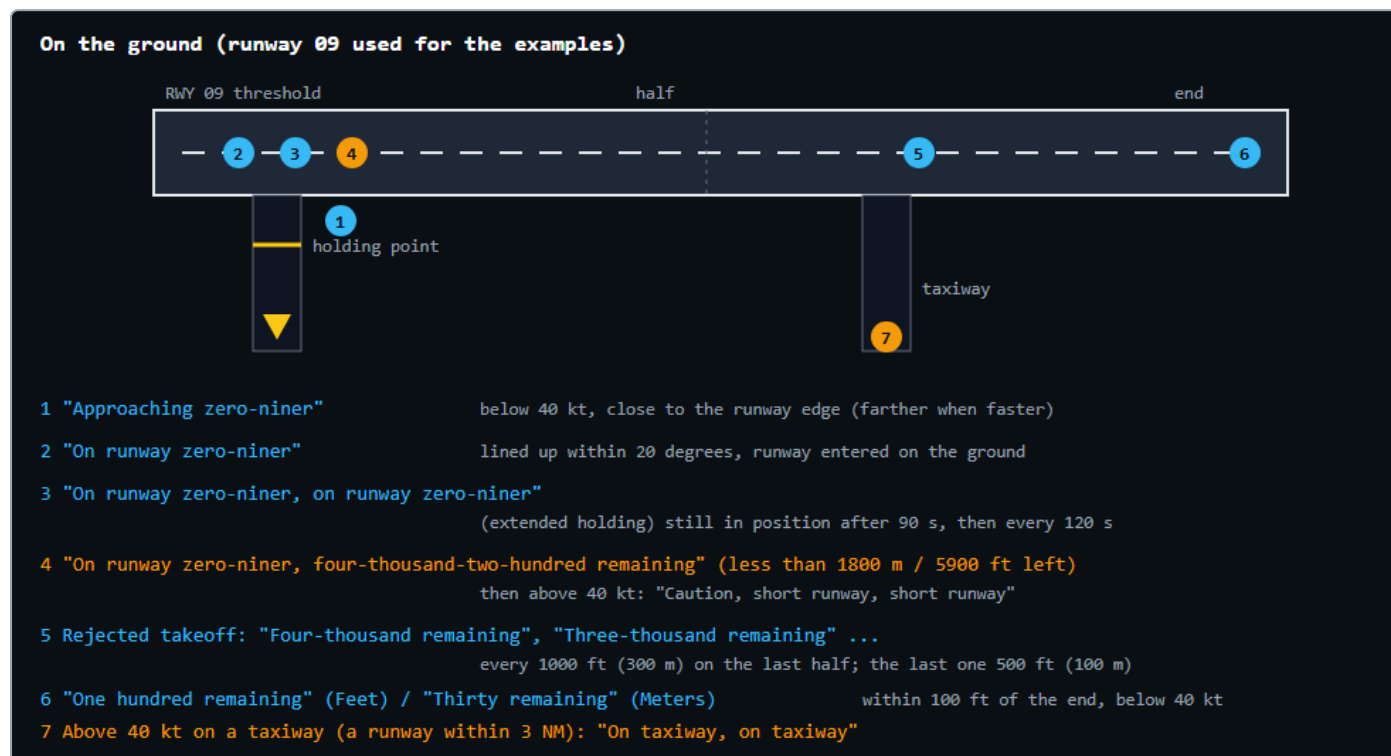


Figure 3. The ground callouts, in blue the advisories, in amber the cautions.

Diagram drawn for this manual: not representative of the in-game UI, reference for the explanation only.

Situation	What you should see
Taxiing towards a runway (below 40 kt)	"Approaching" and the runway end nearest to where you will enter; "Approaching runways" when two at once.
Lining up	"On runway" and the runway, once, when the runway was entered on the ground.
Lining up with less than 1800 m (5900 ft) ahead	The length remaining is added; at the start of the roll (above 40 kt), "Caution, short runway".
Holding on the runway for 90 s	"On runway ..." twice, then every 120 s while you stay.
Starting a takeoff roll on a taxiway (above 40 kt)	"On taxiway, on taxiway".
Rejected takeoff (ground speed down 7 kt from its maximum)	The distance remaining on the last half of the runway.
Slow near the end (within 100 ft, below 40 kt)	"One hundred remaining" (Feet) or "Thirty remaining" (Meters).

4. Approach and landing

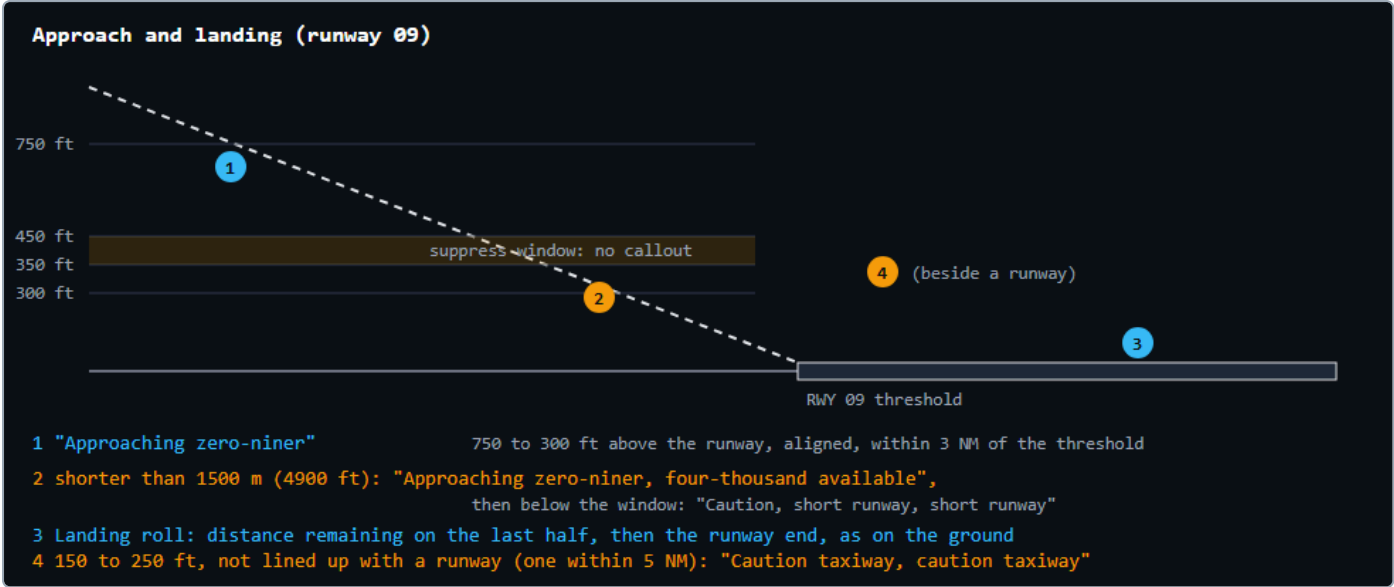


Figure 4. The approach callouts and the suppress window.

Diagram drawn for this manual: not representative of the in-game UI, reference for the explanation only.

Situation	What you should see
Between 750 and 300 ft, lined up, within 3 NM	"Approaching" and the runway, once. Not between 450 and 350 ft (the Airbus suppress window, where the crew is busy with the auto callouts).
The runway is shorter than 1500 m (4900 ft)	The available length is added; below the window: "Caution, short runway".
150 to 250 ft, not lined up with any runway	"Caution taxiway, caution taxiway".
Landing roll	The distance remaining on the last half: every 1000 ft (Feet) or 300 m (Meters), the last one at 500 ft / 100 m; then the runway end.

5. Limitations and why

Limitation	Why	What to do
Only runways known to the sim.	The runways come from the MSFS airport database, not from an airport map: no taxiway geometry, so "on taxiway" is judged as "not aligned with a runway".	-
One synthetic voice.	The clips are generated with the Piper text-to-speech engine (voice trained on the public-domain LJ Speech dataset); the licence of the voice model is to be confirmed before the PR is opened.	-
No RAAS on the A380X.	The A380X uses the OANS runway alerts instead (see the PR 19 manual).	-

A380X RWY AHEAD without an airport map

The OANS RWY AHEAD advisory now works without the Navigraph airport map, from the runways of the sim's own airport database

Aircraft	A380X
Where to find it	PFD and ND, automatic while taxiing
PR branch	pr/19-a380x-oans-rwy-ahead-without-map (stacked on PR 18: it uses the runway module of the RAAS PR)
Status	Built with 4 unit tests; used in the sim during development (September 2026).
Pictures	Captain ND in the local browser harness, running the build installed in the sim; the advisory is triggered by the harness (the harness has no sim airport database).

1. What it is for
2. The rule
3. What it looks like
4. What to expect
5. Limitations and why

1. What it is for

While taxiing, the A380 OANS warns the crew before the aircraft enters a runway: **RWY AHEAD** with the runway name, on the PFD and the ND. In FBW this advisory came from the Navigraph airport map (AMDB): without a Navigraph subscription, or at an airport without a map, it never came. Now, while no airport map is loaded, the OANS uses the runways of the sim's own airport database with the same rule.

2. The rule

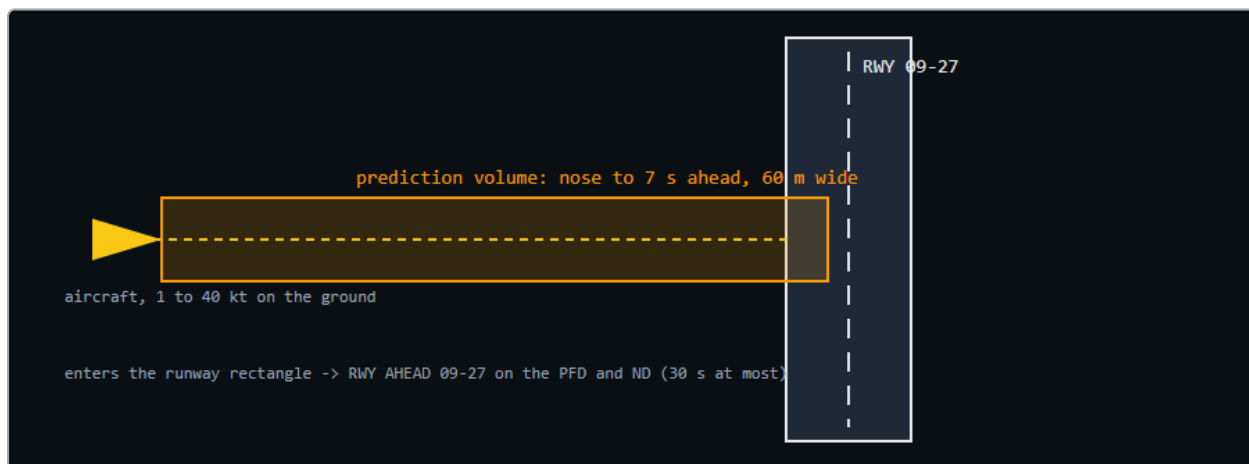


Figure 1. The prediction volume: from the nose to 7 seconds ahead at the current ground speed, 60 m wide. When it reaches a runway, RWY AHEAD appears for that runway.

Diagram drawn for this manual: not representative of the in-game UI, reference for the explanation only.

- On the ground, from 1 to 40 kt.
- The advisory stays at most 30 s, and goes when the aircraft is on the runway.
- With an airport map loaded, nothing changes: the map is used as before.

3. What it looks like

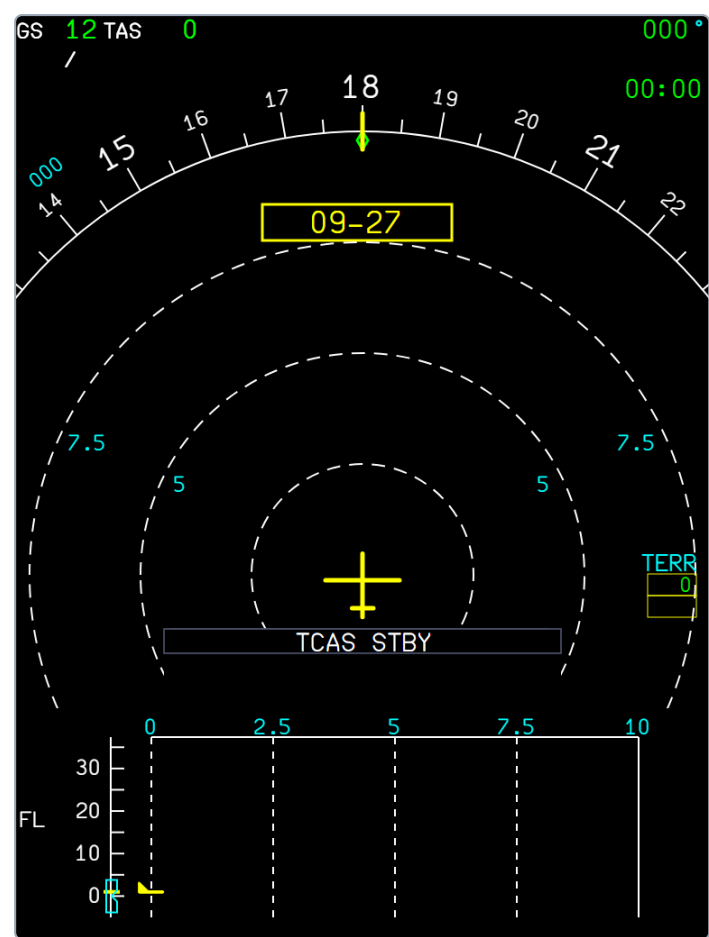


Figure 2. The ND advisory: the runway name in a yellow box at the top (here 09-27, when the aircraft taxis south towards runway 09/27). The PFD shows RWY AHEAD too.

4. What to expect

Situation	What you should see
Taxiing towards a runway, no Navigraph map	RWY AHEAD with the runway name, up to 30 s.
Taxiing towards a runway, with the Navigraph map	RWY AHEAD from the map (as before).
Stopped, or above 40 kt	No advisory.
On the runway	The advisory goes.

5. Limitations and why

Limitation	Why	What to do
Runway rectangles only.	The sim database gives each runway as a rectangle (threshold, length, width); displaced thresholds and runway shapes of the airport map are not known.	-
Waits for PR 18.	It shares the runway module of the RAAS PR.	-

A380X computed alternate fuel, EXTRA time and OFP fuel figures

The ALTN fuel and time of FUEL&LOAD computed as the FCOM defines them, instead of a fixed 6.5 t; the EXTRA time next to the EXTRA fuel

Aircraft	A380X
Where to find it	MFD > ACTIVE > FUEL&LOAD (ALTN line) and ACTIVE > F-PLN > ALTERNATE
PR branch	pr/20-a380x-alt-n-fuel-computation (stacked on PR 10)
Status	Built with 5 unit tests; ALTN checked in the sim during the fuel formula audit (September 2026); EXTRA time checked in the harness only.
Pictures	Captain MFD in the local browser harness (no flight plan, so the values are dashes). The table in section 3 comes from the predictor of this feature run on its own.

1. What it is for

2. Where you see it

3. How much to expect

4. The computation
5. EXTRA time

6. Fuel figures of the company flight plan

7. What to expect

8. Limitations and why

1. What it is for

The block fuel includes the fuel to divert to the alternate. FBW used a fixed 6.5 t, whatever the alternate. The A380 FCOM (DSC-22-FMS-20-30, FUEL&LOAD page) defines the ALTN fuel as the fuel (and time) for the trip from the primary destination to the landing at the alternate, computed with **CI 0**, at **FL220** when the alternate route is shorter than 200 NM and **FL310** otherwise. This is what the FMS computes now.

2. Where you see it

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/FUEL&LOAD

GW

CG

FOB

ZFW

ZFWCG

BLOCK

FUEL PLANNING *

TAXI

PAX NBR

TRIP

CI

RTE RSV

JTSN GW

ALTN

TOW

FINAL

LW

UTC

EFOB

MIN FUEL AT DEST

DEST

ALTN

EXTRA

RETURN

MSG LIST

Figure 1. FUEL&LOAD: the ALTN line gets its fuel and its time; the arrival at the alternate (destination + ALTN time) is under ALTN in the UTC column.

- The distance is the alternate flight plan distance, without its missed approach; without an alternate route, the direct distance.
- The weight is the weight at the destination (ZFW + MIN FUEL AT DEST), with the ALTN wind of the WIND page.
- No ZFW: no default ALTN fuel. The value is recomputed only when its inputs change.

3. How much to expect

Results of the predictor for an alternate close to sea level, fuel on board 12.5 t at the destination (fuel in tonnes / time in minutes / cruise level):

Alternate distance	ZFW 300 t	ZFW 344 t	ZFW 344 t, 30 kt headwind	ZFW 344 t, 30 kt tailwind
60 NM	2.3 t / 13 min / FL140	2.3 / 13 / FL120	2.5 / 14	2.1 / 12
100 NM	3.7 / 19 / FL220	3.7 / 19 / FL200	4.0 / 21	3.4 / 18
150 NM	4.9 / 26 / FL220	5.1 / 27 / FL220	5.5 / 29	4.8 / 25
199 NM	6.1 / 34 / FL220	6.3 / 34 / FL220	6.8 / 37	5.9 / 32
200 NM	6.3 / 32 / FL310	6.5 / 33 / FL310	6.9 / 36	6.2 / 31
300 NM	8.5 / 46 / FL310	8.8 / 46 / FL310	9.4 / 50	8.3 / 44
400 NM	10.8 / 59 / FL310	11.1 / 59 / FL310	11.8 / 64	10.4 / 56

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/F-PLN/ALTERNATE

DATA FROM

TO ALTN

CO RTE

DIR TRK

DIST

EXTRA

SELECTED

DATABASE

NO ALTN

OTHER ALTN

RETURN

MSG LIST

Figure 2. ALTERNATE page: the fuel to each alternate (EXTRA column) uses the same computation.

500 NM	13.0 / 72 / FL310	13.3 / 73 / FL310	14.2 / 78	12.5 / 68
--------	-------------------	-------------------	-----------	-----------

- At 200 NM the level jumps from FL220 to FL310 (the FCOM rule): the time drops a little, the fuel rises a little.
- Short alternates (60-100 NM) fly lower than FL220: the cruise level is lowered until the climb and the descent fit in the distance.
- The old fixed value, 6.5 t, is about right only for an alternate 200 NM away.

4. The computation

- Climb at climb thrust, cruise, idle descent, with the aircraft's VNAV prediction models (the same as the FMS predictions).
- Climb speeds of the A380 FMS: 290 kt climb and cruise, 288 kt descent, M0.84, 250 kt below FL100.

5. EXTRA time

The EXTRA time next to the EXTRA fuel (FCOM: the fuel, and time, that remains after landing at the destination, minus the MIN FUEL AT DEST) was never computed and always showed --:--. It is now the time the EXTRA fuel lasts at the holding fuel flow of the FINAL fuel default, **0.2 t/min** (30 min = 6 t): 7.4 t of EXTRA shows 00:37, 12.0 t shows 01:00.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/FUEL&LOAD

GW

CG

FOB

ZFW

ZFWCG

BLOCK

FUEL PLANNING *

TAXI

PAX NBR

TRIP

CI

RTE RSV

JTSN GW

ALTN

TOW

FINAL

LW

TIME

EFOB

MIN FUEL AT DEST

DEST

ALTN

EXTRA

RETURN

MSG LIST

Figure 3. FUEL&LOAD with 7.4 t of EXTRA fuel: EXTRA time 00:37. The harness has no flight plan, so the EXTRA fuel is set by hand for this picture; the other values stay as dashes.

- The FCOM leaves the FINAL default to the airline fuel policy (AMI file) and gives no rule to turn the EXTRA fuel into a time: the same holding flow keeps FINAL and EXTRA consistent on the page.

- No EXTRA fuel (no flight plan predictions, or a SEC flight plan): --:-- , as the fuel.

6. Fuel figures of the company flight plan

Before engine start, performance data come with the company flight plan (FCOM COMPANY F-PLN REQUEST). When the flight plan received from SimBrief is inserted (INIT CPNY F-PLN REQUEST, then INSERT), FUEL&LOAD takes from the OFP, as if typed by the crew: **TAXI**, **RTE RSV** (the OFP contingency, the % is cleared), **ALTN** (only with an alternate) and **FINAL** (the OFP final reserve).

- The **BLOCK is not taken**: press FUEL PLANNING, which computes it from these figures and the FMS TRIP.
- After engine start the flight plan comes in without its fuel figures.
- The FINAL time is computed from the FINAL fuel at 0.2 t/min (4.8 t shows 00:24).
- Any value can be overwritten as usual.

7. What to expect

Situation	What you should see
No ZFW entered	ALTN fuel shown as dashes (no default).
Alternate route entered	ALTN fuel and time from its distance; the arrival time at the alternate in UTC.
No alternate route, alternate airport set	Computed on the direct distance.
ALTN wind entered	More fuel with a headwind, less with a tailwind.
EXTRA fuel shown	EXTRA time = EXTRA fuel at 0.2 t/min (6 t = 30 min).
Company flight plan inserted before engine start	TAXI, RTE RSV, ALTN and FINAL from the OFP; BLOCK empty until FUEL PLANNING or an entry.

8. Limitations and why

Limitation	Why	What to do
One mean wind for the alternate trip.	The FCOM computation uses the ALTN wind of the WIND page.	-
No holding or approach fuel in ALTN.	The FCOM definition ends at the landing at the alternate; FINAL covers the reserve.	-
EXTRA time at a fixed holding flow.	The FCOM gives no fuel-to-time rule for EXTRA; the flow is the FINAL default one, not a computed holding at the actual weight.	-

A380X company datalink reply time

Company requests (flight plan, winds, takeoff data) answer after a realistic datalink delay, with REQUEST PENDING... and NO COMPANY REPLY as in the FCOM

Aircraft	A380X (the A32NX MCDU uplink stays instant)
Where to find it	flyPad > Settings > Realism > <i>Company Datalink Reply Time</i> ; MFD INIT, WIND and PERF company request buttons
PR branch	pr/21-a380x-company-datalink-delay (stacked on PR 12)
Status	Built and checked in the harness; used in the sim during development.
Pictures	flyPad and captain MFD in the local browser harness, running the build installed in the sim. The harness has no SimBrief access (blocked), so requests stay pending.

1. What it is for
2. The setting
3. Requesting the flight plan
4. Winds and takeoff data
5. What to expect
6. Limitations and why

1. What it is for

On the A380 the flight plan, the winds and the takeoff data come from the airline by datalink (FCOM DSC-22-FMS-10-40-90). A request takes time: the button shows **REQUEST PENDING...** until the answer, and **NO COMPANY REPLY** appears when nothing comes within 4 minutes. FBW answered instantly; now the answer arrives after a reply time you choose.

2. The setting

Company Datalink Reply Time (F-PLN, Winds, T.O Data)

Instant

Fast

Real

Figure 1. flyPad > Settings > Realism: Company Datalink Reply Time, Real by default.

Setting	Answer to a request	Message sent without a request (e.g. SEND TO FMS)
Instant	immediately	immediately
Fast	5 to 15 s	1 to 3 s
Real (default)	60 to 120 s	5 to 15 s

Airbus gives no figure: these are typical airline values, below the 4 minute limit of the FCOM.

3. Requesting the flight plan

1. INIT > **CPNY F-PLN REQUEST** opens COMPANY F-PLN REQUEST (flight number, company route, free text).
2. Press **SEND F-PLN REQUEST**. The button becomes **REQUEST PENDING...**, here and on INIT; one request at a time.
3. After the reply time the SimBrief flight plan is delivered to the FMS, as before.
4. If SimBrief does not answer (no network, no flight plan), **NO COMPANY REPLY** appears after 4 minutes.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

COMPANY F-PLN REQUEST

FLT NBR

CPNY RTE

FREE TEXT

REQUEST PENDING...

RETURN

MSG LIST

Figure 2. Request sent: REQUEST PENDING... (greyed until the answer).

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

ACTIVE/INIT

FLT NBR

ACFT STATUS

REQUEST PENDING...

FROM

TO

ALTN

CPNY RTE

RTE SEL

ALTN RTE

ALTN RTE SEL

CRZ FL

CRZ TEMP

CI

TROPO

TRIP WIND

WIND

CPNY WIND REQUEST

IRS

DEPARTURE

NAVAIDS

FUEL&LOAD

T.O. PERF

RTE SUMMARY

CPNY T.O. REQUEST

MSG LIST

Figure 3. INIT shows the pending request too.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

COMPANY F-PLN REQUEST

FLT NBR

CPNY RTE

FREE TEXT

SEND F-PLN REQUEST * *

RETURN

CLEAR INFO

NO NAVIGRAPH USER

Figure 4. Without a Navigraph account or a SimBrief ID in the flyPad settings: NO NAVIGRAPH USER, no request.

4. Winds and takeoff data

- **CPNY WIND REQUEST** (INIT or WIND page): the winds are delivered after the reply time.
- **Takeoff data** (PERF T.O > CPNY T.O REQUEST, see the PR 11 manual): the answer of the flyPad Send to FMS reaches the FMS not before the reply time of the request; without a request, after the transit time only.

5. What to expect

Situation	What you should see
Real, request sent	REQUEST PENDING... for one to two minutes, then the data.
Instant	The data at once, as before this change.
A second request while one is pending	Not possible: the button stays REQUEST PENDING....
No answer	NO COMPANY REPLY after 4 minutes.
No Navigraph user / SimBrief ID	NO NAVIGRAPH USER; nothing is sent.

6. Limitations and why

Limitation	Why	What to do
The reply times are typical values.	Airbus publishes no figure for the airline answer.	Choose Fast or Instant if you prefer.
The A32NX MCDU uplink stays instant.	This change is for the A380 FMS; the A320 uplink has its own messages.	-

A380X ATC COM pages and SD mailbox

Datalink with ATC (CPDLC logon, requests, reports, emergency, ATIS) on the MFD ATC COM pages of the FCOM, and the ATC mailbox on the SD

Aircraft	A380X
Where to find it	MFD system menu ATC COM (or the KCCU ATC COM key); the MAILBOX at the bottom of the SD
PR branch	pr/22-a380x-mfd-atccom (stacked on PR 10)
Status	Tested in the sim on 2026-09-27 (ATIS, REQUEST / REPORT / EMERGENCY frames, fixes of the test included); CPDLC dialogues with a live ATC network are still to test.
Pictures	Captain MFD and SD in the local browser harness, running the build installed in the sim; the messages come from the harness (no ATC network).

1. What it is for
2. Logging on: CONNECT
3. Asking ATC: REQUEST
4. Reports: REPORT & MODIFY
5. The SD MAILBOX
6. MSG RECORD and ATIS
7. Emergency: EMER
8. What to expect
9. Limitations and why

1. What it is for

The A380 talks to ATC by datalink: the crew logs on to a center, sends requests and reports, receives clearances and answers them, and gets the airport ATIS as text. The FCOM describes the MFD **ATC COM** pages (DSC-46-10-20-30) and the SD **MAILBOX** (DSC-46-10-10-70). They are all built on their FCOM figures and connected to the FBW ATSU, the same datalink system as the A32NX DCDU.

NOTE

The network is chosen in the flyPad: Settings > ATSU / AOC (Hoppie, BeyondATC or SayIntentions). The datalink needs VHF3 in DATA mode on the RMP.

Page (header)	Use
CONNECT	Logon (NOTIFICATION), CONNECTION STATUS with ADS, MAX UPLINK DELAY.
REQUEST	Build a request from frames (climb, direct, offset, weather deviation, speed, clearances...) and send it to the mailbox.
REPORT & MODIFY	Position report, other reports, modify a prepared answer.
MSG RECORD	The record of the closed dialogues.
ATIS	Request and read the ATIS of three airports.
EMER	MAYDAY / PANPAN and the emergency reports.

2. Logging on: CONNECT

1. CONNECT > NOTIFICATION: type the **ATC CENTER** (4 letters) and press **NOTIFY**. The page shows NOTIFYING, then the center in NOTIFIED TO CENTERS (with the time), or NOTIFICATION FAILED.
2. When ATC accepts, CONNECTION STATUS shows the ACTIVE ATC (and the NEXT ATC after a transfer). DISCONNECT ends the connection.

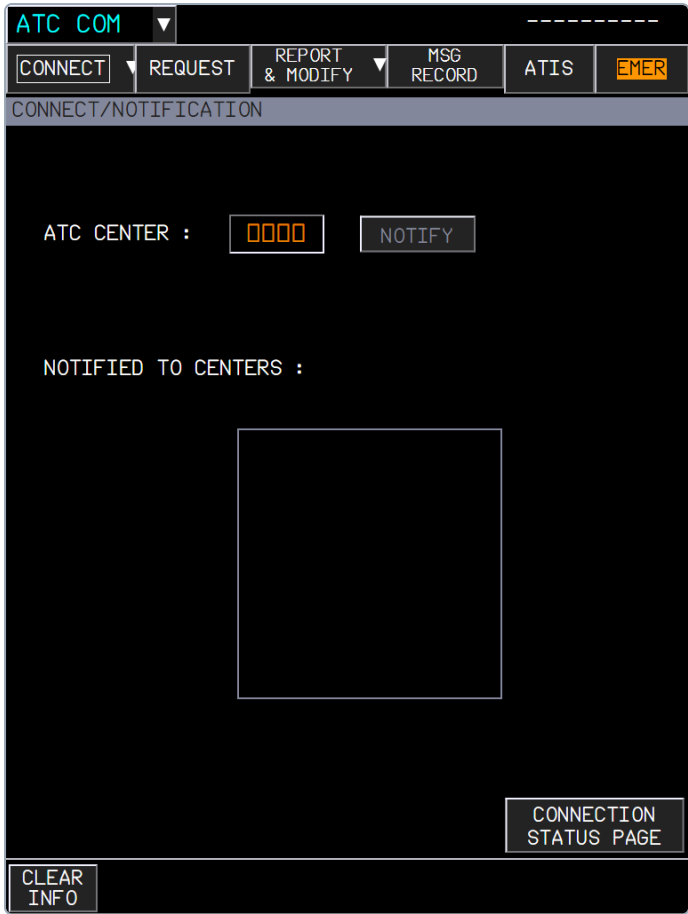


Figure 1. CONNECT / NOTIFICATION.

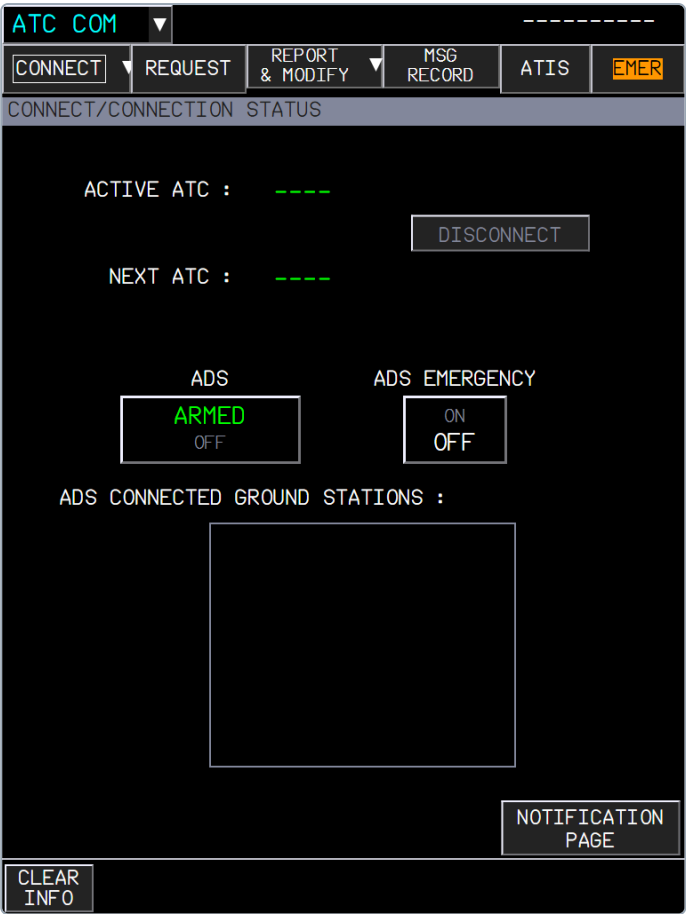


Figure 2. CONNECTION STATUS: active and next ATC, ADS ARMED / OFF, ADS EMERGENCY.

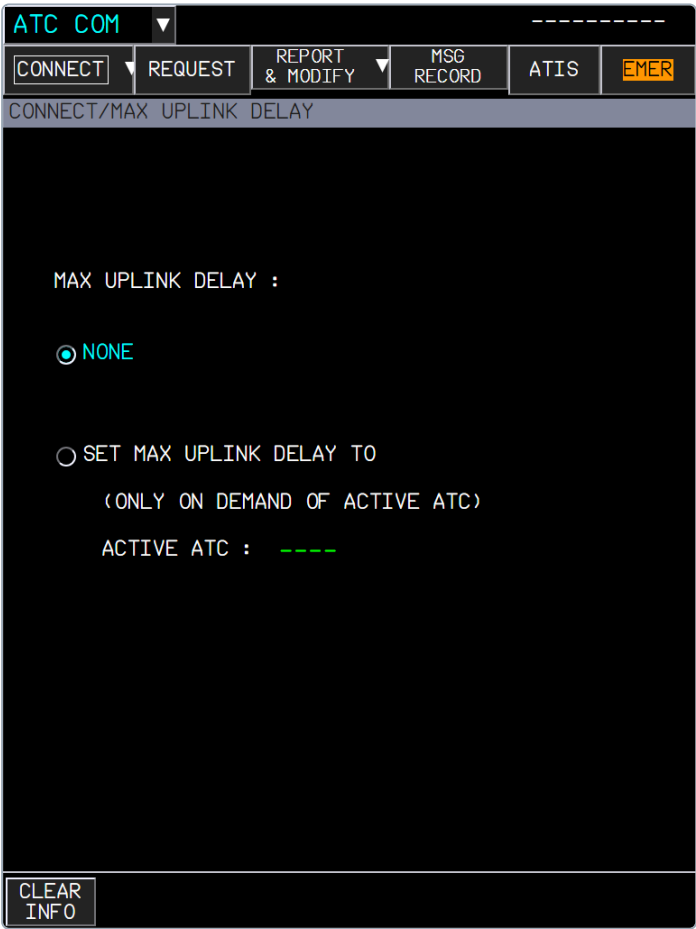


Figure 3. MAX UPLINK DELAY: NONE, or a delay set on the demand of the active ATC.

3. Asking ATC: REQUEST

- 1. Open the menus on the right (VERTICAL, LATERAL, SPEED, CLEARANCE, WHEN CAN WE EXPECT, OTHER) and pick the frames you need; up to 5 frames, each with its fields.
- 2. Fill the fields (FCOM entry formats; amber boxes are mandatory). ADD TEXT adds free text. The bin deletes a frame.
- 3. Press **XFR TO MAILBOX**: the request goes to the SD mailbox, where you **SEND** it.

ATC COM

CONNECTREQUESTREPORT & MODIFYMSG RECORDATISEMER

REQUEST

REQUEST DIR TO

TADEX

REQUEST CLB TO

FL360

AT

REQUEST OFFSET

0000

AT

REQUEST DEVIATION 20NM LEFT

DUE TO WEATHER

VERTICAL

LATERAL

SPEED

CLEARANCE

WHEN CAN WE EXPECT

OTHER

ADD TEXT

CANCEL

XFR TO MAILBOX

CLEAR INFO

Figure 4. A request of 5 frames: DIR TO TADEx, CLB TO FL360, OFFSET (amber: to fill), free text, DUE TO WEATHER.

ATC COM

CONNECTREQUESTREPORT & MODIFYMSG RECORDATISEMER

REQUEST

DEPARTURE ARPT

0000

DESTINATION

0000

GATE

A/C TYPE

A388

ATIS CODE

0

VERTICAL

LATERAL

SPEED

CLEARANCE

WHEN CAN WE EXPECT

OTHER

ADD TEXT

CANCEL

XFR TO MAILBOX

CLEAR INFO

Figure 5. CLEARANCE > DEPARTURE: the departure clearance request (DCL): departure airport, destination, gate, aircraft type, ATIS code.

4. Reports: REPORT & MODIFY

ATC COM

CONNECTREQUESTREPORT & MODIFYMSG RECORDATISEMER

REPORT/AUTO & MANUAL POSITION

AUTO POSITION REPORT

ON
OFF

OVHD	MIPAK	UTC 0946 Z	ALT 32000 FT
PPOS	TAVUN	1005 Z	32000 FT
TO	PUT	1022 Z	
NEXT	DALAN		

CAS	GROUND	VERTICAL
SPEED 310 KT	320 KT	0 FT/MIN
HEADING 135 °T	TRACK 132 °T	

WIND -----	SAT -40 C
ICING TRACE	TURBULENCE LIGHT

ETA DEST 1133 Z	ENDURANCE 4 H30 MIN
--------------------	------------------------

☐DEVIATING☐CLIMBING TO☐DESCENDING TO

ERASE
ALL FIELDS

REFRESH
DATA

ADD
FREETEXT

XFR
TO MAILBOX

CLEAR
INFO

Figure 6. AUTO & MANUAL POSITION: the position report filled by the FMS (overhead, present position, to, next, times, levels, speeds, wind, SAT, ETA, endurance); AUTO POSITION REPORT ON sends it at each reporting point.

ATC COM

CONNECTREQUESTREPORT & MODIFYMSG RECORDATISEMER

REPORT/OTHER REPORTS

BACK ON ROUTE

BACK
ON ROUTE

PASSING
POSITION

LEAVING
ALTITUDE

LEVEL

REACHING
ALTITUDE

REACHING
BLOCK

ADD
FREETEXT

CANCEL

XFR
TO MAILBOX

CLEAR
INFO

Figure 7. OTHER REPORTS: back on route, passing position, leaving / reaching altitude, level, reaching block, free text.

ATC COM

CONNECT

REQUEST

REPORT & MODIFY

MSG RECORD

ATIS

EMER

REPORT/MODIFY

PRESENT ALTITUDE 12500 FT

CANCEL

ADD FREETEXT

XFR TO MAILBOX

CLEAR INFO

NO FMS DATA

Figure 8. MODIFY: a prepared answer (here PRESENT ALTITUDE 12500 FT) that the crew can change before sending.

5. The SD MAILBOX

Every message to and from ATC goes through the mailbox at the bottom of the SD. It shows the active ATC, the ADS status, the message and the answer buttons the message expects.

SAT

XX °C

10:24: 00 GPS

GWCG

XX %

XX KG

FOB

XX KG

ACTIVE ATC : LFBO

ADS OFF

RECALL

Figure 9. Mailbox without a message: ACTIVE ATC LFBO, ADS OFF.

SAT XX °C		10:24: 00 GPS	GW XX KG GWCG XX % FOB XX KG	
 MSG 1/3  RECALL	1425Z FROM KZAK CTL OPEN CLIMB TO REACH FL340 BY BERGI		WILCO STANDBY UNABLE PRINT	

Figure 10. An uplink: CLIMB TO REACH FL340 BY BERGI, OPEN, with WILCO / STANDBY / UNABLE. MSG 1/3: three messages are waiting.

SAT XX °C		10:24: 00 GPS	GW XX KG GWCG XX % FOB XX KG	
 RECALL	TO KZAK CTL REQUEST DEVIATION 20NM LEFT OF ROUTE DUE TO CB AHEAD WILL ADVISE WHEN CLEAR OF WEATHER THANKS		SEND CANCEL PRINT	

Figure 11. A request ready to go: SEND or CANCEL.

The message expects	Buttons
WILCO / UNABLE (a clearance)	WILCO, STANDBY, UNABLE
AFFIRM / NEGATIVE (a question)	AFFIRM, STANDBY, NEGATIVE
ROGER (information)	ROGER
A departure or oceanic clearance	ACK / REFUSE

UNABLE and NEGATIVE can add a reason (DUE TO WEATHER, A/C PERFORMANCE, TURBULENCE) and free text; MODIFY opens the answer on the MFD. The message colours follow the FCOM status legend.

6. MSG RECORD and ATIS

ATC COM

CONNECTREQUESTREPORT & MODIFYMSG RECORDATISEMER

MSG RECORD/LIST

1250ZFROM LFBO CTL
CLIMB TO FL380

1247ZFROM LFBO CTL
WILCO

1246ZFROM LFBO CTL
ROGER

1245ZFROM LFBO CTL
UNABLE

1244ZFROM LFBO CTL
NEGATIVE

1243ZFROM LFBO CTL
AFFIRM

1242ZFROM LFBO CTL
STANDBY

1241ZFROM LFBO CTL
WILCO

ERASE ALL

PRINT ALL

CLEAR INFO

Figure 12. MSG RECORD: the closed dialogues with their time and the answer given; ZOOM shows one in full; ERASE ALL, PRINT ALL.

ATC COM

CONNECTREQUESTREPORT & MODIFYMSG RECORDATISEMER

ATIS/LIST

LFB0DEPR 1145Z

OPTIONS

LFB0 DEP ATIS R 1145Z EXP ARR RWY 32R AND 21L /DEP /SID 5 B. 5 P WIND CALM VIS 3500 M - CB FEW 500 FT BKN 26000 FT /

LFP0ARR

SEND REQUEST

AUTO UPDATE

LFPGARRP 1200Z

SEND AGAIN

OPTIONS

LFPG ARR ATIS P 1200Z ATIS INFO ARR/DEP APCH ILS APCH LDG 27L AND 26L / RWY 26 L WET TL 40 BIRD HAZARD AFTER VACATING RWY 26L HOLD SHORT OF RWY 26R TKOF 27L AND 26R SID 8A 8B 8Y 8Z QNH 1013 TEMP 12 DEWPOINT 8

PRINT ALL

UPDATE ALL

CLEAR INFO

Figure 13. ATIS: three request areas (departure, arrival, alternate, following the FMS airports), each with its ATIS text, AUTO UPDATE, and SEND REQUEST / SEND AGAIN.

ATC COM

CONNECT

REQUEST

REPORT
& MODIFY

MSG
RECORD

ATIS

EMER

ATIS/RECEIVED

LFB0 DEP R 1145Z
LFB0 DEP ATIS R 1145Z EXP ARR RWY 32R AND
21L /DEP /SID 5 B. 5 P WIND CALM VIS 3500 M
- CB FEW 500 FT BKN 26000 FT /

RETURN
TO LIST

CLEAR
INFO

Figure 14. ATIS RECEIVED: the full text of an ATIS, RETURN TO LIST.

7. Emergency: EMER

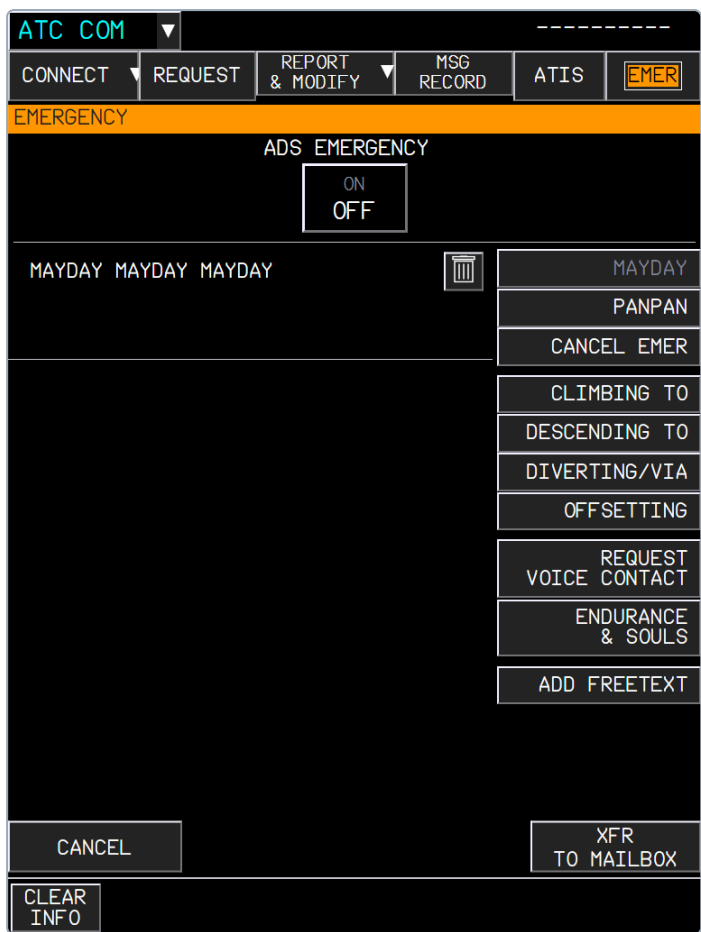


Figure 15. EMERGENCY (orange title): MAYDAY MAYDAY MAYDAY frame, PANPAN, CANCEL EMER and the emergency reports (climbing / descending to, diverting, offsetting, voice contact, endurance and souls). MAYDAY also switches the ADS to EMERGENCY.

8. What to expect

Situation	What you should see
NOTIFY with no network set	NOTIFICATION FAILED.
Request with an empty mandatory field	XFR TO MAILBOX stays greyed.
Uplink received	Mailbox message with its answer buttons.
Answer sent	The dialogue closes and goes to MSG RECORD.
ATIS requested	The text appears in its area; a newer version replaces it (AUTO UPDATE).
MAYDAY sent	ADS EMERGENCY ON.

9. Limitations and why

Limitation	Why	What to do
Needs an online datalink network.	ATC messages come from Hoppie, BeyondATC or SayIntentions through the flyPad ATSU settings.	Set the network in the flyPad.
No real ADS contracts.	The ATSU networks do not handle ADS-C; ADS ARMED / EMERGENCY is shown locally.	-

MSG RECORD and mailbox PRINT say PRINTER NOT AVAIL.	Printing needs the pedestal printer, which has no PR yet.	-
CPDLC dialogues with a live network are not tested yet.	Only the harness messages and the ATIS were checked in the sim.	Report any wrong answer element.

flyPad landing calculator

Landing distances, maximum landing weight, approach and go-around speeds for the A380 (new) and the A320 (reworked), on one screen laid out like the A380 OIS LDG PERF application

Aircraft	A380X and A32NX
Where to find it	flyPad > Performance (calculator icon) > Landing tab
PR branch	pr/23-efb-landing-calculator (stacked on PR 12)
Status	Built, deployed and checked in the local flyPad harness; the in-sim test is still to do.
Pictures	flyPad pages in the local browser harness, running the build installed in the sim. The harness has no navigation database, so the runway is entered by hand (in the sim, the runway list fills the RUNWAY section).

1. What it is for
2. The screen
3. A normal in-flight calculation (A380)
4. Braking modes
5. Runway condition and reverse thrust
6. BTV exits (A380)
7. Dispatch
8. When the runway is too short or the aircraft too heavy
9. The A320 version
10. What to expect
11. Limitations and why

1. What it is for

Before the descent, the crew checks that the aircraft can land on the planned runway: the landing distance for the runway condition and braking, the margin left, the approach speed VAPP and the go-around performance. On the real A380 this is the LDG PERF application of the OIS (FCOM PER-LND); on the A320 it is the QRH (in flight) and the FCOM tables (dispatch). This page does both, with the data of each aircraft.

- **IN-FLIGHT** (default): the actual landing distance for the braking mode you will use, as in the QRH / OIS, with the stop margin.
- **DISPATCH**: the regulatory (factored) required landing distance, to check a runway before departure.

2. The screen

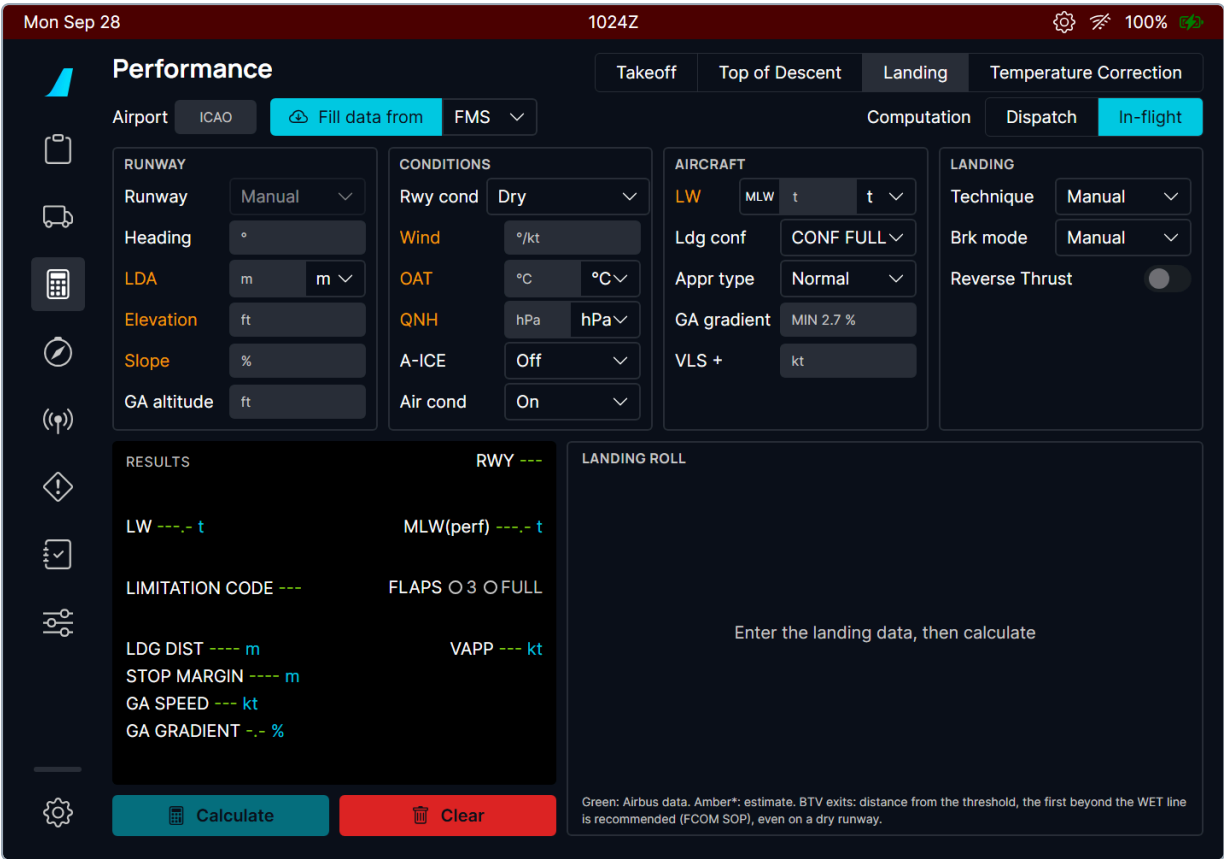


Figure 1. The A380 landing screen before any input. Amber labels are the inputs still missing for a calculation.

Part	Contents
Airport, Fill data from	The ICAO code; Fill data from imports the data from the FMS (A380 FMC / A320 MCDU), the OFF (SimBrief) or the METAR .
Computation	DISPATCH or IN-FLIGHT.
RUNWAY	Runway from the list (or Manual), heading, landing distance available (LDA), elevation, slope, and on the A380 the go-around altitude.
CONDITIONS	Runway condition, wind (direction/speed), OAT, QNH; on the A380 the anti-ice and air conditioning (they change the go-around gradient).
AIRCRAFT	Landing weight (MLW button = the maximum landing weight), landing configuration, and on the A380 the approach type, the required go-around gradient and the VLS increment.
LANDING	Manual landing or autoland, braking mode, reverse thrust, and on the A320 the overweight procedure.
RESULTS	The results in the OIS layout (below).
LANDING ROLL	A picture of the runway with the touchdown, the stop point of each braking mode, the stop margin, and on the A380 the BTV lines and the runway exits.

Figure 2. Fill data from: FMS (destination, runway, predicted landing weight and the PERF APPR QNH, temperature, wind and landing configuration), OFF or METAR.

3. A normal in-flight calculation (A380)

1. In flight, with the destination and the arrival runway in the flight plan and PERF APPR filled: choose **FMS** and press **Fill data from**. Otherwise type the airport, choose the runway and fill the conditions.
2. Check the landing weight (the FMS prediction) and the configuration.
3. Choose the braking mode you plan to use (manual, autobrake LO / 2 / 3 / HI).
4. Press **Calculate**.

Figure 3. Dry runway, 3000 m, 380 t, CONF FULL, manual braking: landing distance 1130 m, stop margin 1870 m, VAPP 141 kt (VLS 136 + 5). The landing roll shows where each braking mode would stop.

Reading the results

Line	Meaning
LW	The landing weight used.

MLW(perf)	The highest landing weight the runway and the go-around allow : the lower of the weight whose landing distance just fits the LDA and the weight that still gives the required go-around gradient. It is a performance limit, not the structural maximum landing weight (395 t on the A380): it can be above it. Above the structural MLW the page shows OVERWEIGHT LANDING.
LIMITATION CODE	WGT : your weight is below MLW(perf), nothing limits you. LDA (red): the landing distance limits the weight. ACG : the go-around gradient limits it.
FLAPS	The landing configuration used (with AUTO CONF: the one the page chose).
LDG DIST / REG LDG DIST	In flight: the actual landing distance for the braking mode. Dispatch: the required (regulatory) distance.
STOP MARGIN	LDA minus the landing distance. Negative (red): the aircraft does not stop on the runway.
GA SPEED, GA GRADIENT	The go-around speed and the climb gradient of the go-around configuration.
VAPP	VLS of the landing configuration + the VLS increment (the wind increment, or what you enter).
Bottom line	The actual landing distance (ALD), the wind taken into account, the go-around configuration, and whether reverse thrust was credited.

Colours follow the flyPad takeoff calculator: **green** values come from Airbus data; values in **amber with a star (*)** are estimates of a model fitted on that data (see Limitations).

4. Braking modes

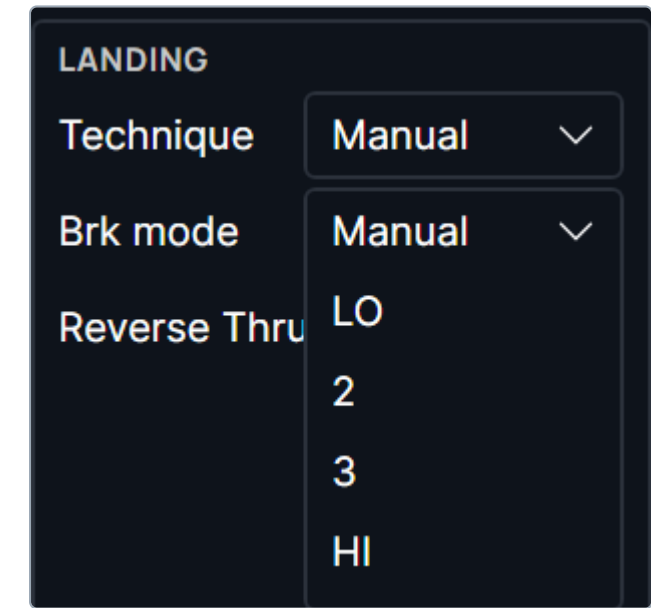


Figure 4. A380 braking modes: Manual, autobrake LO, 2, 3, HI. BTV is not a mode here: the page gives its DRY / WET lines and exit advice.

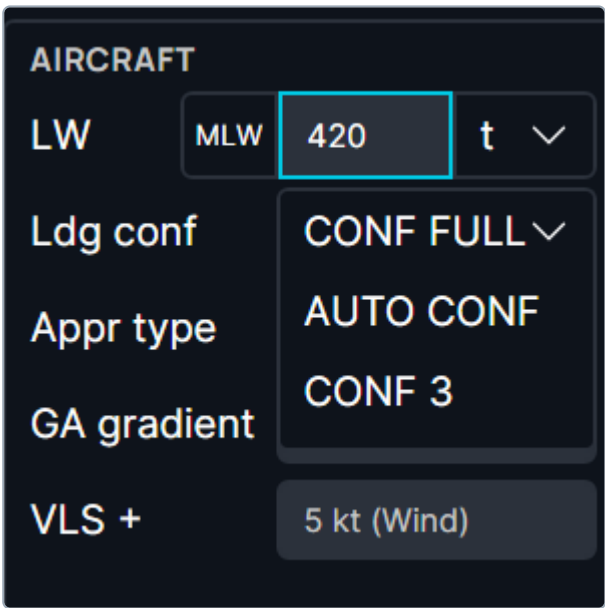


Figure 5. Landing configuration: CONF FULL, CONF 3 or AUTO CONF (the page chooses, as the OIS does).

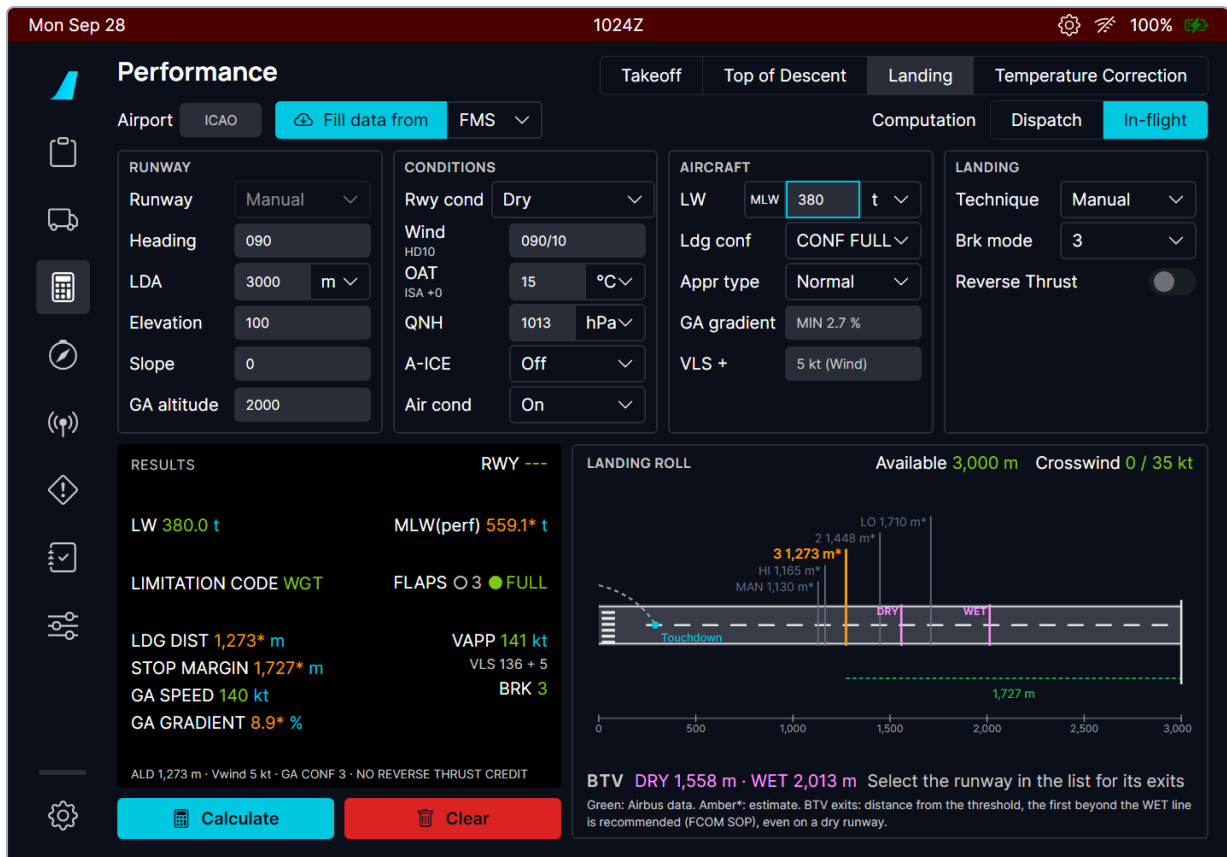


Figure 6. Same landing with autobrake 3: 1507 m. The stop point of the chosen mode is highlighted; the others stay visible for comparison.

AUTO CONF (A380, FCOM PER-LND): CONF FULL when its go-around in CONF 3 gives the required gradient, otherwise CONF 3 with a go-around in CONF 2; for an overweight landing, CONF 3 with a go-around in CONF 1+F.

5. Runway condition and reverse thrust

The runway condition follows the runway condition codes (RWYCC): dry, wet, water and slush 1/4" or 1/2", compacted snow, ice. The crosswind limit of the condition is shown at the top of the landing roll.

- **Reverse thrust** is credited only on a **contaminated** runway (FCOM: no credit on dry and wet runways); the result line says *REVERSE THRUST CREDIT* or *NO REVERSE THRUST CREDIT*.
- On the A380, **BTV is prohibited on a contaminated runway** (FCOM LIM-32-30): the BTV line says so and advises autobrake 3.
- With 1/2" of water or slush, a message reminds you that Airbus does not recommend these distances: a deeper contaminant gives a shorter distance in the data.

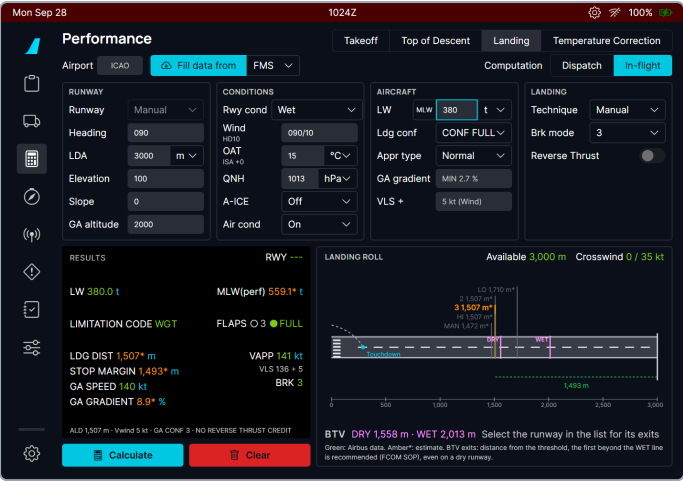


Figure 7. Wet runway, autobrake 3.

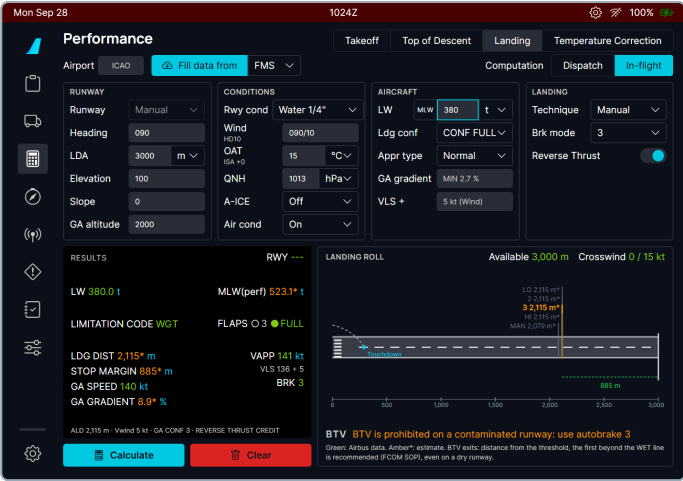


Figure 8. Water 1/4" with reverse thrust: credited, BTV prohibited, crosswind limit 15 kt.

6. BTV exits (A380)

On the A380, the page shows the **BTV DRY** and **WET** lines: where the aircraft stops with the BTV decelerations (FCOM DSC-32-10-30-20), from 400 m after the threshold plus 5 s of roll. When the runway comes from the list and a Navigraph account is linked, the runway exits of the airport map are drawn and rated:

Exit	Meaning
Recommended	The first exit beyond the WET line (the FCOM SOP choice, even on a dry runway).
Beyond WET	Achievable wet or dry.
Dry only	Between the DRY and WET lines: only on a dry runway.
Not achievable	Before the DRY line.

Without the runway list or the Navigraph data, the line says "Select the runway in the list for its exits". This is information for your briefing: the BTV selection itself is made on the OANS, as on the real aircraft.

7. Dispatch

DISPATCH gives the **required landing distance** used to check a runway before departure: the actual distance divided by 0.6 on a dry runway, $\times 1.15$ on a wet one, and on a contaminated runway the higher of the wet value and $ALD \times 1.15$ (A380 FCOM PER-LND). Braking is manual (the braking mode is greyed out). The orange bar is the required distance.

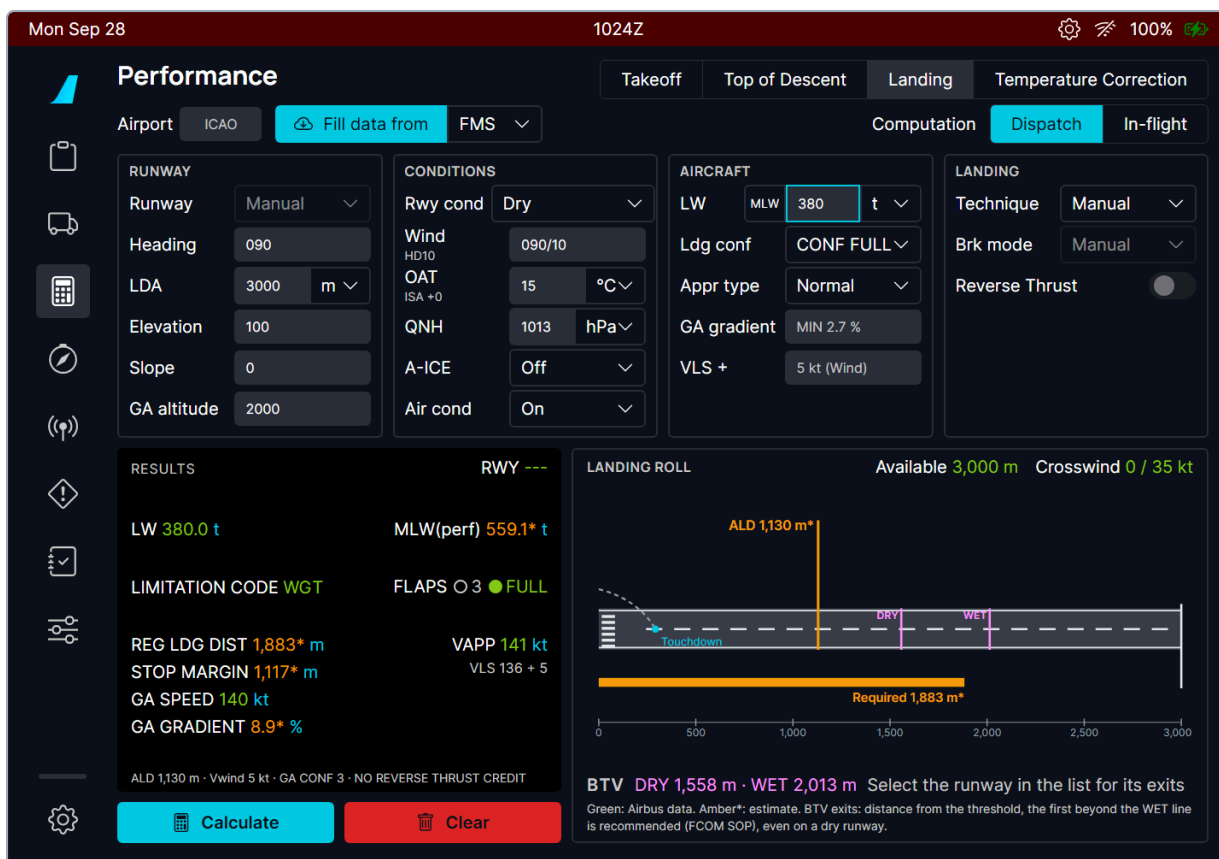


Figure 9. Dispatch on the same runway: required landing distance 1883 m (REG LDG DIST).

8. When the runway is too short or the aircraft too heavy

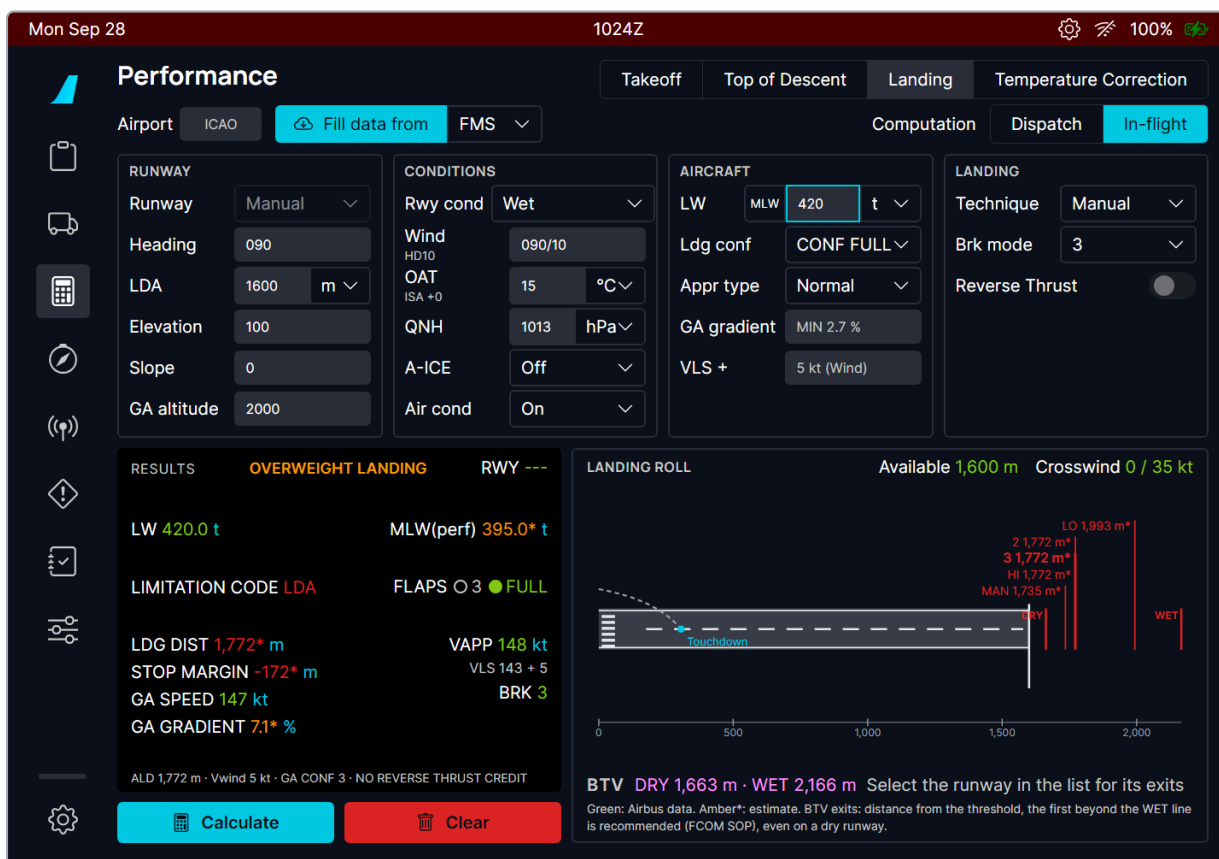


Figure 10. 1600 m wet runway at 420 t: OVERWEIGHT LANDING (above the 395 t MLW), limitation code LDA in red, negative stop margin (-172 m): the aircraft does not stop on this runway with autobrake 3.

9. The A320 version

The A320 uses the same screen with its own data: the **QRH in-flight landing distances** (shown $\times 1.15$, the QRH margin) and the **FCOM dispatch tables** (PER-LDG-DIS-RLD) with the autoland correction and the wet minimum. Braking modes are Manual, LO and MED; the **Overweight Procedure** switch applies the QRH overweight landing corrections. There is no go-around section.

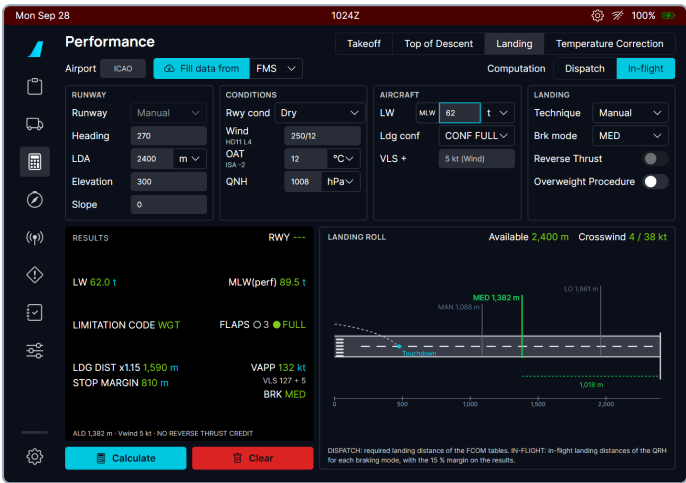


Figure 11. A320 in flight: 62 t, 2400 m dry, autobrake MED: 1382 m (1590 m with the 15 % margin).

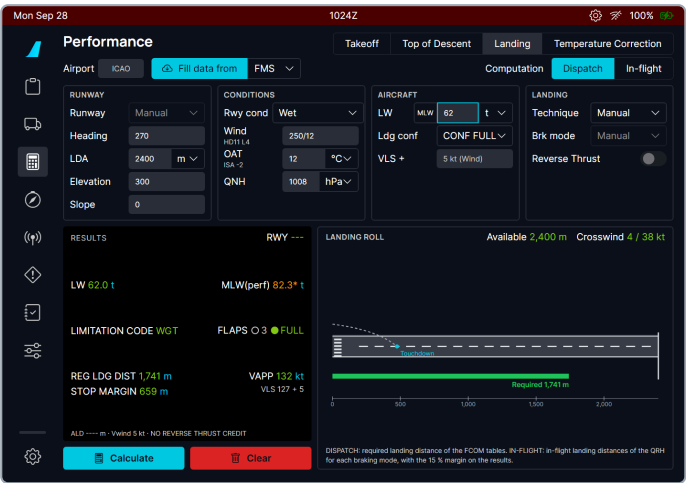


Figure 12. A320 dispatch on a wet runway: required 1741 m from the FCOM tables (green: Airbus data).

10. What to expect

Situation	What you should see
Inputs missing	Amber labels; the Calculate button stays dim until they are filled.
Fill data from FMS without a flight plan	Only what the FMS has is filled; complete the rest by hand.
Dry or wet runway, reverse thrust on	NO REVERSE THRUST CREDIT: the FCOM gives none there.
Contaminated runway	Longer distances, lower crosswind limit; A380: BTV prohibited.
Landing weight above the MLW	OVERWEIGHT LANDING in the RESULTS; A380 AUTO CONF lands in CONF 3.
LW above MLW(perf)	Limitation code LDA or ACG in red; the stop margin may be negative.
Change of an input after Calculate	Press Calculate again.

11. Limitations and why

Limitation	Why	What to do
Most A380 values are estimates (amber *).	The only public A380 landing data is the landing field length chart of the Airbus A380 Aircraft Characteristics (dry, manual, CONF FULL). The other cases use a model fitted on it (0.8 % error on the chart) and the FCOM rules.	Use the results for training and planning, not as certified data.
Autobrake and BTV distances are estimates.	They come from the autobrake decelerations of the FCOM, not from Airbus tables.	-
BTV exits need the runway from the list and a Navigraph account.	The exits come from the Navigraph airport map (the OANS data).	Choose the runway in the list; link Navigraph.

No BTV on the A320.	The A320 has no BTV.	-
The go-around gradient is calibrated on one FCOM example.	No A380 go-around tables are public.	-

flyPad descent calculator

Top of descent, time, distance and fuel of the descent with the FMS performance model, and a live check of where to start down: the IN-FLT PERF DES module of the A380 OIS on both aircraft

Aircraft	A380X and A32NX
Where to find it	flyPad > Performance > Top of Descent tab
PR branch	pr/24-efb-descent-calculator (stacked on PR 23)
Status	Built, deployed and checked in the local flyPad harness; the in-sim test is still to do.
Pictures	flyPad pages in the local browser harness, running the build installed in the sim (A380X). The harness has no FMS, so the inputs are typed; in the sim, Fill data from FMS fills them.

1. What it is for
2. The screen
3. Standard descent
4. Descent check
5. Given V/S, emergency and ECON
6. Anti-ice, fuel factor and estimates
7. What to expect
8. Limitations and why

1. What it is for

The FMS computes the top of descent (T/D) for the managed descent. The crew still needs other answers: how long an emergency descent takes, where to start a descent at a given V/S, whether the aircraft is late on the profile after a shortcut, and a cross-check of the FMS T/D. The A380 OIS has an in-flight descent module for this (FCOM PER-IFT-DES); this page does the same on both aircraft, and replaces the old Top of Descent calculator.

It uses the **same flight and engine models as the FBW FMS predictions**, so its T/D agrees with the one drawn on the ND for the same inputs.

2. The screen

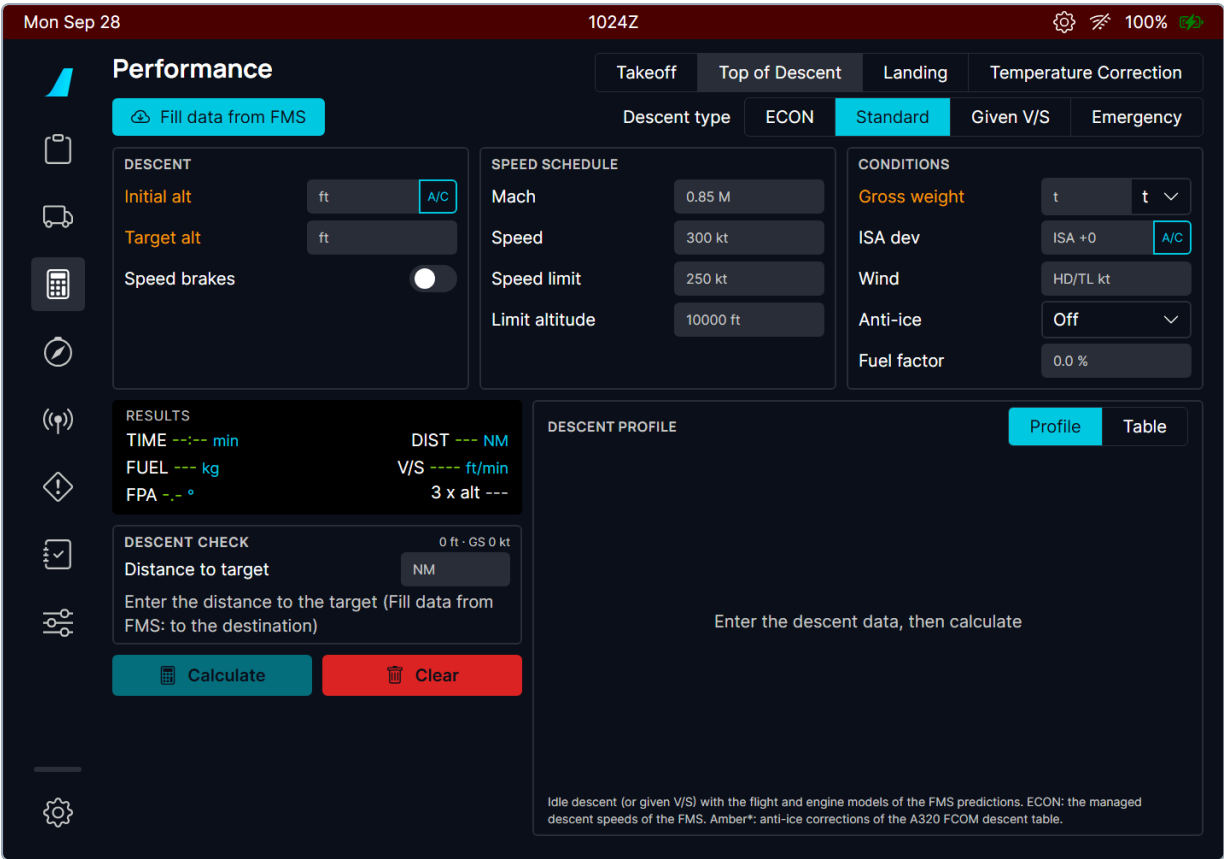


Figure 1. The descent calculator. Amber labels: inputs still needed.

Part	Contents
Fill data from FMS	Cruise level (or the current altitude when lower), destination elevation + 1500 ft, gross weight, the FMS speed limit, the ECON speeds and the distance to the destination.
Descent type	ECON : the managed descent speeds of the FMS. Standard : the FCOM speed schedule (A380 M.85 / 300 kt / 250 kt below FL100; A320 M.78 / 300 / 250). Given V/S : a constant vertical speed. Emergency : MMO / VMO with the speed brakes.
DESCENT	Initial altitude (A/C : the current one) and target altitude; speed brakes.
SPEED SCHEDULE	Mach, speed, speed limit and its altitude. Empty = the standard values shown in grey.
CONDITIONS	Gross weight, ISA deviation (A/C : from the current temperature), wind component (positive = headwind, negative = tailwind), anti-ice (off / engine / total), fuel factor.
RESULTS	Time, distance, fuel, mean V/S and flight path angle, and the "3 × altitude" rule of thumb for comparison; the crossover altitude and where the aircraft slows to the speed limit.
DESCENT CHECK	With the distance to the target: when to start down, or how late you are.
DESCENT PROFILE	Profile chart or table every 5000 ft.

3. Standard descent

- 1. Press **Fill data from FMS** in cruise with a flight plan, or type the initial altitude, the target altitude and the weight.
- 2. Choose the descent type and, if needed, the wind component and anti-ice.

3. Press **Calculate**.

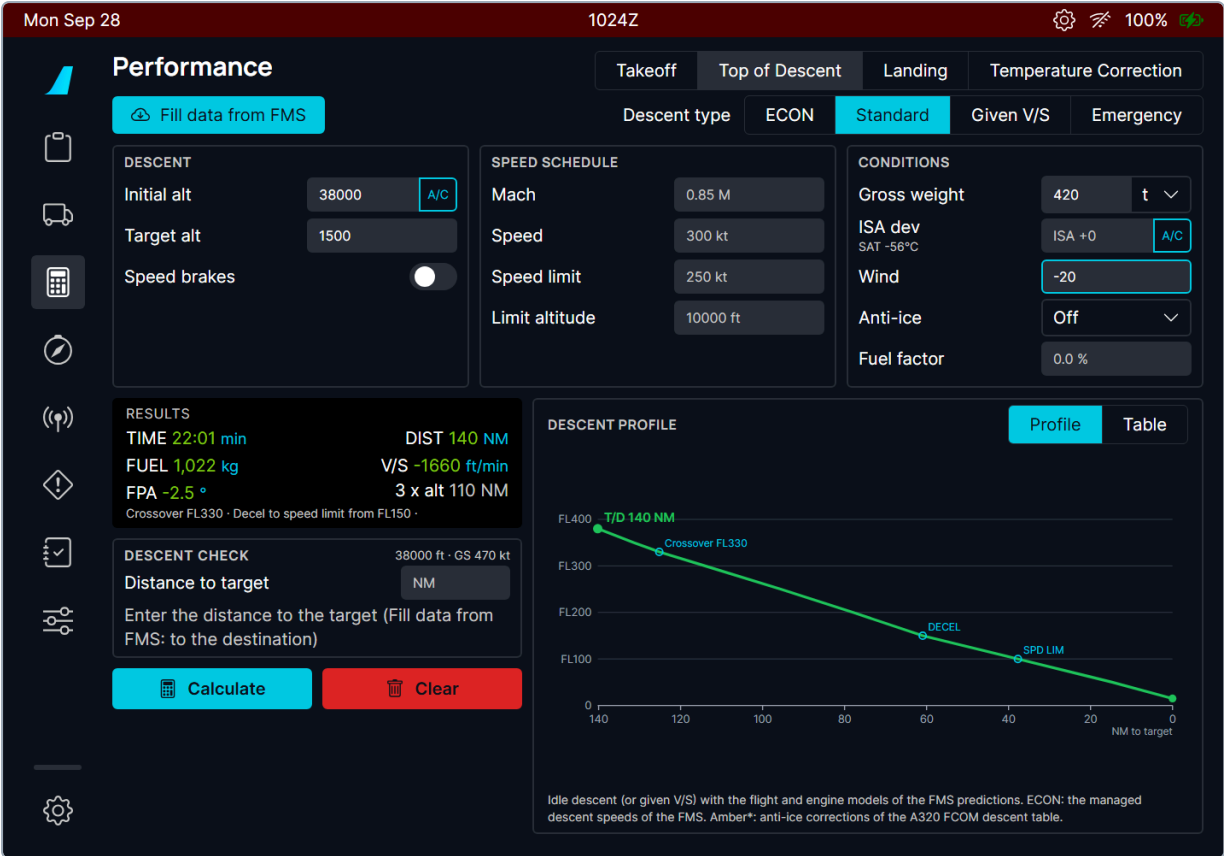


Figure 2. FL380 to 1500 ft, 420 t, 20 kt tailwind, standard speeds: 22 min, 140 NM, 1022 kg, -1660 ft/min on average (-2.5°). The chart shows the T/D, the crossover altitude (Mach to CAS), the deceleration (DECEL) and the speed limit (SPD LIM).

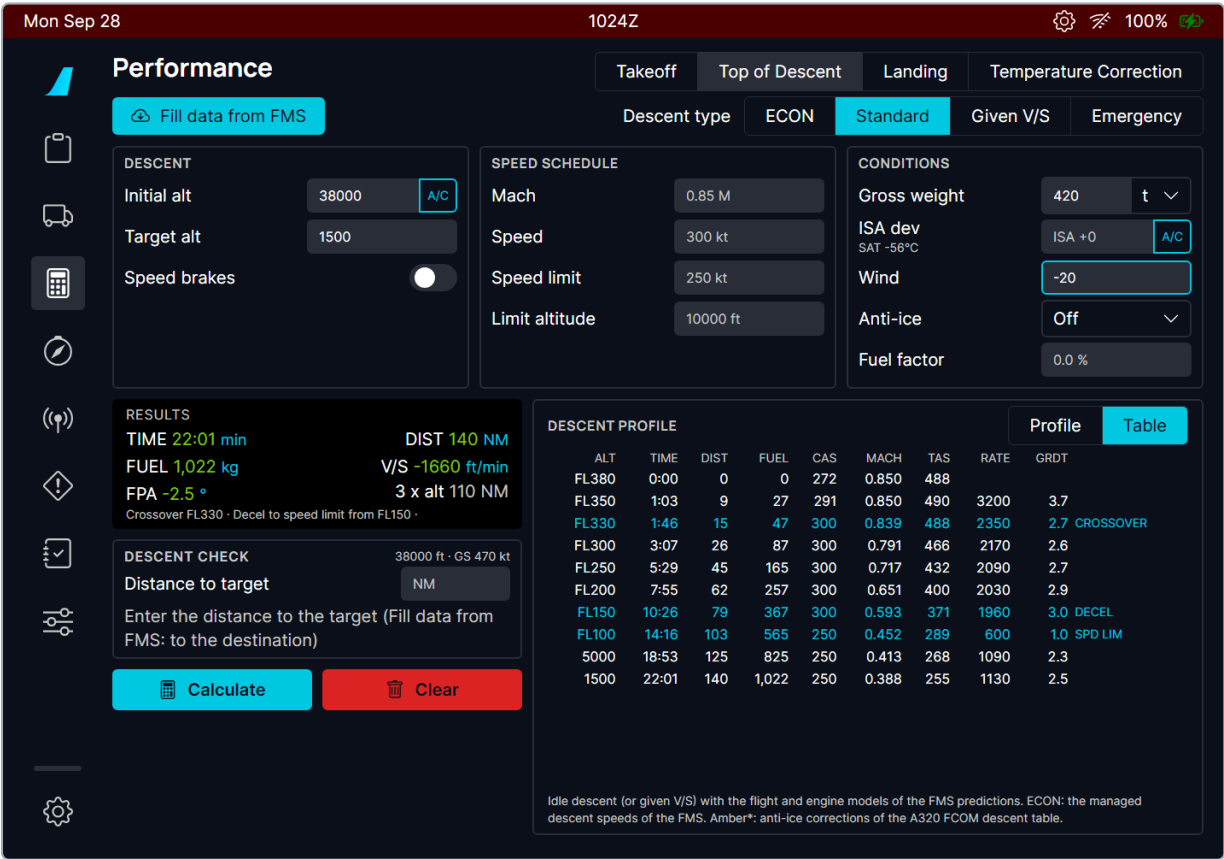


Figure 3. Table view: time, distance, fuel, CAS, Mach, TAS, RATE (rate of descent, ft/min) and GRDT (descent gradient, degrees) every 5000 ft, named and signed like the FCOM table results (PER-IFT-DES-DSR: both positive in descent); the crossover, DECEL and SPD LIM lines in blue. The RESULTS line gives V/S and FPA negative, as set on the FCU.

The deceleration to the speed limit is flown on the -1° idle path of the FMS descent strategy, which is why the profile flattens between DECEL and SPD LIM.

4. Descent check

Enter the **distance to the target** (Fill data from FMS enters the distance to the destination). The page reads the aircraft altitude and ground speed (shown top right of the box) and compares the aircraft with the profile. The aircraft is the triangle on the chart.

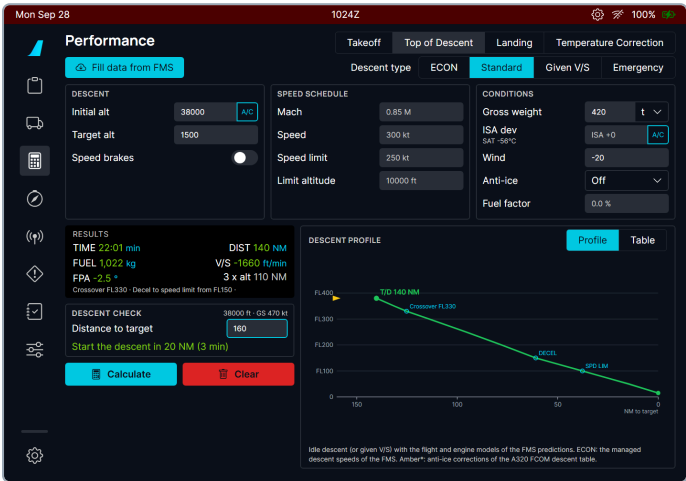


Figure 4. 160 NM to go, T/D at 140 NM: "Start the descent in 20 NM (3 min)".

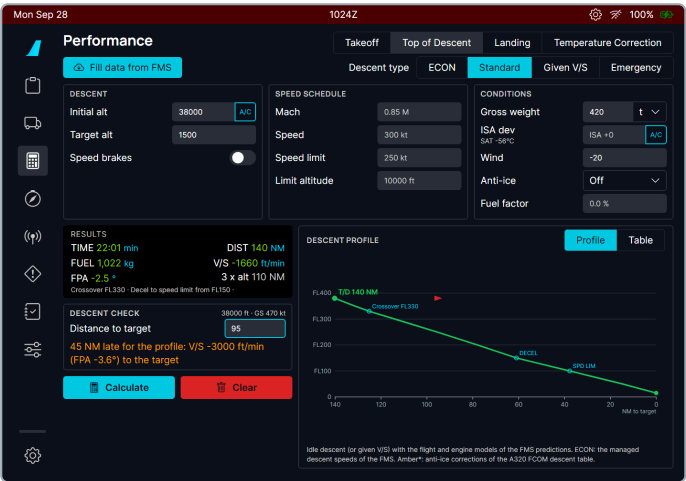


Figure 5. 95 NM to go at FL380: "45 NM late for the profile: V/S -3000 ft/min (FPA -3.6°) to the target".

TIP

After a shortcut, the late message gives the V/S (or FPA) that still makes the target, to set in V/S or FPA mode, or tells you to use the speed brakes.

5. Given V/S, emergency and ECON

Given V/S adds a V/S field. The page flies that V/S with the thrust it needs. Where the V/S is steeper than an idle descent can give, the page flies idle there and says so in amber (the aircraft cannot descend faster at idle without speed brakes).

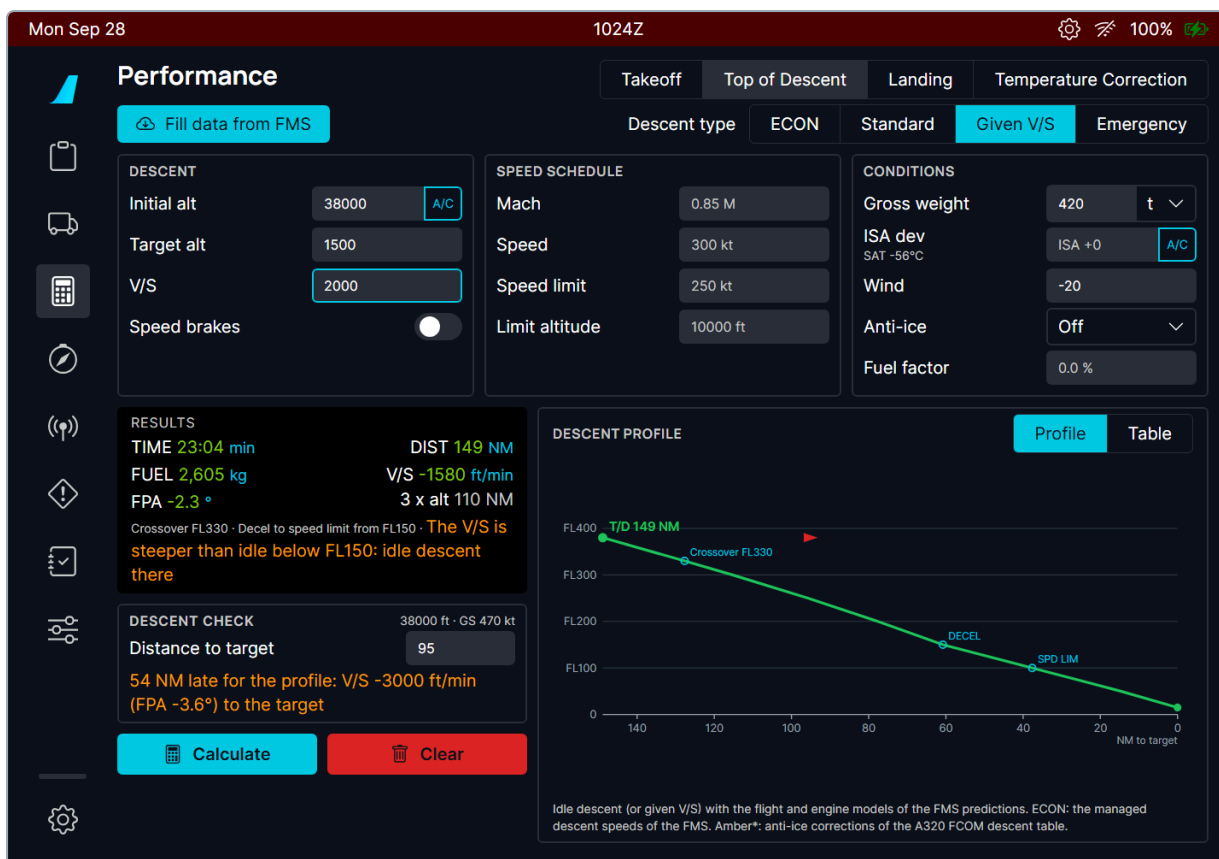


Figure 6. 2000 ft/min: shallower than idle at high level (149 NM, more fuel: 2605 kg), and steeper than idle below FL150, where the page flies idle.

Emergency uses MMO / VMO with the speed brakes extended (the schedule fields turn grey).

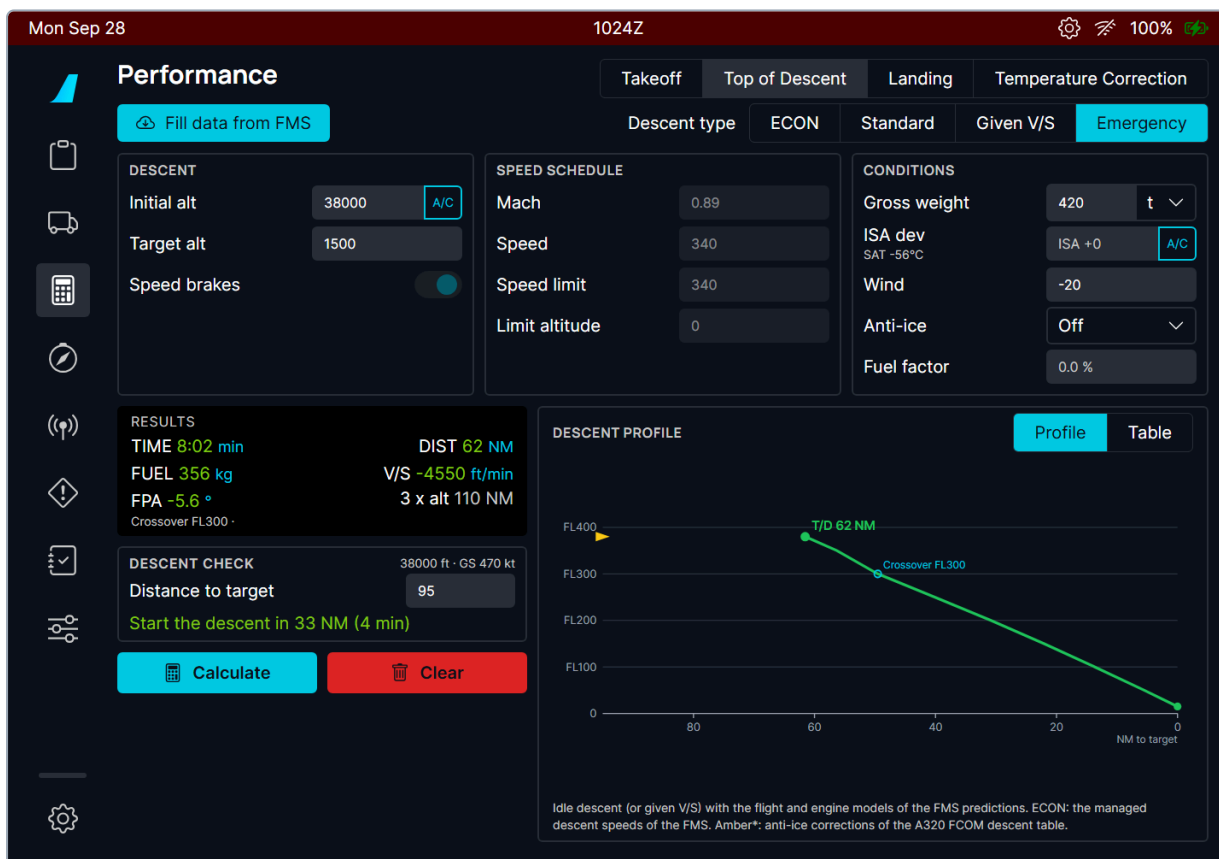


Figure 7. Emergency descent from FL380: 8 min, 62 NM, -4550 ft/min on average.

ECON needs the FMS: it asks the FMS for its managed descent speeds. Without a flight plan in the FMS, a message says so and the page stays on the previous type.

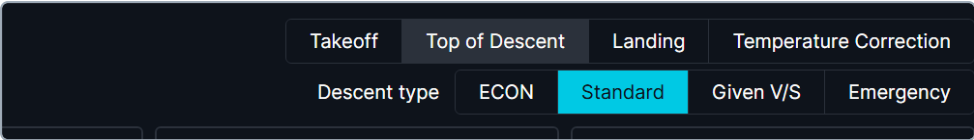


Figure 8. ECON without FMS data.

6. Anti-ice, fuel factor and estimates

Anti-ice (engine or total) raises the idle thrust and flattens the descent. Its effect uses the corrections of the A320 FCOM descent table (no A380 table is public): the values it changes turn **amber with a star**. The fuel factor (a percentage) scales the fuel burn, like the FMS PERF FACTOR.

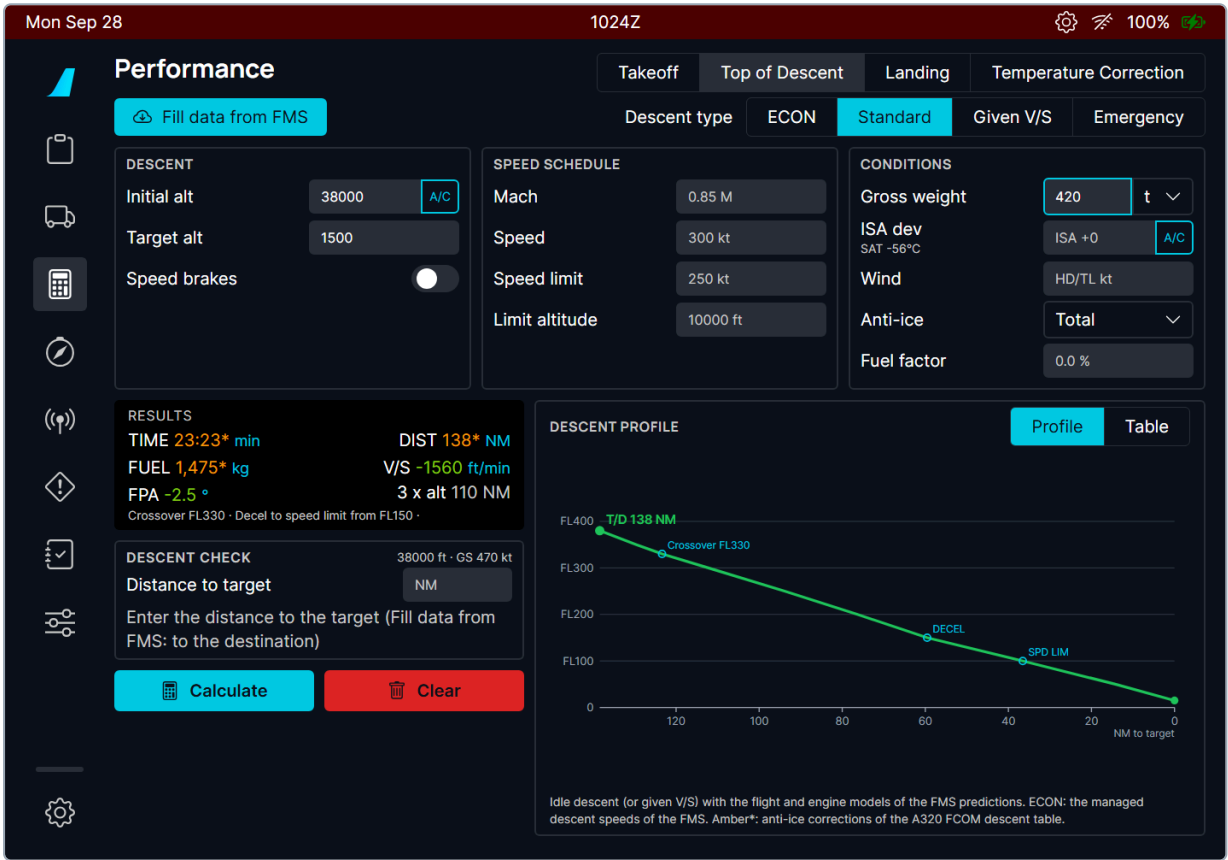


Figure 9. Total anti-ice: 138 NM, 23 min, 1475 kg, in amber (estimate).

7. What to expect

Situation	What you should see
Inputs missing	Amber labels; no results.
Any input changed	The results are cleared: press Calculate again.
Target above the initial altitude	Message "The target altitude must be below the initial altitude".
Initial altitude above the aircraft ceiling, or weight out of range	A message gives the limit (maximum altitude, or the weight range from the empty weight to the MTOW).
Given V/S steeper than idle	Amber line: idle descent below the altitude it names.
ECON without FMS flight plan	"No FMS data: enter a flight plan in the FMS".

Distance to target larger than the descent	Green "Start the descent in X NM (Y min)".
Distance shorter than the descent	Amber "N NM late for the profile" with the V/S and FPA needed.

8. Limitations and why

Limitation	Why	What to do
The descent is a still-air profile plus a mean wind.	The wind is one component for the whole descent, as in the FCOM and OIS; the FMS uses the wind of each altitude.	Use the mean wind of the descent.
About 8 % longer than the A320 FCOM table.	The FCOM table is for the CFM56 A320; the FBW A320neo engine model idles differently.	Compare with the FMS T/D, which uses the same model.
Anti-ice values are estimates.	Only the A320 FCOM has anti-ice descent corrections.	-
No descent path constraints or approach.	The page computes a clean descent to one target altitude, not the FMS path with its constraints.	Use it as a cross-check of the FMS.

A380X OANS default airport

The airport map of the ND (OANS) shows the right airport by itself: the current airport on ground, the origin, destination or alternate in flight, as the A380 FCOM describes the moving airport map

Aircraft	A380X
Where to find it	ND in ARC or ROSE-NAV at the ZOOM range (the OANS), or PLAN; the ARPT SEL panel of the OANS
PR branch	pr/25-a380x-oans-default-airport (independent)
Status	Tested in the sim on 2026-09-27 (user-confirmed).
Pictures	Captain ND in the local browser harness, running the build installed in the sim, with a small test airport (XTST, no buildings) standing in for the Navigraph airport map.

1. What it is for

2. The rules

3. What it looks like
4. What to expect

5. Limitations and why

1. What it is for

The OANS draws the airport around the aircraft on the ND. The FCOM (DSC-34-10-70-20, MOVING AIRPORT MAP) says no manual selection is needed: the OANS shows a **default airport**. FBW loaded the nearest airport on ground, but in flight only the destination (within 50 NM, at any height), and a manual selection stopped the automatic choice for 10 minutes. Now the FCOM rules apply.

2. The rules

ND mode	Default airport
ARC, ROSE-NAV (on ground)	The airport where the aircraft is.
ARC, ROSE-NAV (in flight)	The origin, destination or alternate of the FMS while the aircraft is inside its virtual cylinder: 20 NM radius, 5000 ft above the airport. The OANS keeps it while the aircraft stays in the cylinder.
PLAN (in flight)	The origin if it is closer than 50 NM, otherwise the destination; the destination when origin and destination are less than 300 NM apart.
Manual (ARPT SEL)	Only in PLAN: another airport can be displayed. It stays until the ND leaves PLAN, or the aircraft lands or takes off. In ARC / ROSE-NAV the OANS says SET PLAN MODE.

Only one airport is displayed on both NDs; the PLAN rule applies when the other ND is not in ARC or ROSE-NAV, so the two sides never fight.

3. What it looks like



Figure 1. In flight, 12 NM from the destination XTST at 3000 ft: the destination loads by itself (name top right; the arrow points to the airport, still outside the zoom range).



Figure 2. 30 NM away at 9000 ft: outside the cylinder, no airport is loaded.

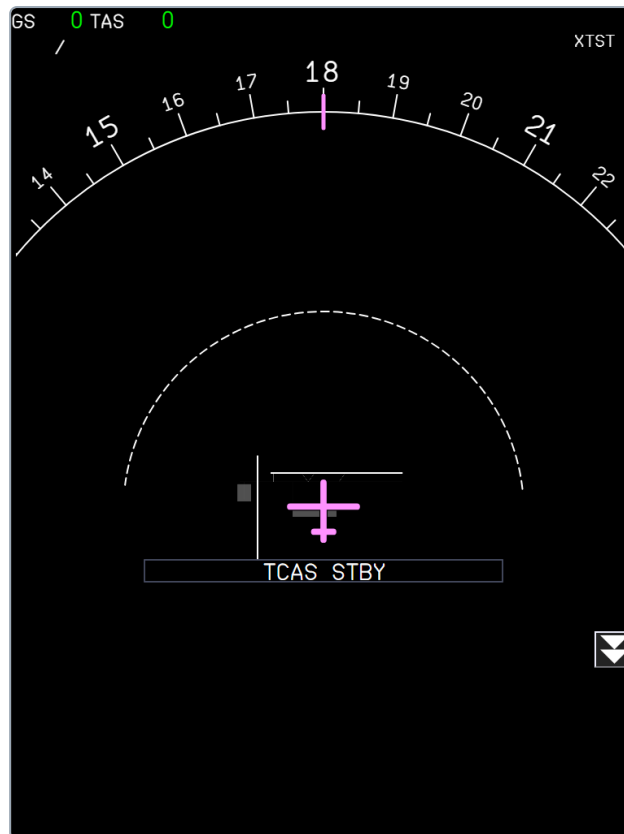


Figure 3. On ground at XTST: the current airport, with the aircraft at its gate.

4. What to expect

Situation	What you should see
On ground	The airport where you are, whatever the flight plan.
Climbing out of the origin	The origin until you leave its 20 NM / 5000 ft cylinder, then nothing (or the alternate / destination when inside theirs).
Approach to the destination	The destination from 20 NM and 5000 ft above it.
Diversion to the alternate	The alternate from 20 NM and 5000 ft above it.
ARPT SEL in PLAN	Your airport until you leave PLAN, land or take off.
No present position	No automatic airport.

5. Limitations and why

Limitation	Why	What to do
Needs the Navigraph airport database.	The OANS map is the Navigraph AMDB.	Link Navigraph in the flyPad.
One displayed airport for both NDs.	The real OANS has one per side; FBW shares one map.	-
The destination arrow (5 NM range, map out of range) is FBW's.	Not part of this change.	-

flyPad taxi route

Ground > Taxi: see the taxi route of your ATC clearance on the airport map, from the gate to the runway holding point and from the runway exit to the gate

Aircraft	A32NX and A380X (the same flyPad page on both)
Where to find it	flyPad > Ground (truck icon) > Taxi tab
PR branch	pr/26-efb-taxi-route
Status	Tested in the sim at RJTT (Tokyo Haneda) on 2026-09-28. The OANS flags (section 9) are built and deployed but not committed or tested in the sim yet.
Pictures	flyPad pages in the local browser harness, running the build installed in the sim, on a small test airport (XTST) made for the pictures. In the sim, the page uses the Navigraph airport map of the real airport.

1. What it is for
2. Before you start
3. Departure: from the gate to the runway
4. The ATC clearance
5. Runway crossings
6. Arrival: from the runway exit to the gate
7. From the aircraft position
8. The map
9. On the A380: the route on the OANS
10. What to expect
11. Limitations and why
12. Troubleshooting

1. What it is for

Before taxiing, the crew gets a taxi clearance from ATC such as "*taxi to holding point runway 09 via M, K, A*". On an unknown airport, following it on the chart is where mistakes happen: a wrong turn, a runway crossed without clearance. This page draws the route of the clearance on the airport map, lists the runways it crosses and stops it at the runway holding point.

The real A380 has no taxi routing: its OANS (the airport map on the ND) only lets the crew place green flags and white crosses (FCOM DSC-34-10-70-20), and MSFS has no datalink taxi clearance (D-TAXI). So the route is built on the flyPad, and the cockpit displays stay as the FCOM describes them. On the A380, the accepted route is marked on the OANS with those green flags (section 9).

- **Departure** (default): from a gate, a ramp or cargo stand, or the aircraft position, to a runway entry.
- **Arrival**: from the runway exit you plan to take (or the aircraft position) to your gate.
- The page **suggests** the shortest route; you type the taxiways of the ATC clearance and **accept** it.

2. Before you start

- **Navigraph account** linked in the flyPad settings (third party options). The airport map comes from the Navigraph airport database, the same data as the A380 OANS. Without it the page says "*No airport database*".
- **Airport**: type the ICAO code, or press **ORIG** / **DEST** to take the airports of the SimBrief flight plan loaded in the flyPad. The line below shows how many gates, ramp and cargo stands the map has.
- An airport that is not in the Navigraph airport database shows "*Not in the Navigraph airport database*".

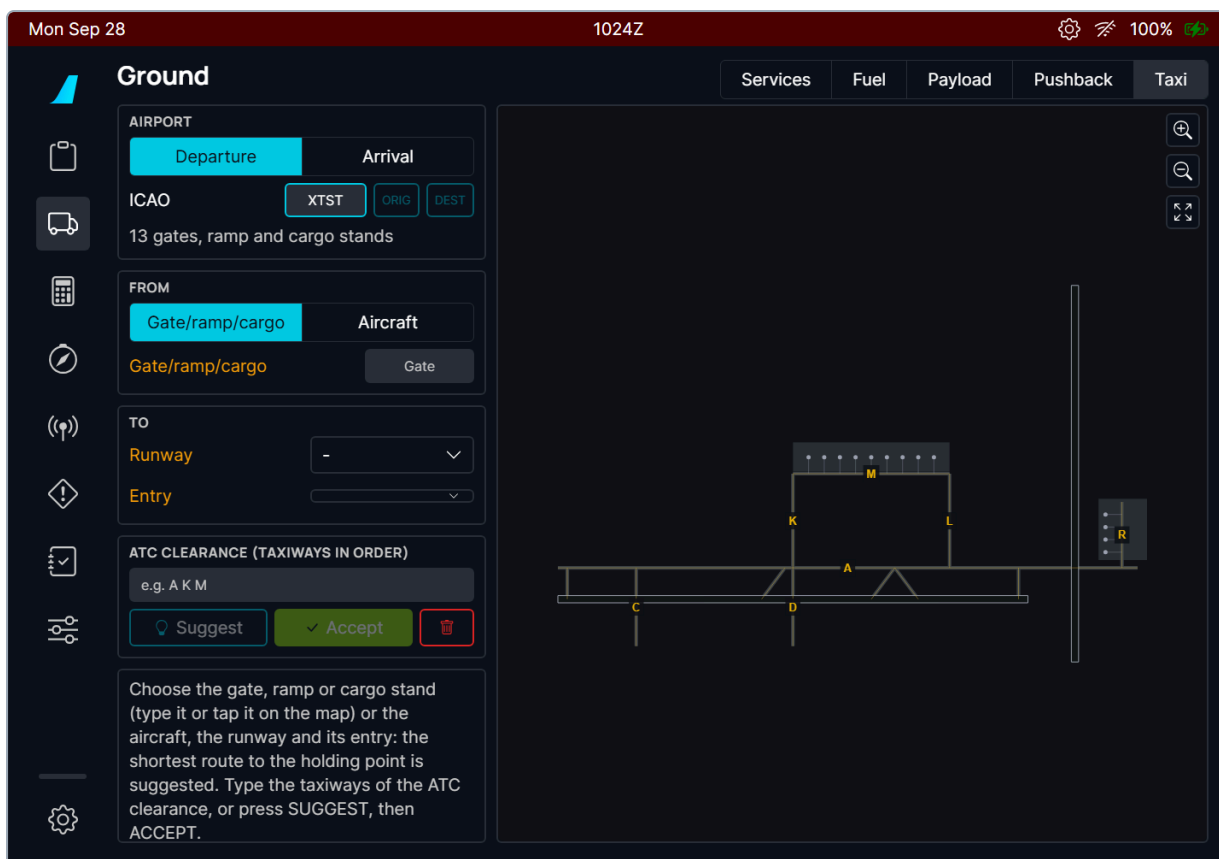


Figure 1. The Taxi page after entering the airport: the airport map on the right, the route inputs on the left. Amber labels show what is still missing.

Part of the page	What it does
AIRPORT	Departure or Arrival, the ICAO code, ORIG / DEST buttons, the number of stands on the map.
FROM	Where the taxi starts: a gate / ramp / cargo stand (departure), a runway exit (arrival), or the aircraft.
TO	Where it ends: the runway and its entry (departure), or the gate (arrival).
ATC CLEARANCE	The taxiways of the clearance, in order. SUGGEST fills it with the shortest route, ACCEPT confirms it, the red bin clears the page.
Route box	The route as a clearance reads, its length, the taxiways and the runway crossings in red.
Map	The airport with the route, the runway entries (cyan), the holding point (amber bar), the aircraft (yellow). Drag to move, buttons to zoom and to fit the route.

3. Departure: from the gate to the runway

1. Keep **Departure** selected and enter the airport.
2. In FROM, keep **Gate/ramp/cargo** and type your gate. While you type, the gates that match appear as buttons under the field: tap one. You can also **tap the gate on the map** (zoom in to see the names).
3. In TO, choose the **runway**. The **entry** is set to the full length entry; choose another one for an intersection departure.
4. The shortest route appears on the map as a **dashed magenta line** ending at the **HOLD** bar of the runway holding point.

FROM

Gate/ramp/cargo Aircraft

Gate/ramp/cargo 4

40 41 42 43 44 45 46 47

Figure 2. Typing "4" lists the gates 40 to 47: tap one.

FROM

Gate/ramp/cargo Aircraft

Gate/ramp/cargo 99

No such name on the airport map: tap the gate on the map (zoom in for the names)

Figure 3. A name that is not on the map: the amber line says so.

Gate names are found even when they are written differently on the map: "**16**" finds "**016**", spaces and dashes do not matter. If nothing matches, the airport map uses another name (for example a letter prefix): tap the gate on the map instead.

Mon Sep 28 1024Z 100%

Ground

AIRPORT

Departure Arrival

ICAO XTST ORIG DEST

13 gates, ramp and cargo stands

FROM

Gate/ramp/cargo Aircraft

Gate/ramp/cargo T1 42

TO

Runway 09

Entry S1 full length

ATC CLEARANCE (TAXIWAYS IN ORDER)

e.g. A K M

Suggest Accept

Suggested route (shortest)

Gate 42 → M → K → A → S1 → Holding point RWY 09

2.6 km · M K A S1

HOLD 09 S1 C B1 D

A K

Figure 4. Gate 42 to runway 09, full length: the suggested route (dashed) along M, K and A to entry S1, stopped at the holding point (HOLD 09). The route box reads it as a clearance: Gate 42 → M → K → A → S1 → Holding point RWY 09, 2.6 km.

Runway entries

The **Entry** list shows every taxiway that enters the runway, in order from the threshold. The first one is the **full length** entry; the others show the **runway length remaining** from them, to check an intersection departure against your takeoff performance. On the map, the entries of the selected runway are the cyan dots with their names.

TO

Runway 09 ▼

Entry S1 full length ▼

ATC CLEARANCE (TAXIWAYS IN ORDER)

e.g. A K M

💡 Suggest ✓ Accept 🗑️

Suggested route (shortest)
Gate 42 → M → K → A → Holding point RWY 09
2.6 km · M K A S1

C 2500 m

B1 1700 m

D 1500 m

B2 1000 m

B3 700 m

S2 60 m

Figure 5. The entries of runway 09: S1 full length, then C (2500 m left), B1 (1700 m), D (1500 m) and so on.

Mon Sep 28 1024Z 100%

Ground

AIRPORT

Departure Arrival

ICAO XTST ORIG DEST

13 gates, ramp and cargo stands

FROM

Gate/ramp/cargo Aircraft

Gate/ramp/cargo T1 42

TO

Runway 09 ▼

Entry D 1500 m ▼

ATC CLEARANCE (TAXIWAYS IN ORDER)

e.g. A K M

💡 Suggest ✓ Accept 🗑️

Suggested route (shortest)
Gate 42 → M → K → D → Holding point RWY 09
1.1 km · M K D

Services Fuel Payload Pushback Taxi

40 41 42 43 44

M K A

HOLD 09

Figure 6. Intersection departure from D: the route stops at the holding point of D, with 1500 m of runway ahead.

4. The ATC clearance

Type the taxiways of your clearance in the order ATC gives them, separated by spaces (commas and dashes work too). The route then follows them strictly, in that order:

- Short unnamed connecting lines between two cleared taxiways are allowed.
- The runway entry can be left out: "via M, K, A" to a holding point still ends on the entry you chose.
- While the typed route is not accepted, the box reads "Route along the clearance (not accepted)" and the line stays dashed.

SUGGEST copies the taxiways of the suggested route into the clearance field: a quick start when ATC clears you along the obvious way. **ACCEPT** confirms the route: the line turns **solid**. Any change (gate, runway, entry, clearance) takes

the acceptance away.

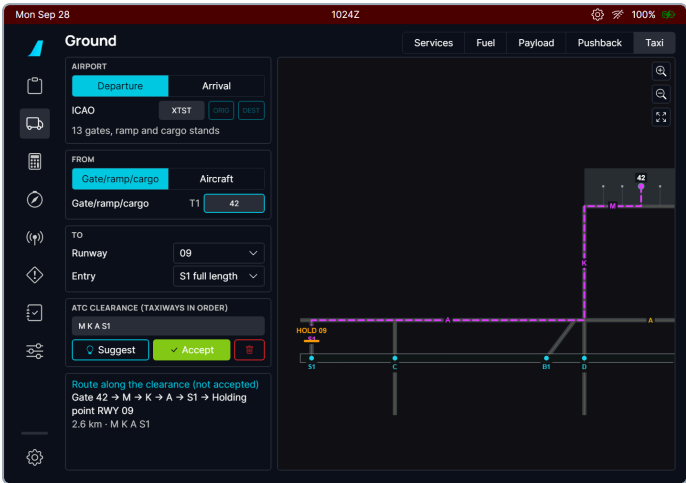


Figure 7. After SUGGEST: the clearance field holds M K A S1, the route is a preview.

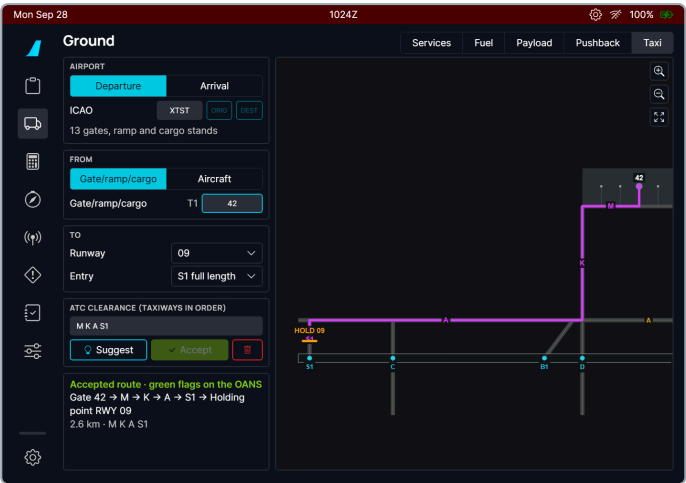


Figure 8. After ACCEPT: solid line, "Accepted route". On the A380 the route is also flagged on the OANS.

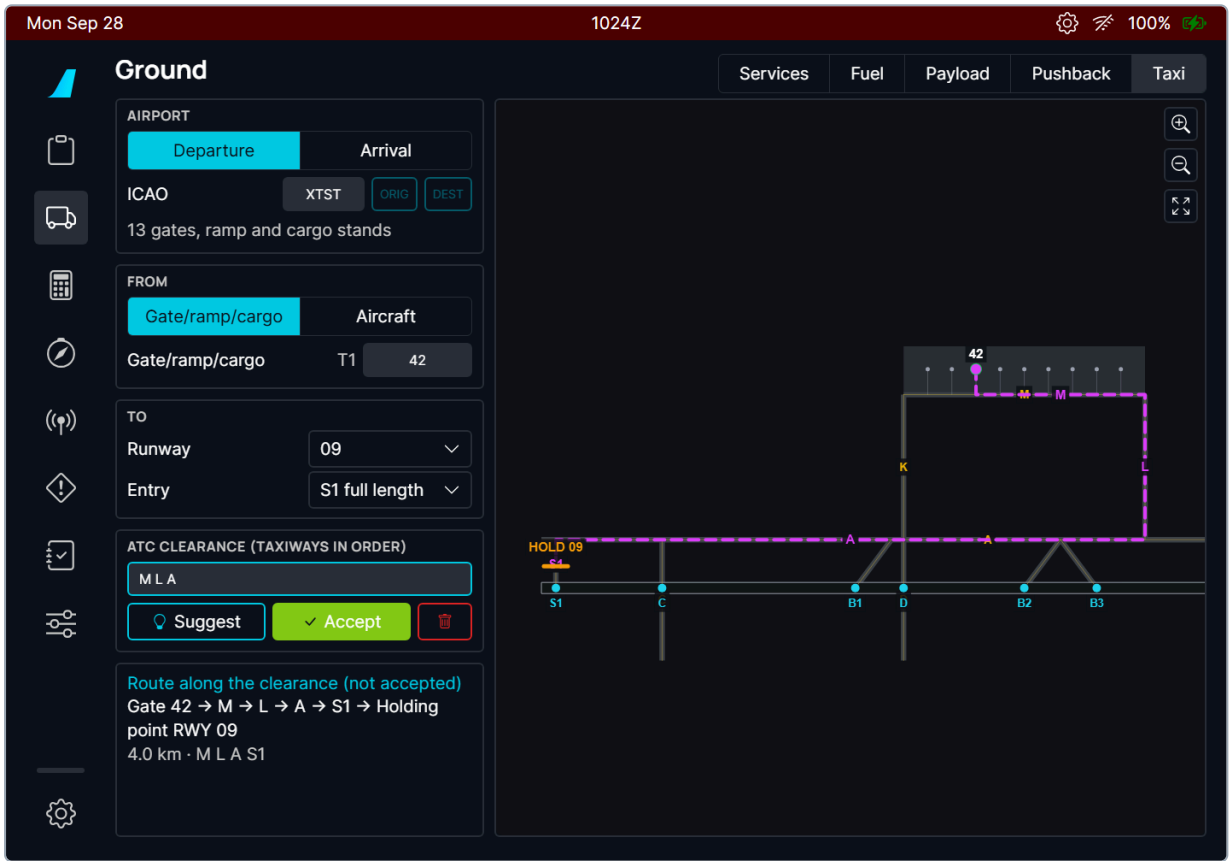


Figure 9. A different clearance, "M L A": the route goes the long way round via L (4.0 km instead of 2.6 km), as ATC asked.

3. In TO, type or tap the gate.
4. Check the route, type the clearance if ATC gives another one, and ACCEPT.

FROM

Runway exit Aircraft

Runway 09

Exit B1 1300 m

TO

Gate/ramp/cargo

ATC CLEARANCE (TAXIWAYS IN ORDER)

e.g. A K M

Suggest Accept

Figure 12. The exits of runway 09 with their distance from the threshold.

TIP

Prepare the arrival route during the descent briefing with the exit from the landing calculator: after landing, the page is ready. If you vacate by another exit, choose it (or use the aircraft position) and the route updates.

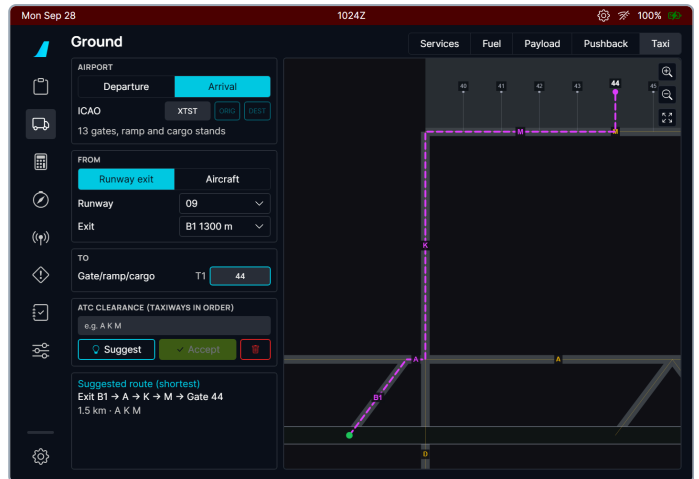


Figure 13. Exit B1 to gate 44: Exit B1 → A → K → M → Gate 44. The green dot is the start.

7. From the aircraft position

Choose **Aircraft** in FROM to start the route where the aircraft is, for example after a pushback, or when ATC changes the route while you taxi. The page takes the position when you select it; press **Update** to take it again. The route starts on the taxi line nearest to the aircraft and **prefers its heading**: it does not start by turning back if there is a way ahead.

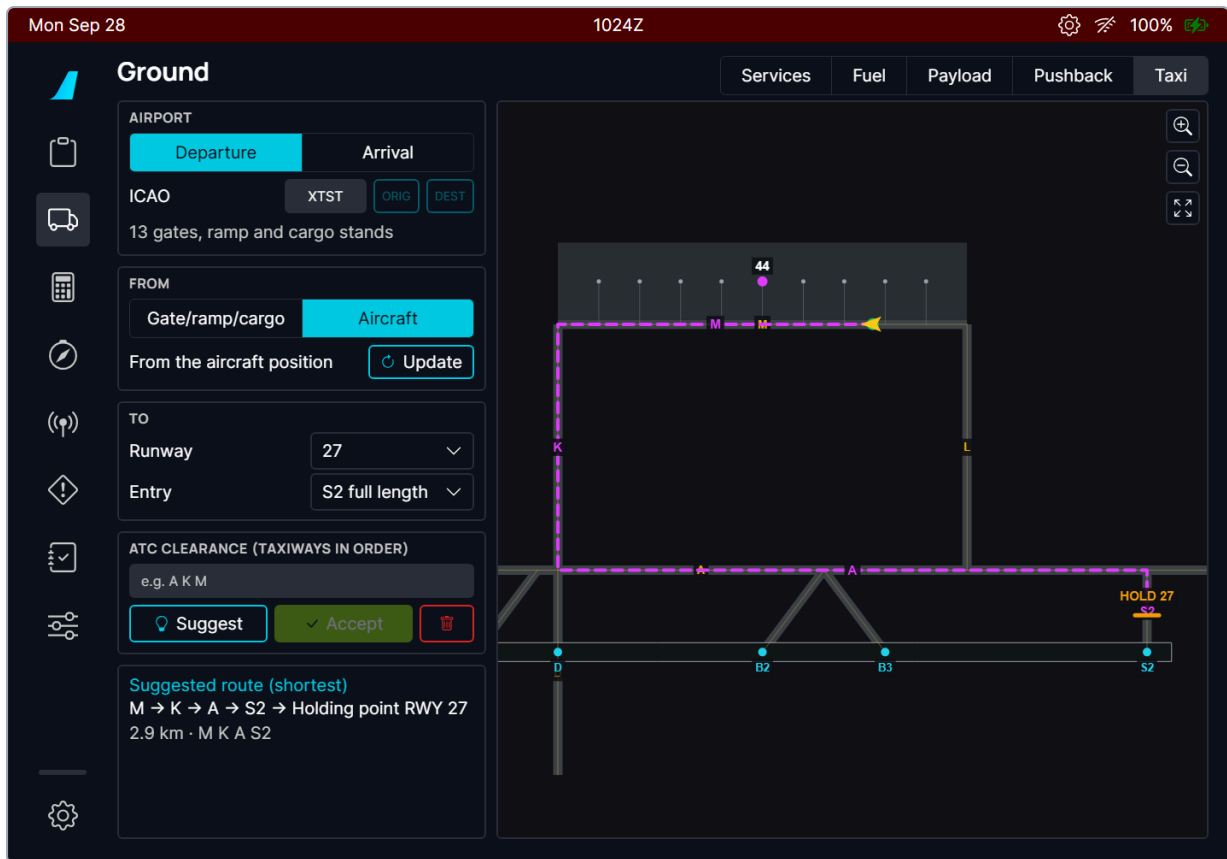


Figure 14. The aircraft (yellow) on M facing west, for runway 27: the route goes ahead via K and A (2.9 km) rather than turning back on M.

"The aircraft is not at this airport" means the aircraft is more than 8 km from the airport of the page.

8. The map

- **Drag** the map to move it; + / - zoom; the **frame** button fits the route (or the whole airport).
- The map is north up. Aprons are lighter grey, taxiways dark grey, runways black with a white edge. The thin yellow lines are the taxi guidance lines the route follows.
- Taxiway names are shown in yellow; the taxiways of the route in magenta. Gate names appear when you zoom in.
- **Tap a gate** on the map to choose it (when the page asks for a gate).

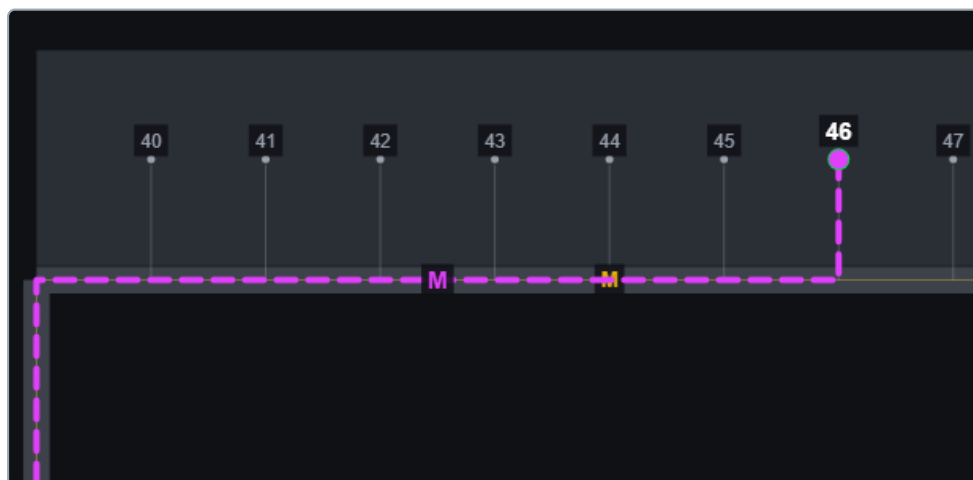


Figure 15. Zoomed on the apron: the gate names, the route to gate 46 along M.

9. On the A380: the route on the OANS

When you **accept** the route on the A380 flyPad, green flags appear on the OANS (the airport map on the ND) at the start of each taxiway of the route and at the gate or holding point. They are the crew flags of the FCOM ("*to locate or mark a given point on the airport*"), so you can remove them like any flag: click one and DELETE, or ERASE ALL SYMBOLS in the ND menu. Flags you placed yourself are never touched.

- The flags appear when the OANS displays that airport (they come back if the OANS reloads it).
- A new accepted route replaces the flags of the previous one; a route that is no longer accepted removes them.
- The A320 has no OANS: nothing else changes on it.

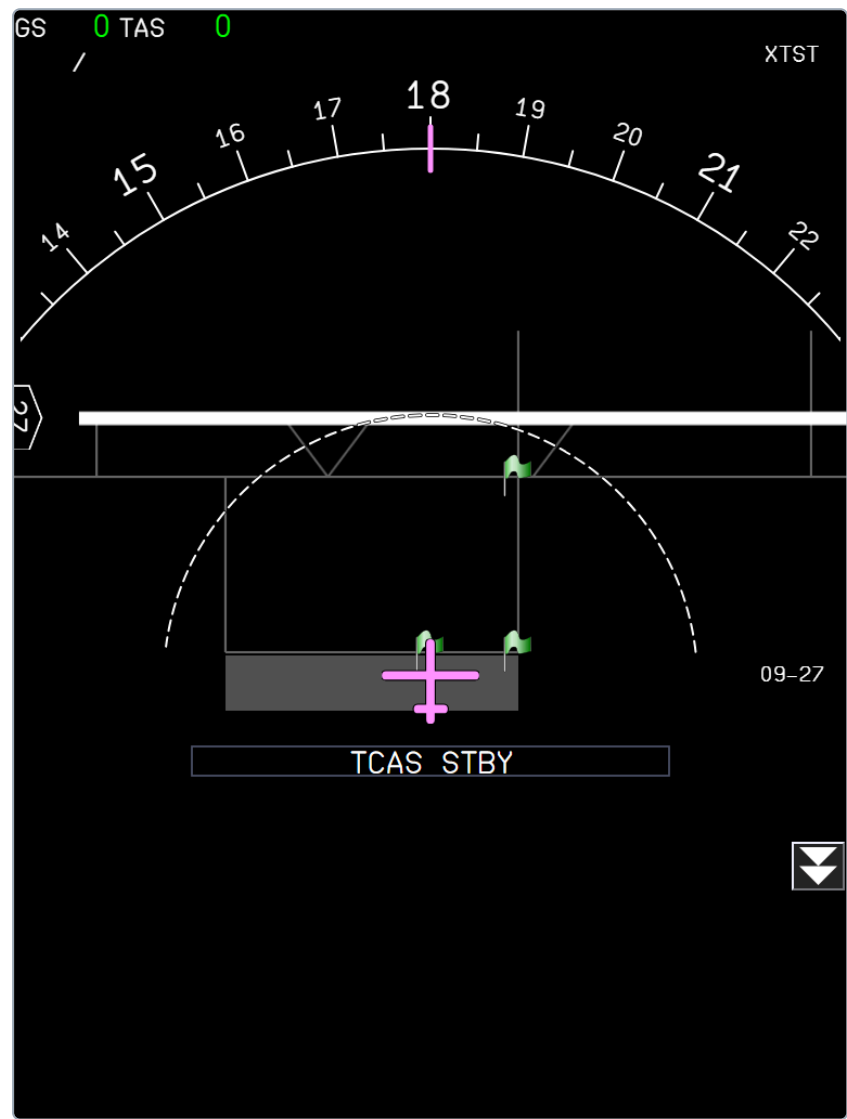


Figure 16. The A380 ND (OANS, ARC, zoom range) at gate 42 facing south, after ACCEPT of the route of figure 6: green flags where M, K and A start (S1 and the holding point are further west, off the picture). Harness picture: the test airport has no buildings.

NOTE
This part is built and deployed but not committed yet, and not tested in the sim.

10. What to expect

Situation	What you should see
Airport entered, nothing else	The airport map, amber labels on the inputs still missing, the help text in the route box.

Start and end chosen	The suggested route, dashed, with its clearance text, length and crossings.
SUGGEST	The clearance field holds the taxiways of the suggested route; the route is a preview (dashed).
Clearance typed	The route follows the typed taxiways in order (dashed until accepted), or the red message "No route along these taxiways, in this order".
ACCEPT	Solid line, "Accepted route" in green (A380: green flags on the OANS).
Any input changed after ACCEPT	Back to a preview: accept again.
Departure to a runway with holding position lines	The route stops at the HOLD bar before the runway.
Two entries behind the same HOLD bar (e.g. RJTT 34R C2 and C3)	The same route to the holding point: the entries only differ after it.
Gate whose lead-in is another stand's line (e.g. RJTT gate 5 behind 5R)	The route leaves along that line: "Gate 5 → Gate 5R → W ...".
Runway without holding position lines on the map	The route ends on the runway entry itself and the box says "Runway 09" instead of "Holding point RWY 09".
The route crosses a runway	Red circle with the runway name on the map, red "Crossing runway" line.
Aircraft start, aircraft far from the airport	"The aircraft is not at this airport"; no route.
Leaving the page and coming back	Everything is kept, except an Aircraft start: press Aircraft or Update again.

11. Limitations and why

Limitation	Why	What to do
Needs a Navigraph account.	The airport map (taxi lines, stands, holding points) exists only in the Navigraph airport database; MSFS does not give it to the aircraft.	Link your account in the flyPad settings.
Closed taxiways, one-way taxiways and NOTAMs are unknown.	The airport database has the lines painted on the ground, not their status of the day.	Type the taxiways of the ATC clearance: the route follows them.
The suggestion is the shortest route, not the usual one.	Airports have preferred flows (ATC habits, one-way loops) that are not in the data.	Always check against the clearance and the airport chart.
Some real airports have gaps in their taxi lines.	A lead-in line may stop a few metres short of the taxi line. Gaps up to 15 m between parts that are otherwise not connected are joined; larger ones are not.	If "No taxi line of the airport map connects the start to the end" appears, choose another start (the nearest taxiway, or the aircraft) or end.
Gate names follow the airport database.	The database may use names that differ from the gate signs.	Tap the gate on the map.
No route on the A380 OANS, only flags.	The A380 FCOM has no taxi route display; the flags are the FCOM way to mark points on the airport map.	-
The aircraft position is taken once.	The route is a plan to check, not a moving map that follows you.	Press Update after a change.

12. Troubleshooting

What you see	Why	What to do
SUGGEST and ACCEPT are greyed	There is no route yet: the gate is not found (amber label, no terminal next to the name), or the runway / entry / exit is missing.	Pick the gate from the suggestion buttons or tap it on the map; check the amber labels.
"No such name on the airport map"	No stand of the map has this name.	Tap the gate on the map.
"No taxi line of the airport map connects the start to the end"	The start and the end are on parts of the map that are not connected.	Start from the nearest taxiway (Aircraft), or choose another gate or entry.
"No route along these taxiways, in this order"	A taxiway of the clearance is misspelt, missing, or they do not connect in that order.	Check the clearance; press SUGGEST to see the names the map uses.
"Not in the Navigraph airport database"	Navigraph has no airport map for this airport.	-